FIT PM12 twelve-channel test campaign

FIT hardware acceptance bench

November 13, 2026

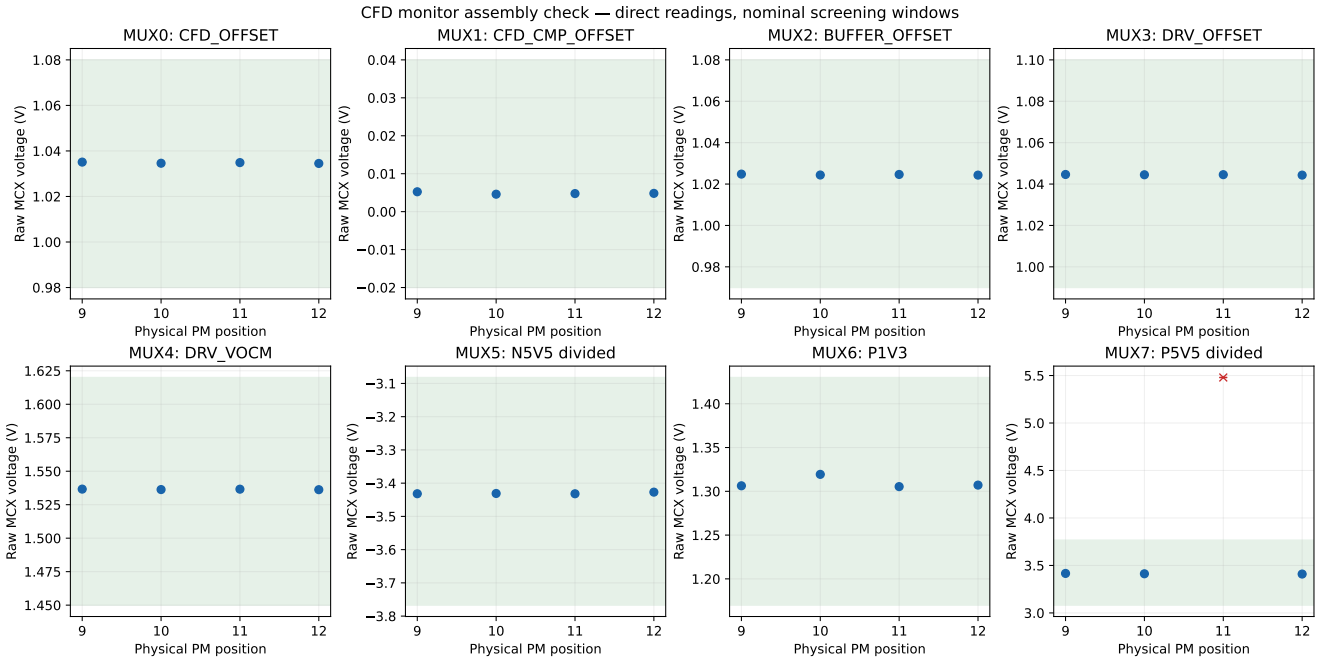
1 Assembly verification: current fitted group9–12

The present campaign is a functional assembly check. Unmeasured fixtures or missing stimuli do not receive PASS. The current group has two observed discrepancies; this document is not an unconditional board release.

Case	Observed result	Disposition
CFD monitor9/10/12	All eight inputs pass	PASS
CFD monitor11	Raw MUX7=5.479V; other modules about3.41V	FAIL: fixture branch/variant
JTAG/static links	168/168 in each independent path	PASS: scoped static check
CPU/FPGA and peripherals	32walking patterns restored; IDs/PGOOD correct	PASS
I2C rails and telemetry	Eight populated rail voltages in nominal windows	PASS; nominal currents
Ethernet/IPbus	393216ADC payload bytes checked; stable ADC/TDC transfers	PASS: functional transport
HDMI data lanes	Walking, slow/fast PRBS and idle frames correct	Data checks PASS
HDMI shared SPI release	SYS resets after pad release following reads0/1/2	Overall HDMI FAIL
Analog channel9	USB remains on10	NOT MEASURED; owner move pending

1.1 CFD monitor discrepancy is on the tester

The emulator at PM position11 (provisional CFD02) gives about5.479V directly at MUX7. Independent revisits reproduce it; position12 gives about3.410V. Check CFD SIM R3=33kOhm to ground, R4=20kOhm from P5V5 and their joints, or confirm an intended assembly variant. The nominal53/33 correction would yield8.80V, but that is not a measured8.80V PM supply. The independent PM INA bus reading is5.496V.



PM position	Mux	Signal	Raw MCX(V)	Nominal equivalent(V)	Screen
9	0	CFD_ OFFSET	1.035097	1.035097	within window
10	0	CFD_ OFFSET	1.034600	1.034600	within window
11	0	CFD_ OFFSET	1.034880	1.034880	within window
12	0	CFD_ OFFSET	1.034510	1.034510	within window
9	1	CFD_CMP_ OFFSET	0.005240	0.005240	within window
10	1	CFD_CMP_ OFFSET	0.004597	0.004597	within window
11	1	CFD_CMP_ OFFSET	0.004763	0.004763	within window
12	1	CFD_CMP_ OFFSET	0.004817	0.004817	within window
9	2	BUFFER_ OFFSET	1.024807	1.024807	within window
10	2	BUFFER_ OFFSET	1.024370	1.024370	within window
11	2	BUFFER_ OFFSET	1.024637	1.024637	within window
12	2	BUFFER_ OFFSET	1.024320	1.024320	within window
9	3	DRV_ OFFSET	1.044630	1.044630	within window
10	3	DRV_ OFFSET	1.044490	1.044490	within window
11	3	DRV_ OFFSET	1.044540	1.044540	within window
12	3	DRV_ OFFSET	1.044317	1.044317	within window

PM position	Mux	Signal	Raw MCX(V)	Nominal equivalent(V)	Screen
9	4	DRV_VOCM	1.536640	1.536640	within window
10	4	DRV_VOCM	1.536307	1.536307	within window
11	4	DRV_VOCM	1.536590	1.536590	within window
12	4	DRV_VOCM	1.536213	1.536213	within window
9	5	N5V5 divided	-3.431790	-5.511663	within window
10	5	N5V5 divided	-3.431030	-5.510442	within window
11	5	N5V5 divided	-3.431913	-5.511861	within window
12	5	N5V5 divided	-3.427043	-5.504039	within window
9	6	P1V3	1.306307	1.306307	within window
10	6	P1V3	1.319360	1.319360	within window
11	6	P1V3	1.305370	1.305370	within window
12	6	P1V3	1.307083	1.307083	within window
9	7	P5V5 divided	3.416020	5.486335	within window
10	7	P5V5 divided	3.412337	5.480419	within window
11	7	P5V5 divided	5.479303	8.800093	OUTSIDE WINDOW
12	7	P5V5 divided	3.409380	5.475671	within window

Nominal equivalent uses53/33 only for the divided rails. On the failed branch that assumption is suspect; use the direct reading for diagnosis. This is an assembly window check, not traceable calibration or a measurement of analog off-resistance.

1.2 HDMI reset isolation and restored operating state

Both full integrated runs retain the unexpected reset. Controlled isolation observes no reset during an idle-only enable/release or during the signature reads themselves. The counter clears when FPGA pads are released after SPI reads, before the DIO direction restoration. The exact electrical/MCU interaction remains unresolved; it must not be called a PM assembly defect solely from this test. Full HDMI remains FAIL until resolved. No arbitrary SPI writes were used.

Isolation event	Raw64-bit SYS uptime
initial	18902352588
before FPGA pad release	18902357244
after FPGA pad release	18903171692
outputs enabled	18918616796
idle inside	18918621884
before FPGA pad release	18918626578
after FPGA pad release	18919443448
after idle direction restore	18930711667
before FPGA pad release	18930722089
after FPGA pad release	18931546991
outputs enabled	18947031189
before read 0	18947035621
after read 0	18947072587
before read 1	18947077350
after read 1	18947107457
before read 2	18947110108
after read 2	18947137010
before FPGA pad release	18947139746
after FPGA pad release	40569538
after SPI direction restore	51888040

Isolation event	Raw64-bit SYS uptime
before FPGA pad release	51894323
after FPGA pad release	52713683

After isolation, fresh ADC and TDC transport checks passed with no SYS reset. The JTAG wrapper retains its ERROR from the combined HDMI guard; JTAG PASS here comes from independently rebuilt static vectors and checksum-verified restored SRAM followed by those stable checks. The separate HDMI failure remains visible.

Firmware still prints the old R15-not-fitted label for P3V8. The owner confirmed R15 fitted; the schematic specifies1mOhm. Host conversion of the retained signed shunt code at40uV/LSB gives about3.12A, uncalibrated. The original firmware text is retained. No MCU firmware was changed; route6/channel3 is unpopulated and excluded. Power-sequence fault injection is not performed.

Final state: qualified c7c42d28 diagnostic in FPGA SRAM, private PM bridge active, HDMI/CFD outputs released, DIO cards input, ADC bankA physical capture and frozen quiet TDC. SMU OFF/HIZ at0A, all four monitor muxes inhibited and TCA masks0. USB cables stay on10. Prior analog measurements retain their own acquisition settings; they are not silently remeasured after reconfiguration.

2 Scope and recorded coverage

Board PM12-K7-3A17A1B402212223. Campaign 5ef2e2b4-c04a-4fad-b86f-a23be621d7a7.

Recorded status: INCOMPLETE. Missing tests are NOT MEASURED. Complete selected-protocol coverage is not a calibrated whole-board release.

Channel	cfd-monitor	cfd-dacs	gate-phase	tdc	adc-dc	adc-lines	integrator
1	NM	NM	NM	NM	NM	NM	NM
2	NM	NM	NM	NM	NM	NM	NM
3	NM	NM	NM	NM	NM	NM	NM
4	NM	NM	NM	NM	NM	NM	NM
5	NM	NM	NM	NM	NM	NM	NM
6	NM	NM	NM	NM	NM	NM	NM
7	NM	NM	NM	NM	NM	NM	NM
8	NM	NM	NM	NM	NM	NM	NM
9	PASS	PASS	PASS	PASS	NM	NM	NM
10	PASS	PASS	PASS	PASS	PASS	PASS	PASS
11	FAIL	PASS	PASS	PASS	PASS	PASS	PASS
12	PASS	PASS	PASS	PASS	PASS	PASS	PASS

NM = not measured; INC = incomplete; ERR = acquisition error. No missing series is invented.

2.1 Board-level tests

Test	Status
Boundary scan and CPLD–FPGA connections	PASS
MCU, CPU–FPGA communication and peripheral IDs	PASS
I2C sensors, currents and voltages	PASS
HDMI digital link	FAIL
Ethernet/bridge diagnostic	PASS

3 Fitted batches and fixture provenance

Batch attempt	PM channel	CFD label	Declared connection	Fitting
ch09-12-attempt001	12	CFD01	slot 7, RJ45 0	Identity/fitting confirmed
ch09-12-attempt001	11	CFD02	slot 7, RJ45 1	Identity/fitting confirmed
ch09-12-attempt001	10	CFD03	slot 7, RJ45 2	Identity/fitting confirmed
ch09-12-attempt001	9	CFD04	slot 7, RJ45 3	Identity/fitting confirmed

Fixture labels CFD01–CFD04 are provisional bench labels. Factory identities and photos remain unknown until supplied; a label is not proof of module identity. Declared cable routes must be checked physically.

Raw inputs and selected result IDs are retained in inputs/. Repeated measurements are never silently substituted.

4 Retained attempts and explicit selection

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-b9126235	adc-dc	12	ch09-12-attempt001	PASS	yes
result-ee0613ea	adc-lines	12	ch09-12-attempt001	PASS	yes
result-217f5cb4	integrator	12	ch09-12-attempt001	PASS	yes
result-b99efe3a	cfd-dacs	9	ch09-12-attempt001	INCOMPLETE	no
result-ef50493e	cfd-dacs	10	ch09-12-attempt001	INCOMPLETE	no
result-c4800a66	cfd-dacs	11	ch09-12-attempt001	INCOMPLETE	no
result-c80f3b9e	cfd-dacs	12	ch09-12-attempt001	INCOMPLETE	no
result-9e64b43e	cfd-dacs	9	ch09-12-attempt001	PASS	yes
result-0fcaadd7	cfd-dacs	10	ch09-12-attempt001	PASS	yes
result-8b6b8aa7	cfd-dacs	11	ch09-12-attempt001	PASS	yes
result-b4c6c497	cfd-dacs	12	ch09-12-attempt001	PASS	yes
result-aa2f2b7c	adc-dc	11	ch09-12-attempt001	PASS	yes
result-fa68f809	adc-lines	11	ch09-12-attempt001	PASS	yes
result-021f9e97	integrator	11	ch09-12-attempt001	PASS	yes

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-317b8d60	adc-dc	10	ch09-12-attempt001	PASS	yes
result-0386d766	adc-lines	10	ch09-12-attempt001	PASS	yes
result-e87c1c5e	integrator	10	ch09-12-attempt001	PASS	yes
result-1642153e	tdc	12	ch09-12-attempt001	PASS	yes
result-5f833ba0	tdc	11	ch09-12-attempt001	PASS	yes
result-cdf2341e	tdc	10	ch09-12-attempt001	PASS	yes
result-5e3a4c64	tdc	9	ch09-12-attempt001	PASS	yes
result-d2e5cd6c	gate-phase	12	ch09-12-attempt001	PASS	yes
result-769a5717	gate-phase	11	ch09-12-attempt001	PASS	yes
result-5b389adb	gate-phase	10	ch09-12-attempt001	PASS	yes
result-a00721bb	gate-phase	9	ch09-12-attempt001	PASS	yes
result-a2a60491	cfd-monitor	9	ch09-12-attempt001	PASS	yes
result-161d19ce	cfd-monitor	10	ch09-12-attempt001	PASS	yes
result-fb9647b0	cfd-monitor	11	ch09-12-attempt001	FAIL	yes
result-88ebf7d4	cfd-monitor	12	ch09-12-attempt001	PASS	yes
result-4f22ebeb	jtag	board	ch09-12-attempt001	PASS	yes
result-5ab3262e	mcu	board	ch09-12-attempt001	PASS	yes
result-0a5dcd9c	i2c	board	ch09-12-attempt001	PASS	yes
result-6231a067	ethernet	board	ch09-12-attempt001	PASS	yes
result-10b214db	hdmi	board	ch09-12-attempt001	FAIL	yes

All attempts, including failures and rejected repeats, remain in the archived inputs. The coverage matrix uses only the explicitly selected attempt.

5 Selected measurement conditions and limits

Test	Physical channel	Confirmed stimulus connection
adc-dc	12	{"dac_channels": [0, 1], "stimulus_channel": 12, "usb_input": "J8"}
adc-lines	12	{"dac_channels": [0, 1], "stimulus_channel": 12, "usb_input": "J8"}
integrator	12	{"dac_channels": [2, 3], "stimulus_channel": 12, "usb_input": "J6"}
cfd-dacs	9	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}
cfd-dacs	10	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}
cfd-dacs	11	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}
cfd-dacs	12	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}
adc-dc	11	{"dac_channels": [0, 1], "stimulus_channel": 11, "usb_input": "J8"}
adc-lines	11	{"dac_channels": [0, 1], "stimulus_channel": 11, "usb_input": "J8"}
integrator	11	{"dac_channels": [2, 3], "stimulus_channel": 11, "usb_input": "J6"}
adc-dc	10	{"dac_channels": [0, 1], "stimulus_channel": 10, "usb_input": "J8"}
adc-lines	10	{"dac_channels": [0, 1], "stimulus_channel": 10, "usb_input": "J8"}
integrator	10	{"dac_channels": [2, 3], "stimulus_channel": 10, "usb_input": "J6"}
tdc	12	{"phase_lanes": [8, 9], "rj45_port": 0, "slot": 7, "stimulus_dio": 0}
tdc	11	{"phase_lanes": [10, 11], "rj45_port": 1, "slot": 7, "stimulus_dio": 4}
tdc	10	{"phase_lanes": [12, 13], "rj45_port": 2, "slot": 7, "stimulus_dio": 8}

Test	Physical channel	Confirmed stimulus connec tion
tdc	9	{”phase_lanes”: [14, 15], ”rj45_port”: 3, ”slot”: 7, ”stimulus_dio”: 12}
gate-phase	12	{”phase_lanes”: [8, 9], ”rj45_port”: 0, ”slot”: 7, ”stimulus_dio”: 0}
gate-phase	11	{”phase_lanes”: [10, 11], ”rj45_port”: 1, ”slot”: 7, ”stimulus_dio”: 4}
gate-phase	10	{”phase_lanes”: [12, 13], ”rj45_port”: 2, ”slot”: 7, ”stimulus_dio”: 8}
gate-phase	9	{”phase_lanes”: [14, 15], ”rj45_port”: 3, ”slot”: 7, ”stimulus_dio”: 12}
cfd-monitor	9	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}
cfd-monitor	10	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}
cfd-monitor	11	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}
cfd-monitor	12	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}
jtag	board	{}
mcu	board	{}
i2c	board	{}
ethernet	board	{}
hdmi	board	{}

5.1 Common hardware configuration

Setting	Recorded value
clock_profile.adc_rate	2x reference
clock_profile.kasli_rate	3x reference
clock_profile.reference_mhz	40.0787
cpld_gate_mode	original INTA/INTB enabled, nomina l20MHz
kasli_build.ABI	CFD4
kasli_build.expected_bit_sha256	a71a08a258620b0e5a52c5d03aefd8c69040 c9d15e8de6391c484ac182b7173b
pm_build	c7c42d28

5.2 adc-dc

Setting	Recorded value
method	”direct-ADC-DC-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

5.3 adc-lines

Setting	Recorded value
method	”adjacent-pair-states-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

5.4 cfd-dacs

Setting	Recorded value
method	”shared-monitor-SMU-BCD-v1”
instrument_identity	”Keysight Technologies,B2910BL,MY6332 0299,5.0.2429.8966”
sense	”Force HI/LO, local”
source_current_A	0
source_range_A	1e-06
voltage_range_V	20
nplc	1
settling_s	0.1
points	101
repeats	3
limits	{”repeat_std_V”: 0.001, ”reverse_step_ V”: 0.0002, ”span_relative_error”: 0.05}
post_scan_verification	”SMU OFF/HIZ, four monitor muxes inhibi ted, explicit TCA masks0”

5.5 cfd-monitor

Setting	Recorded value
method	”eight-monitor-assembly-v1”

Setting	Recorded value
windows_V	[[0.98, 1.08], [-0.02, 0.04], [0.97, 1.08], [0.99, 1.1], [1.45, 1.62], [-6.05, -4.95], [1.17, 1.43], [4.95, 6.05]]
relative_offset_windows_V	[[0.005, 0.016], [0.012, 0.03]]
vocm_ratio_window	[1.45, 1.55]
raw_repeat_std_limit_V	0.002
divider_correction	1.606060606060606
settle_s	0.2
samples_per_input	3
source_current_A	0
sense	”local Force HI/LO”
nplc	1
range_V	20

5.6 ethernet

Setting	Recorded value
method	”board-assembly-ethernet-v1”
scope	”separate static-link/CPU/sensor/network/HDMI cases; shared-SPI clock guard retained”

5.7 gate-phase

Setting	Recorded value
method	”DAC-A-ISERDES-gate-phase-v1”
codes	[0, 655, 1311, 1966, 2621, 3277, 3932, 4587, 5243, 5898, 6554, 7209, 7864, 8520, 9175, 9830, 10486, 11141, 11796, 12452, 13107, 13762, 14418, 15073, 15728, 16384, 17039, 17694, 18350, 19005, 19661, 20316, 20971, 21627, 22282, 22937, 23593, 24248, 24903, 25559, 26214, 26869, 27525, 28180, 28835, 29491, 30146, 30801, 31457, 32112, 32768, 33423, 34078, 34734, 35389, 36044, 36700, 37355, 38010, 38666, 39321, 39976, 40632, 41287, 41942, 42598, 43253, 43908, 44564, 45219, 45875, 46530, 47185, 47841, 48496, 49151, 49807, 50462, 51117, 51773, 52428, 53083, 53739, 54394, 55049, 55705, 56360, 57015, 57671, 58326, 58982, 59637, 60292, 60948, 61603, 62258, 62914, 63569, 64224, 64880, 65535]
sample_hz	961888800.0
gate_period_bins	48
capture_words	4095
minimum_functional_excursion_ns	5

5.8 hdmi

Setting	Recorded value
method	”board-assembly-hdmi-v1”
scope	”separate static-link/CPU/sensor/network/HDMI cases; shared-SPI clock guard retained”

5.9 i2c

Setting	Recorded value
method	”board-assembly-i2c-v1”
scope	”separate static-link/CPU/sensor/network/HDMI cases; shared-SPI clock guard retained”

5.10 integrator

Setting	Recorded value
method	”signed-integration-pair-v1”
dac_channels	[2, 3]
base_V	[1.024, 2.048]
coarse_offsets_V	[-0.09999999999999987, -0.080000000000 00007, -0.05999999999999983, -0.040000 000000000036, -0.019999999999999796, -0.010000000000000009, 0.0, 0.009999999999 999787, 0.02000000000000024, 0.040000 000000000036, 0.060000000000000275, 0.08000000000000007, 0.10000000000000009]
fine_method	”26 points,0.4mV step around coarse estimate”
return_method	”two candidate revisits and +/-5mV offsets;near-zero <=2codes; signed response >4codes”

5.11 jtag

Setting	Recorded value
method	”board-assembly-jtag-v1”
scope	”separate static-link/CPU/sensor/network/HDMI cases; shared-SPI clock guard retained”

5.12 mcu

Setting	Recorded value
method	”board-assembly-mcu-v1”

Setting	Recorded value
scope	”separate static-link/CPU/sensor/network/ HDMI cases; shared-SPI clock guard retain ed”

5.13 tdc

Setting	Recorded value
method	”finite-OSERDES-TDC-phase-v1”
phase_slots	[24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48]
sample_hz	961888800.0
events_per_phase	128
discovery_events	16
response_slope_window	[-60, -20]
maximum_residual_fine_codes	10

Commanded DAC voltages, whole connected chain; not calibrated ADC-only INL/noise or electrical continuity proof.

Effective differential whole-chain balance; waveform-based integration pair, not an absolute gate timestamp or calibrated opamp bias.

Fine modulo128 unwrapped by nearest continuation; coarse fields retained without repair. Progra mmed slot1.0396ns assumes40.0787MHz; not calibrated TDC timing or intrinsic jitter.

Functional assembled-chain transfer check: span within5% of schematic scaling, reverse step <=0.2mV, repeat standard deviation <=1mV. Not traceable calibration or exhaustive16-bit INL/DNL. DAC-A unchanged; exercised B/C/D end at0000..

Functional assembly evidence; no supply fault injection, calibrated current accuracy, Ethernet throug hput/BER or unconnected analog-channel acceptance.

Functional assembly windows and relative reference checks; negative/positive supply dividers correc ted by53/33. Three fresh0A-SMU voltage readings per input; no traceable voltage calibration or analog off-resistance claim.

Occupancy-weighted sampled phase centroid; nominal1.0396ns bins and49.9ns gate period. Neares t-continuation unwrap; discontinuities retained. No calibrated sub-bin precision or independent-edge sqrt(N) uncertainty. DAC A commanded0000 at cleanup, without hardware code readback.

integrator; physical channel 10: The coarse waveform assignment produces an additional positive-going sign crossing near +40mV. It was not accepted as balance. Fine acquisition used the central negati ve-going crossing for this reversed input coordinate, then confirmed it with independent candidate revisi ts and +/-5mV controls. The original coarse data and both controller versions remain archived.

cfd-monitor; physical channel 11: Fixture anomaly on CFD emulator mounted at PM11 (provis ional CFD02): raw MUX7 reads5.479V, versus3.41V on9/12, confirmed by two five-reading revisits. Nominal53/33 scaling would give8.80V but this is NOT a measured8.80V PM supply. Check CFD_SIM R3=33kOhm to ground, R4=20kOhm fromP5V5 and their joints/variant. Other monitor inputs and independent board P5V5 bus about5.496V are normal. This FAIL concerns the tester monitor branch, not proof of PM channel11 failure.

jtag; physical channel board: Legacy combined boundary wrapper reports ERROR because its post-restore HDMI check resets SYS. The independently rebuilt338vectors in each path all PASS and checks um-verified SRAM restoration is confirmed by fresh stable ADC/TDC transfers. This JTAG case is scoped to static links and restoration; the separate HDMI case remains FAIL.

i2c; physical channel board: Firmware still labels R15/P3V8 current as not fitted. Owner confirmed R15 fitted; schematic R15=1mOhm, so host converts the retained signed13-bit shunt code at40uV/LSB.

Original NA/status text is retained and no MCU firmware was changed. Route6/channel3 is unpopu lated and excluded.

hdmi; physical channel board: All four HDMI data lanes pass walking patterns, slow/fast PRBS and empty frames. Signature reads0/1/2 are correct, but SYS uptime resets when FPGA pads are released after SPI reads, before DIO direction restoration. Idle-only output enable/release does not reset. Two integrated runs and controlled isolation retain the failure; shared-MCU SPI release remains unqualified and is not assigned to a PM assembly defect without further isolation.

ADC total RMS includes the complete connected chain and periodic gate feedthrough. The four-phase-removed RMS removes the mean of each sample phase; it is a residual statistic, not an intrinsic ADC noise specification. FFT and time axes use the declared clock frequency, not an independent frequency measurement.

6 Measurements and common channel plots

Test	Channel	Bank	Observable	Value	Unit
adc-dc	12	A	gain	-1784.205440671235	code/V
adc-dc	12	A	zero_mean	2162.0811339198435	code
adc-dc	12	A	zero_total_rms	56.22754588987761	code
adc-dc	12	A	zero_four_phase_removed_rms	0.5926371250728474	code
adc-dc	12	B	gain	-1784.2694869545635	code/V
adc-dc	12	B	zero_mean	2165.8475073313784	code
adc-dc	12	B	zero_total_rms	64.40894678040004	code
adc-dc	12	B	zero_four_phase_removed_rms	0.5625571866355403	code
adc-lines	12	A	adjacent_pair_states	44	state
adc-lines	12	A	visited_codes	3147	code
adc-lines	12	B	adjacent_pair_states	44	state
adc-lines	12	B	visited_codes	3184	code
integrator	12	A	balance_delta	0.006030008721686399	V
integrator	12	A	gain	-2530.411960665549	code/V
integrator	12	A	near_balance_change	0.13016491045236994	code
integrator	12	B	balance_delta	0.007424191555263186	V
integrator	12	B	gain	-2448.0077895617496	code/V
integrator	12	B	near_balance_change	0.11194322217075807	code

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	9		dac_B_minimum_code_V	1.0447266666666666	V
cfd-dacs	9		dac_B_maximum_code_V	1.0046133333333334	V
cfd-dacs	9		dac_B_span_V	-0.04011333333322	V
cfd-dacs	9		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_B_maximum_repeat_std_V	7.023769168570249e-05	V
cfd-dacs	9		dac_B_interior_gain_error_percent	-0.006716857635891138	%
cfd-dacs	9		dac_C_minimum_code_V	1.0349566666666667	V
cfd-dacs	9		dac_C_maximum_code_V	1.0143966666666666	V
cfd-dacs	9		dac_C_span_V	-0.020560000000000134	V
cfd-dacs	9		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_C_maximum_repeat_std_V	9.504384952928754e-05	V
cfd-dacs	9		dac_C_interior_gain_error_percent	0.8518971864797997	%
cfd-dacs	9		dac_D_minimum_code_V	0.0054	V
cfd-dacs	9		dac_D_maximum_code_V	1.06306	V
cfd-dacs	9		dac_D_span_V	1.0576599999999998	V
cfd-dacs	9		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_D_maximum_repeat_std_V	7.000000000000657e-05	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	9		dac_D_interior_gain_error_percent	-0.026880429186237897	%
cfd-dacs	10		dac_B_minimum_code_V	1.04437	V
cfd-dacs	10		dac_B_maximum_code_V	1.00438	V
cfd-dacs	10		dac_B_span_V	-0.039989999999999997	V
cfd-dacs	10		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_B_maximum_repeat_std_V	8.082903768656434e-05	V
cfd-dacs	10		dac_B_interior_gain_error_percent	-0.14222304801190244	%
cfd-dacs	10		dac_C_minimum_code_V	1.0346766666666667	V
cfd-dacs	10		dac_C_maximum_code_V	1.0141433333333334	V
cfd-dacs	10		dac_C_span_V	-0.020533333333292	V
cfd-dacs	10		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_C_maximum_repeat_std_V	9.539392001416794e-05	V
cfd-dacs	10		dac_C_interior_gain_error_percent	0.5828293358783654	%
cfd-dacs	10		dac_D_minimum_code_V	0.004546666666666667	V
cfd-dacs	10		dac_D_maximum_code_V	1.0692933333333334	V
cfd-dacs	10		dac_D_span_V	1.0647466666666667	V
cfd-dacs	10		dac_D_maximum_reverse_step_V	0	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	10		dac_D_maximum_repeat_std_V	7.000000000003828e-05	V
cfd-dacs	10		dac_D_interior_gain_error_percent	0.042824877925129456	%
cfd-dacs	11		dac_B_minimum_code_V	1.0446666666666666	V
cfd-dacs	11		dac_B_maximum_code_V	1.00461	V
cfd-dacs	11		dac_B_span_V	-0.04005666666666663	V
cfd-dacs	11		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_B_maximum_repeat_std_V	7.549834435276095e-05	V
cfd-dacs	11		dac_B_interior_gain_error_percent	-0.1743207749736131	%
cfd-dacs	11		dac_C_minimum_code_V	1.0349033333333333	V
cfd-dacs	11		dac_C_maximum_code_V	1.0143866666666665	V
cfd-dacs	11		dac_C_span_V	-0.02051666666666674	V
cfd-dacs	11		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_C_maximum_repeat_std_V	7.234178138066471e-05	V
cfd-dacs	11		dac_C_interior_gain_error_percent	0.17720494086375105	%
cfd-dacs	11		dac_D_minimum_code_V	0.00485	V
cfd-dacs	11		dac_D_maximum_code_V	1.0636333333333334	V
cfd-dacs	11		dac_D_span_V	1.0587833333333334	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	11		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_D_maximum_repeat_std_V	6.110100926609052e-05	V
cfd-dacs	11		dac_D_interior_gain_error_percent	0.026777441441017125	%
cfd-dacs	12		dac_B_minimum_code_V	1.04463	V
cfd-dacs	12		dac_B_maximum_code_V	1.0044266666666666	V
cfd-dacs	12		dac_B_span_V	-0.040203333333333337	V
cfd-dacs	12		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_B_maximum_repeat_std_V	4.9328828623155544e-05	V
cfd-dacs	12		dac_B_interior_gain_error_percent	0.06510873936658168	%
cfd-dacs	12		dac_C_minimum_code_V	1.0348066666666667	V
cfd-dacs	12		dac_C_maximum_code_V	1.0144333333333333	V
cfd-dacs	12		dac_C_span_V	-0.020373333333333354	V
cfd-dacs	12		dac_C_maximum_reverse_step_V	3.3333333335772153e-06	V
cfd-dacs	12		dac_C_maximum_repeat_std_V	0.00011846237095938458	V
cfd-dacs	12		dac_C_interior_gain_error_percent	-1.3950279374863153	%
cfd-dacs	12		dac_D_minimum_code_V	0.00498	V
cfd-dacs	12		dac_D_maximum_code_V	1.06321	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	12		dac_D_span_V	1.05823	V
cfd-dacs	12		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_D_maximum_repeat_std_V	8.020806 27701036e-05	V
cfd-dacs	12		dac_D_interior_gain_error_percent	-0.037869 6257134159	%
adc-dc	11	A	gain	-1783.308532 4525614	code/V
adc-dc	11	A	zero_mean	2163.592375 366569	code
adc-dc	11	A	zero_total_rms	61.455886 79712764	code
adc-dc	11	A	zero_four_phase_removed_rms	0.539425 9839466975	code
adc-dc	11	B	gain	-1776.859860 9607993	code/V
adc-dc	11	B	zero_mean	2158.097751 710655	code
adc-dc	11	B	zero_total_rms	59.536546 88741656	code
adc-dc	11	B	zero_four_phase_removed_rms	0.512098 9067289364	code
adc-lines	11	A	adjacent_pair_states	44	state
adc-lines	11	A	visited_codes	3175	code
adc-lines	11	B	adjacent_pair_states	44	state
adc-lines	11	B	visited_codes	3237	code
integrator	11	A	balance_delta	0.003963 723409567792	V
integrator	11	A	gain	-2558.714013 0296752	code/V
integrator	11	A	near_balance_change	-0.136157 26802532035	code
integrator	11	B	balance_delta	0.009665 397533032197	V
integrator	11	B	gain	-2553.295843 675655	code/V
integrator	11	B	near_balance_change	0.111521 0552725027	code
adc-dc	10	A	gain	-1777.673493 0206578	code/V
adc-dc	10	A	zero_mean	2159.386119 257087	code

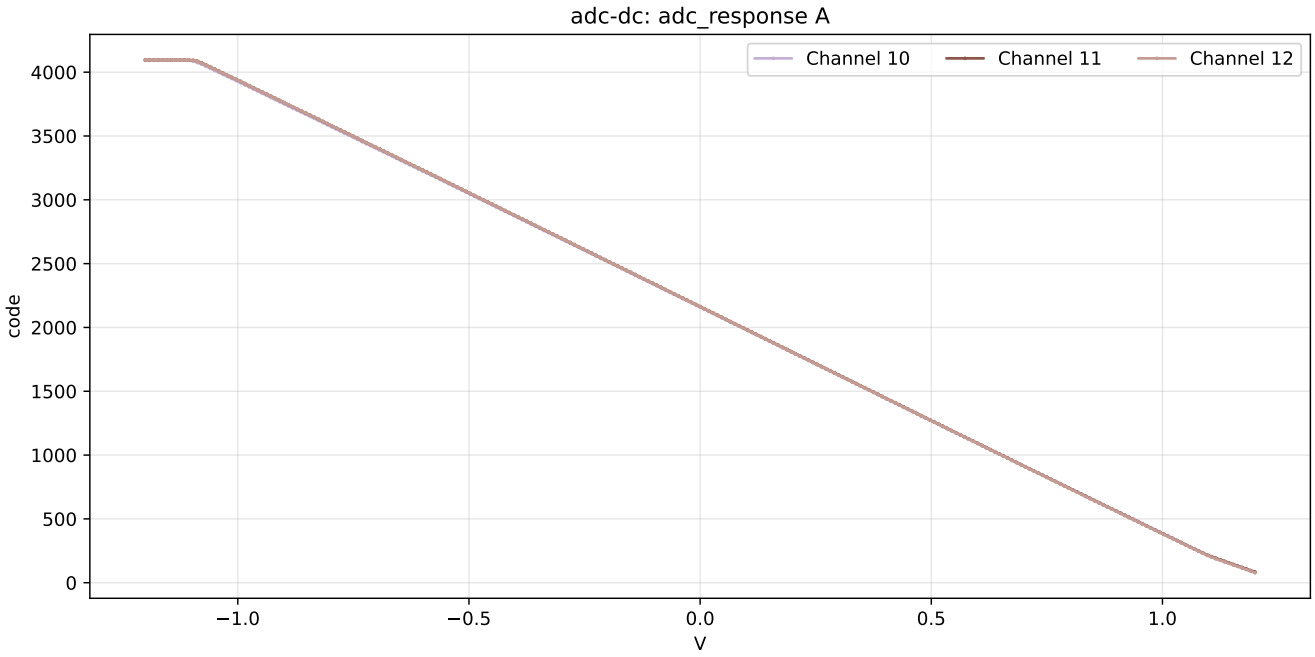
Test	Channel	Bank	Observable	Value	Unit
adc-dc	10	A	zero_total_rms	52.46655577574308	code
adc-dc	10	A	zero_four_phase_removed_rms	0.5128491252775638	code
adc-dc	10	B	gain	-1778.2659098179804	code/V
adc-dc	10	B	zero_mean	2165.9296187683285	code
adc-dc	10	B	zero_total_rms	59.04859019865752	code
adc-dc	10	B	zero_four_phase_removed_rms	0.5082101785128094	code
adc-lines	10	A	adjacent_pair_states	44	state
adc-lines	10	A	visited_codes	3220	code
adc-lines	10	B	adjacent_pair_states	44	state
adc-lines	10	B	visited_codes	3266	code
integrator	10	A	balance_delta	0.0021254808876186398 4	V
integrator	10	A	gain	-2581.6928210142723	code/V
integrator	10	A	near_balance_change	-0.2340184498996449	code
integrator	10	B	balance_delta	0.008891470489998301	V
integrator	10	B	gain	-2528.717823356551	code/V
integrator	10	B	near_balance_change	0.19092461787864753	code
tdc	12		slope	-46.175872088649626	fine code/ns
tdc	12		max_fit_residual	1.7276416408801651	fine code
tdc	12		max_circular_spread	1.0733590480934954	fine code
tdc	12		events	3200	event
tdc	11		slope	-46.175079944169404	fine code/ns
tdc	11		max_fit_residual	1.6139100538513276	fine code
tdc	11		max_circular_spread	1.1070618583155902	fine code
tdc	11		events	3200	event
tdc	10		slope	-46.17427221876848	fine code/ns
tdc	10		max_fit_residual	1.1208303844473448	fine code
tdc	10		max_circular_spread	1.3046642418206191	fine code

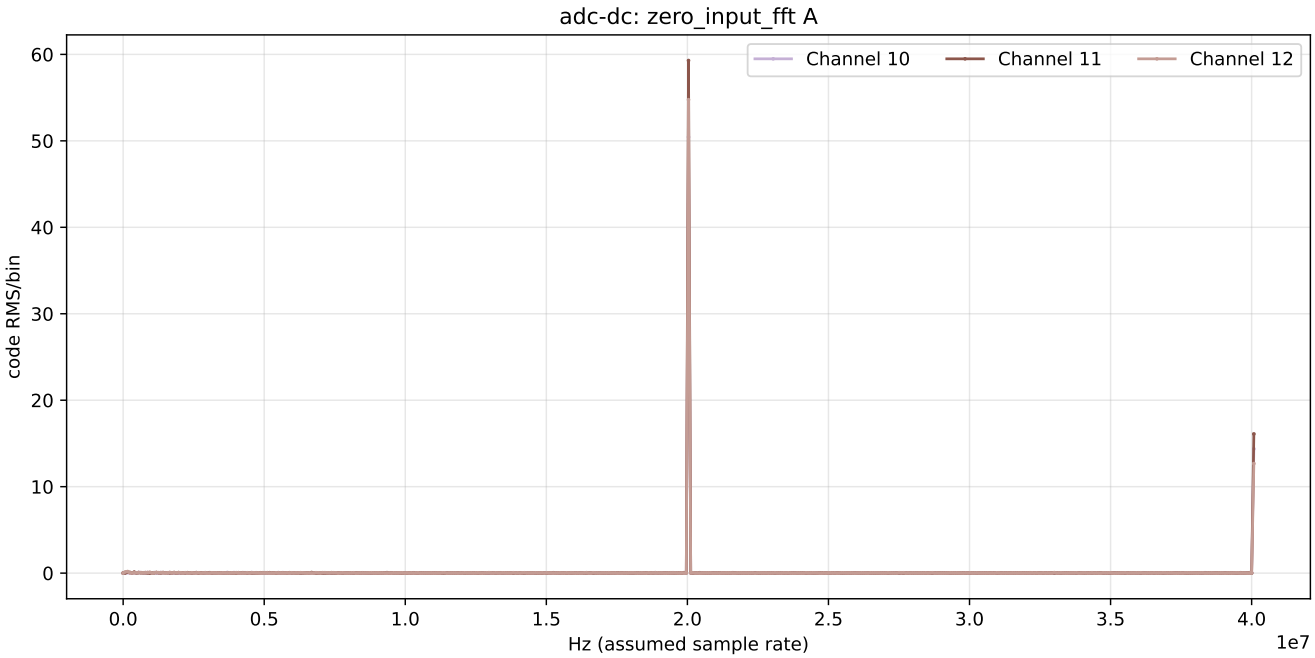
Test	Channel	Bank	Observable	Value	Unit
tdc	10		events	3200	event
tdc	9		slope	-46.171135 68153328	fine code/ns
tdc	9		max_fit_ residual	1.584146 5962344046	fine code
tdc	9		max_circul ar_spread	1.061465 4925174123	fine code
tdc	9		events	3200	event
gate-phase	12	A	valid_points	101	point
gate-phase	12	A	observed_ excursion	19.752803 02671161	ns
gate-phase	12	A	end_to_end	-19.752803 02671161	ns
gate-phase	12	A	largest_step	1.039621 2119321913	ns
gate-phase	12	A	max_spread	0.437643 2508078598	ns
gate-phase	12	B	valid_points	101	point
gate-phase	12	B	observed_ excursion	19.752803 02671161	ns
gate-phase	12	B	end_to_end	-19.752803 02671161	ns
gate-phase	12	B	largest_step	1.039621 2119321913	ns
gate-phase	12	B	max_spread	0.519164 2461901361	ns
gate-phase	11	A	valid_points	101	point
gate-phase	11	A	observed_ excursion	20.905062 554431478	ns
gate-phase	11	A	end_to_end	-20.905062 554431478	ns
gate-phase	11	A	largest_step	1.039621 2119321913	ns
gate-phase	11	A	max_spread	0.517370 8034731153	ns
gate-phase	11	B	valid_points	101	point
gate-phase	11	B	observed_ excursion	20.792424 2386438	ns
gate-phase	11	B	end_to_end	-20.792424 2386438	ns
gate-phase	11	B	largest_step	1.039621 2119321913	ns
gate-phase	11	B	max_spread	0.515059 3173424455	ns
gate-phase	10	A	valid_points	101	point
gate-phase	10	A	observed_ excursion	21.832045 450575993	ns
gate-phase	10	A	end_to_end	-21.832045 450575993	ns
gate-phase	10	A	largest_step	1.039621 211932193	ns
gate-phase	10	A	max_spread	0.519701 0721490758	ns

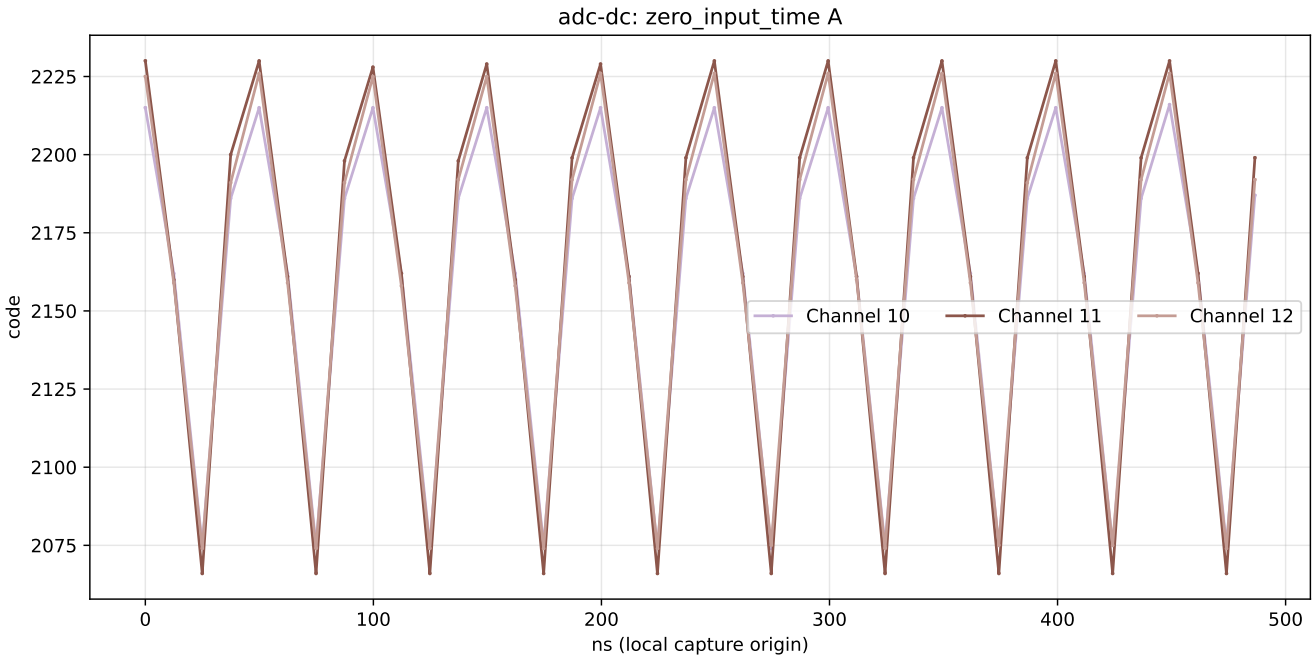
Test	Channel	Bank	Observable	Value	Unit
gate-phase	10	B	valid_points	101	point
gate-phase	10	B	observed_excursion	20.7924242386438	ns
gate-phase	10	B	end_to_end	-20.7924242386438	ns
gate-phase	10	B	largest_step	1.0396212119321913	ns
gate-phase	10	B	max_spread	0.5017937590614953	ns
gate-phase	9	A	valid_points	101	point
gate-phase	9	A	observed_excursion	20.7924242386438	ns
gate-phase	9	A	end_to_end	-20.7924242386438	ns
gate-phase	9	A	largest_step	1.0396212119321913	ns
gate-phase	9	A	max_spread	0.4529892996023046	ns
gate-phase	9	B	valid_points	101	point
gate-phase	9	B	observed_excursion	20.7924242386438	ns
gate-phase	9	B	end_to_end	-20.7924242386438	ns
gate-phase	9	B	largest_step	1.0396212119321913	ns
gate-phase	9	B	max_spread	0.5175945457149657	ns
cfd-monitor	9		CFD_OFFSET	1.0350966666666668	V
cfd-monitor	9		CFD_CMP_OFFSET	0.00524	V
cfd-monitor	9		BUFFER_OFFSET	1.0248066666666666	V
cfd-monitor	9		DRV_OFFSET	1.04463	V
cfd-monitor	9		DRV_VOCM	1.53664	V
cfd-monitor	9		N5V5_PM12	-5.511662727272727	V
cfd-monitor	9		P1V3	1.3063066666666667	V
cfd-monitor	9		P5V5_PM12	5.486335151515151	V
cfd-monitor	10		CFD_OFFSET	1.0346	V
cfd-monitor	10		CFD_CMP_OFFSET	0.0045966666666664	V
cfd-monitor	10		BUFFER_OFFSET	1.02437	V
cfd-monitor	10		DRV_OFFSET	1.04449	V
cfd-monitor	10		DRV_VOCM	1.5363066666666667	V

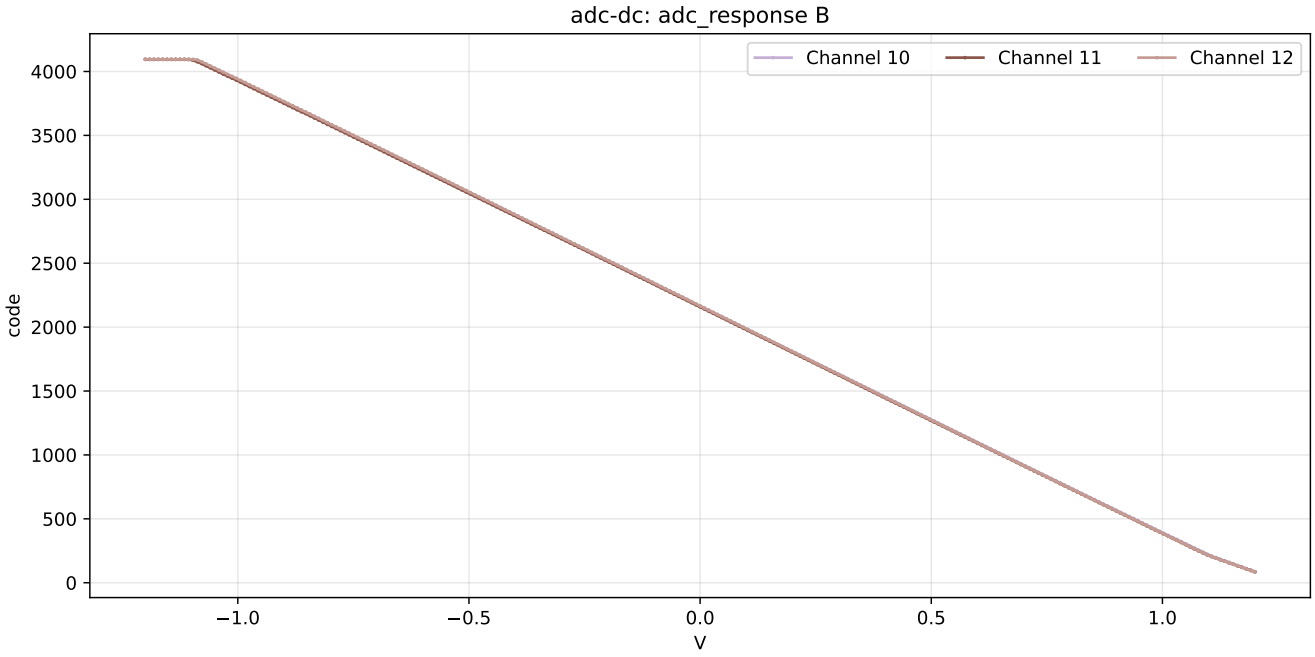
Test	Channel	Bank	Observable	Value	Unit
cfd-monitor	10		N5V5_PM12	-5.510442121212121	V
cfd-monitor	10		P1V3	1.31936	V
cfd-monitor	10		P5V5_PM12	5.480419494949494	V
cfd-monitor	12		CFD_OFFSET	1.03451	V
cfd-monitor	12		CFD_CMP_OFFSET	0.004816666666666	V
cfd-monitor	12		BUFFER_OFFSET	1.02432	V
cfd-monitor	12		DRV_OFFSET	1.0443166666666668	V
cfd-monitor	12		DRV_VOCM	1.5362133333333334	V
cfd-monitor	12		N5V5_PM12	-5.504039292929292	V
cfd-monitor	12		P1V3	1.3070833333333334	V
cfd-monitor	12		P5V5_PM12	5.475670909090909	V
jtag	board		links_per_path	168	connection
jtag	board		vectors_per_path	338	vector
mcu	board		walking_patterns	32	pattern
i2c	board		P1V0_LDO_voltage	1.0	V
i2c	board		P1V0_LDO_nominal_current	0.124	A
i2c	board		P1V2_LDO_voltage	1.2	V
i2c	board		P1V2_LDO_nominal_current	0.092	A
i2c	board		P1V8_LDO_voltage	1.792	V
i2c	board		P1V8_LDO_nominal_current	0.004	A
i2c	board		P3V8_voltage	3.92	V
i2c	board		P3V8_nominal_current	3.12	A
i2c	board		P3V3_voltage	3.328	V
i2c	board		P3V3_nominal_current	4.880000000000001	A
i2c	board		P1V8_voltage	1.776	V
i2c	board		P1V8_nominal_current	0.9600000000000001	A
i2c	board		P1V0_voltage	1.008	V

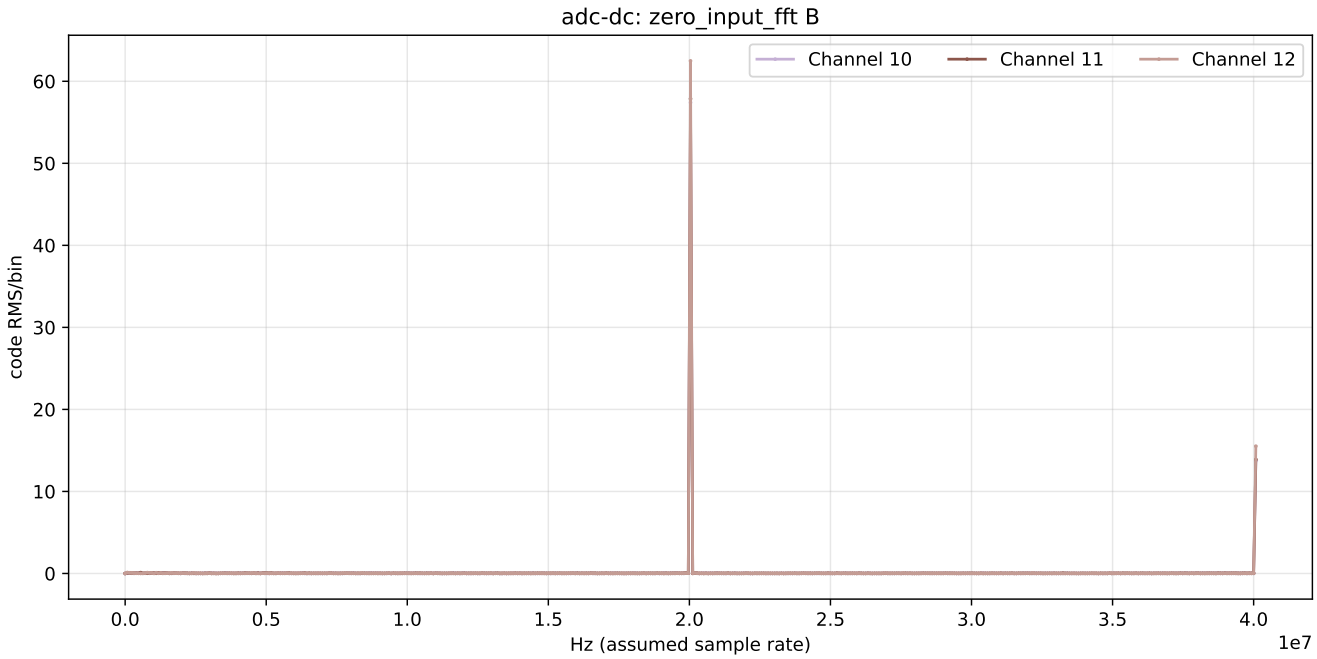
Test	Channel	Bank	Observable	Value	Unit
i2c	board		P1V0_nominal_current	0.32	A
i2c	board		P5V5_voltage	5.496	V
i2c	board		P5V5_nominal_current	0.28	A
ethernet	board		verified_ADC_payload	393216	byte

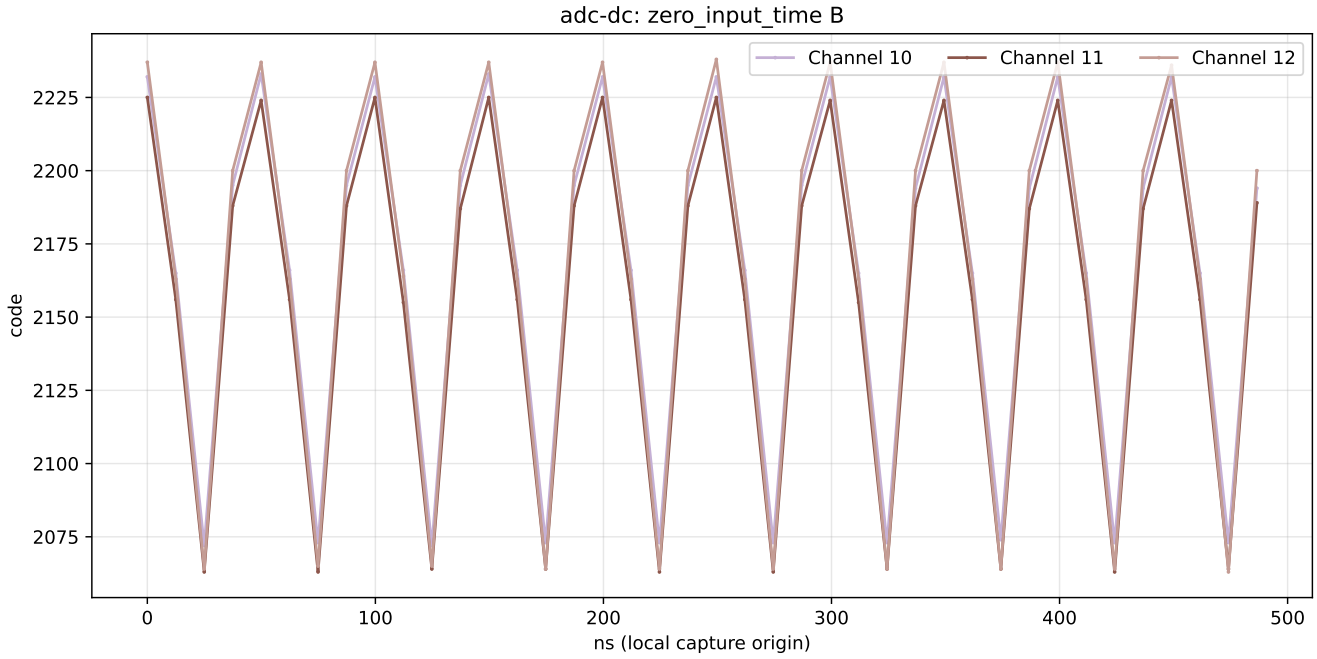


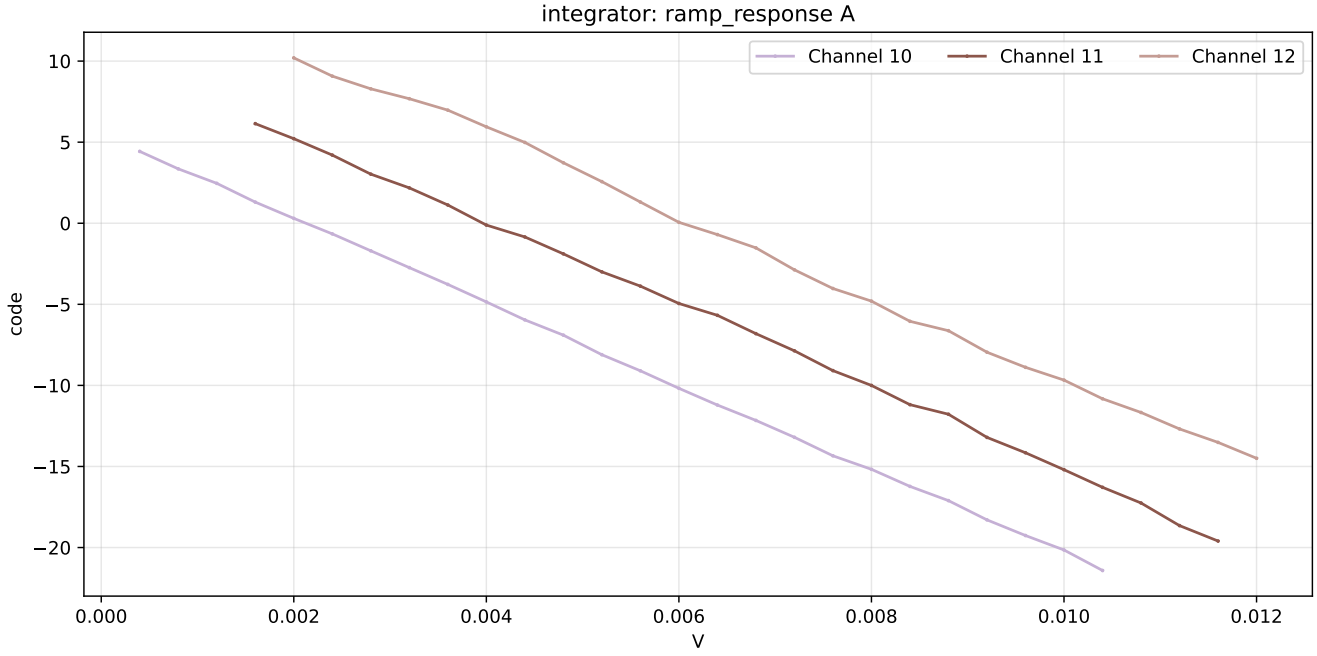


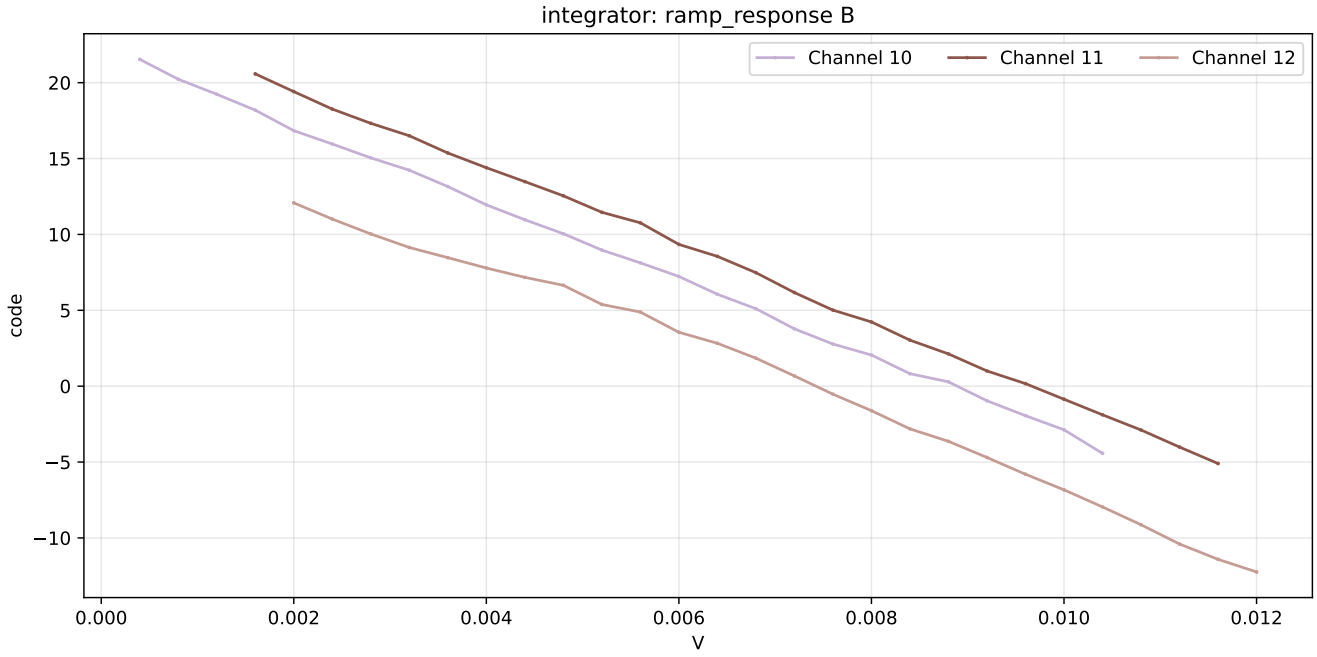

adc-dc: zero_input_time A
code
ns (local capture origin)
Channel 10
Channel 11
Channel 12
2225
2200
2175
2150
2125
2100
2075
0
100
200
300
400
500

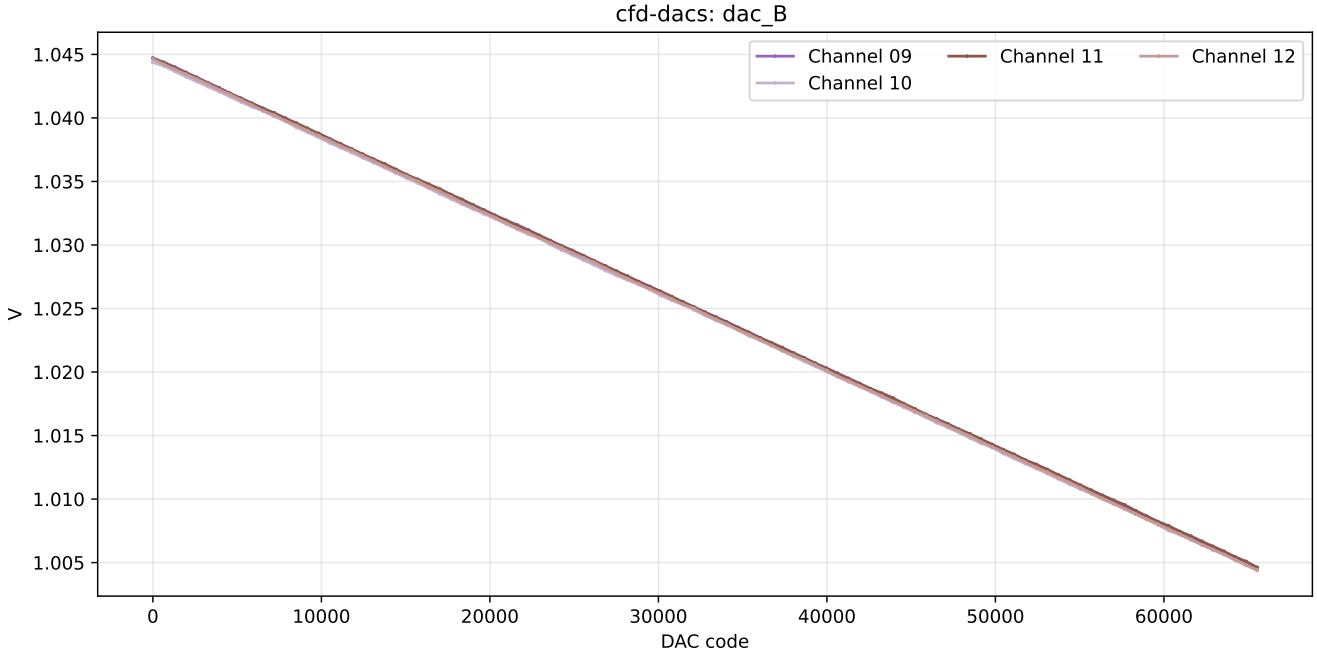

adc-dc: adc_response B
code
V
Channel 10
Channel 11
Channel 12
4000
3500
3000
2500
2000
1500
1000
500
0
−1.0
−0.5
0.0
0.5
1.0

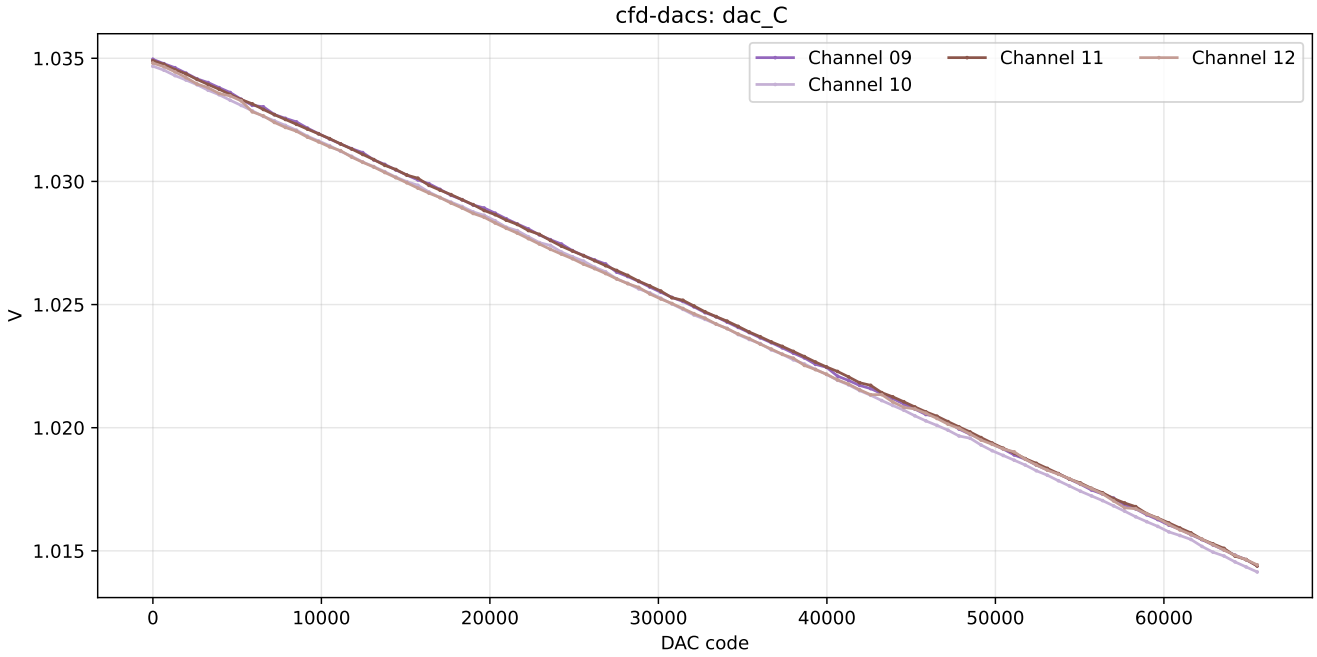

adc-dc: zero_input_fft B
Channel 10
Channel 11
Channel 12
code RMS/bin
0
10
20
30
40
50
60
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1e7
Hz (assumed sample rate)

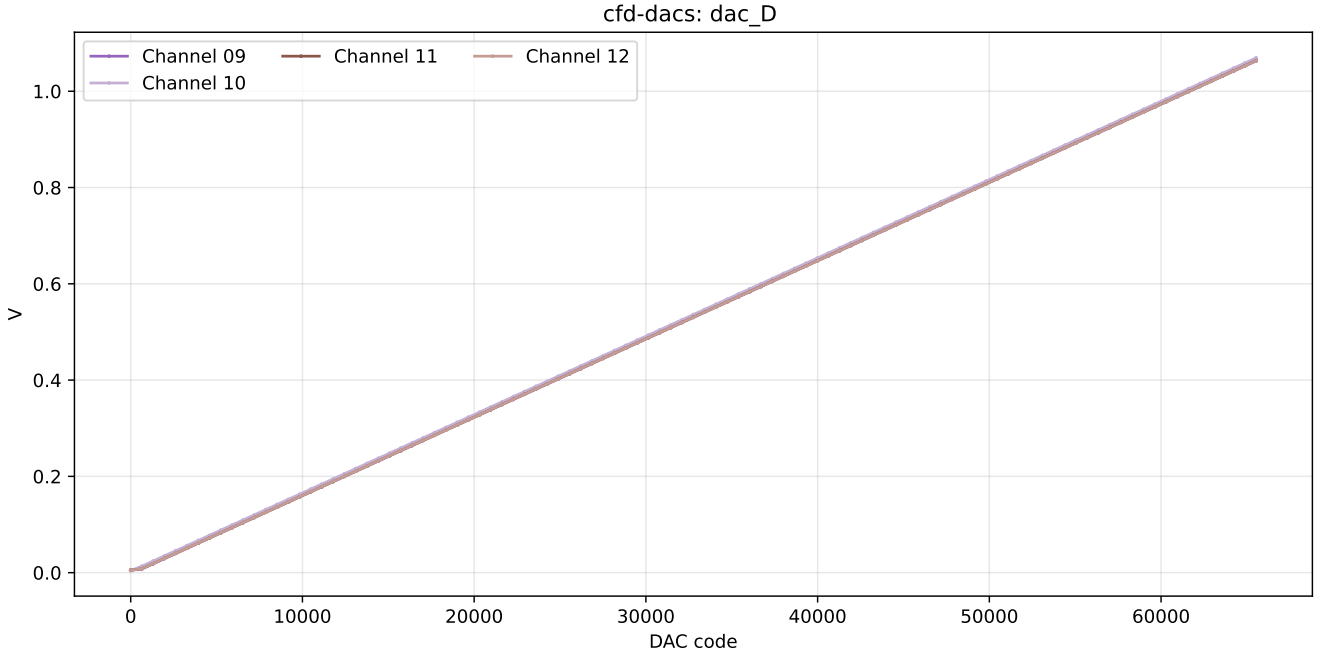

adc-dc: zero_input_time B
Channel 10
Channel 11
Channel 12
code
2075
2100
2125
2150
2175
2200
2225
0
100
200
300
400
500
ns (local capture origin)

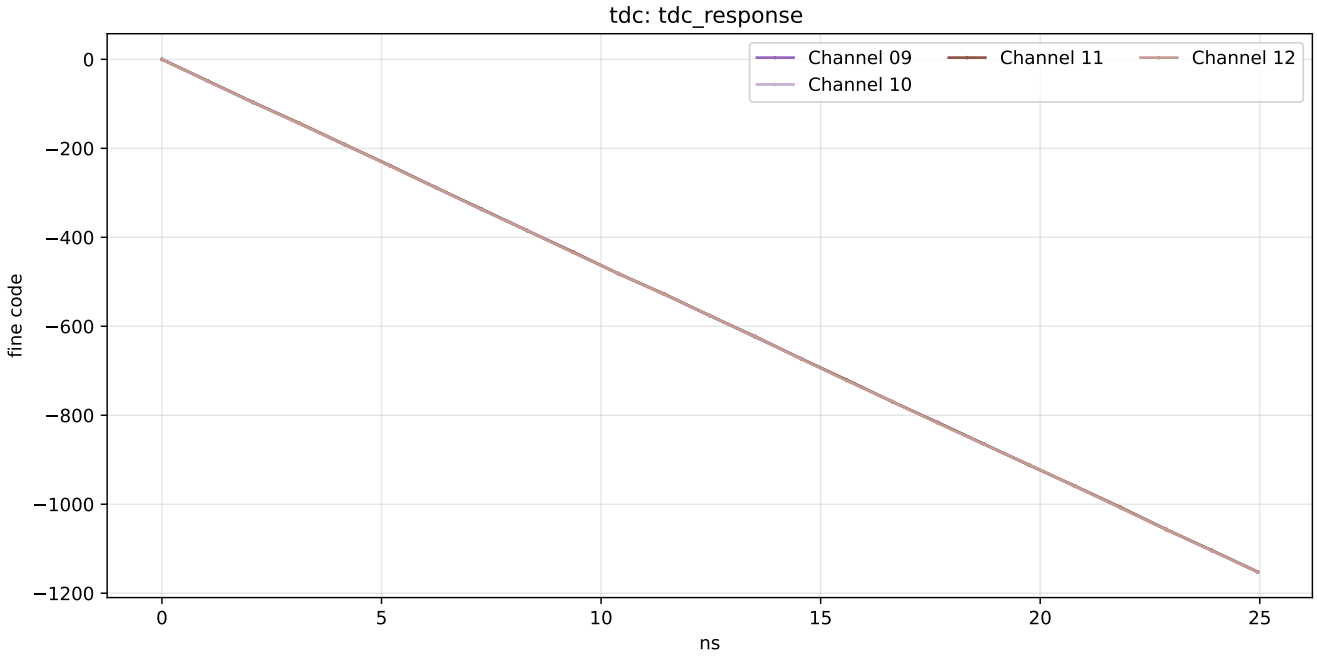

integrator: ramp_response A
Channel 10
Channel 11
Channel 12
code
V
10
5
0
−5
−10
−15
−20
0.000
0.002
0.004
0.006
0.008
0.010
0.012

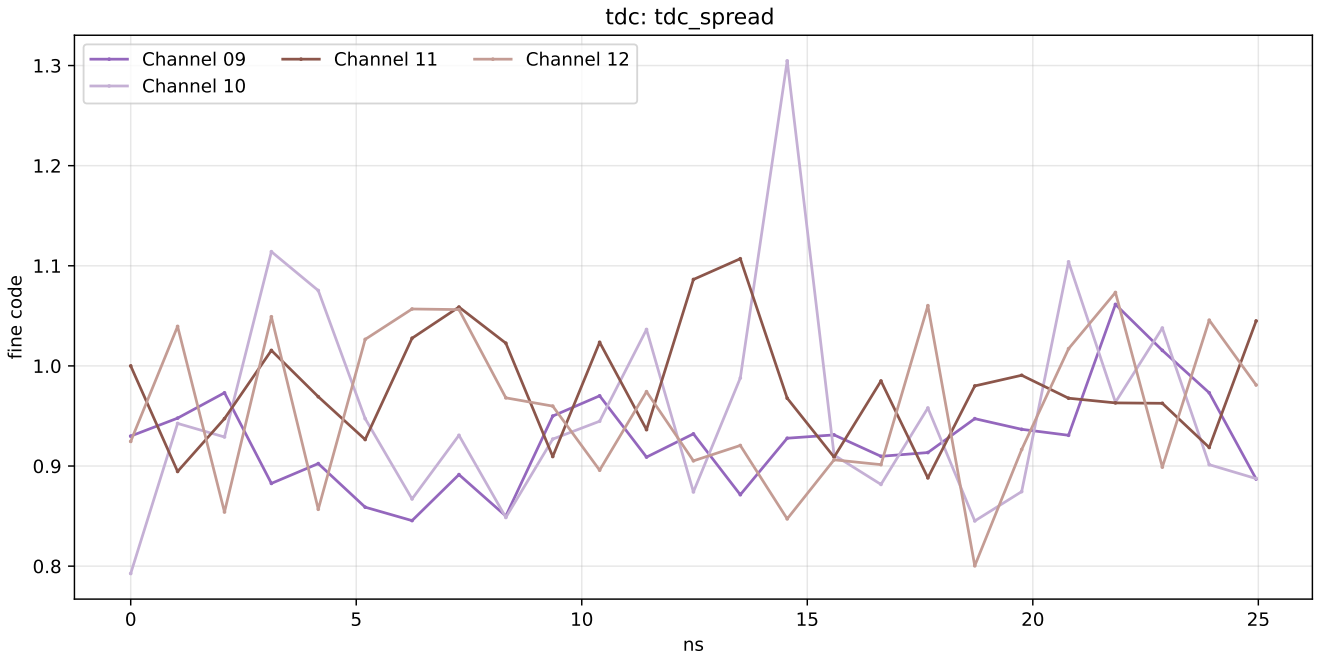

integrator: ramp_response B
Channel 10
Channel 11
Channel 12
code
V
20
15
10
5
0
−5
−10
0.000
0.002
0.004
0.006
0.008
0.010
0.012

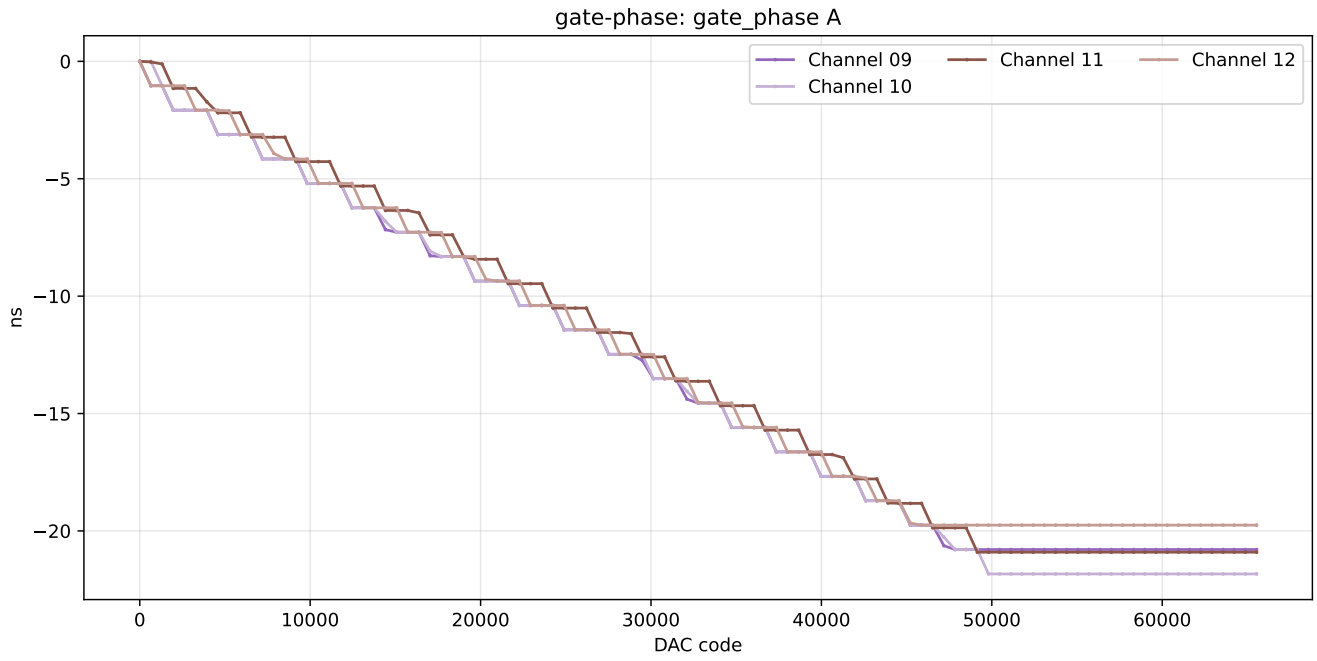

cfd-dacs: dac_B
Channel 09
Channel 10
Channel 11
Channel 12
1.045
1.040
1.035
1.030
1.025
1.020
1.015
1.010
1.005
V
0
10000
20000
30000
40000
50000
60000
DAC code

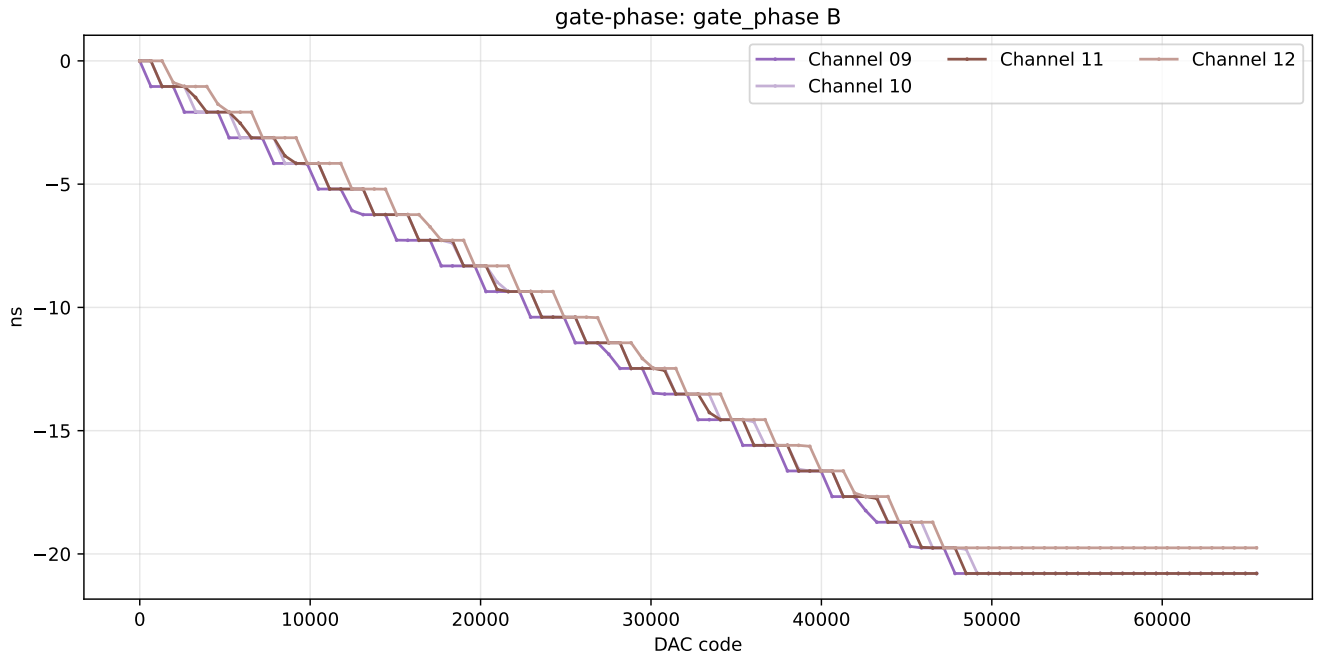

cfd-dacs: dac_C
Channel 09
Channel 10
Channel 11
Channel 12
1.035
1.030
1.025
1.020
1.015
V
0
10000
20000
30000
40000
50000
60000
DAC code

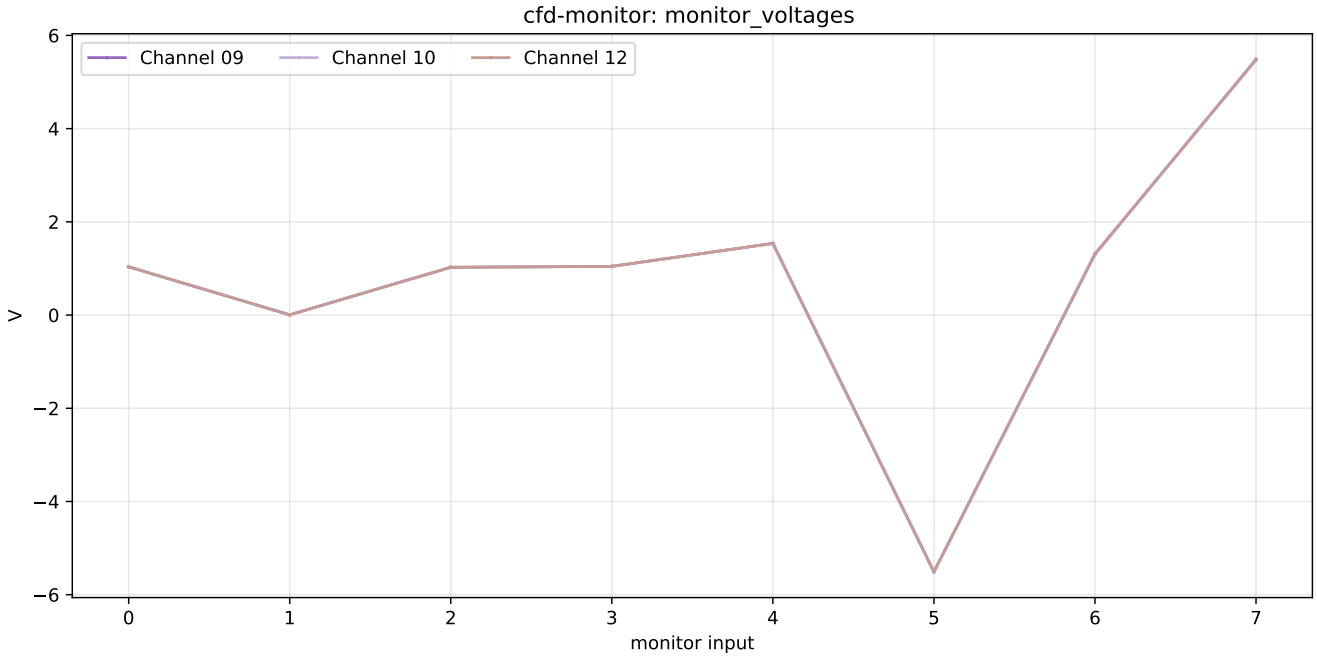

cfd-dacs: dac_D
Channel 09
Channel 11
Channel 12
Channel 10
V
DAC code

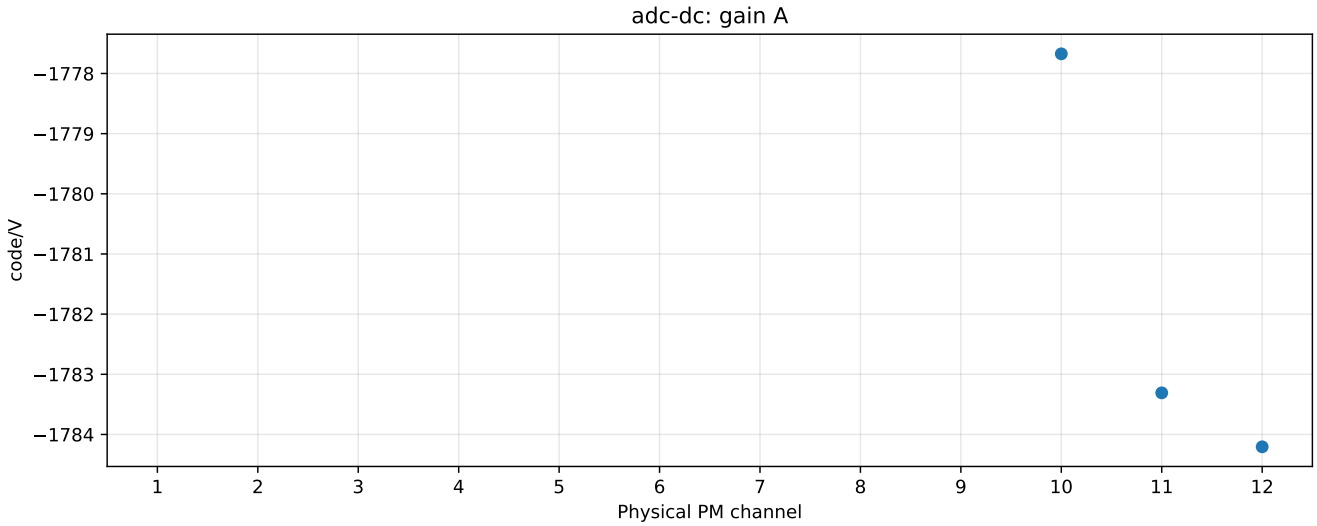

tdc: tdc_response
Channel 09
Channel 11
Channel 12
Channel 10
fine code
ns

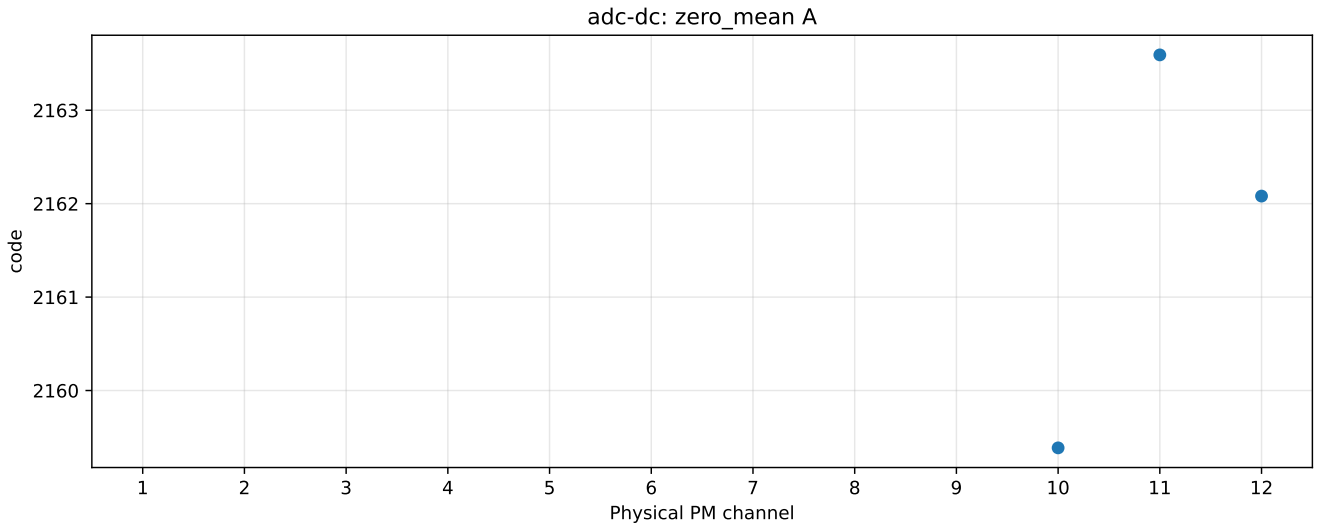

tdc: tdc_spread
Channel 09
Channel 10
Channel 11
Channel 12
fine code
1.3
1.2
1.1
1.0
0.9
0.8
0
5
10
15
20
25
ns

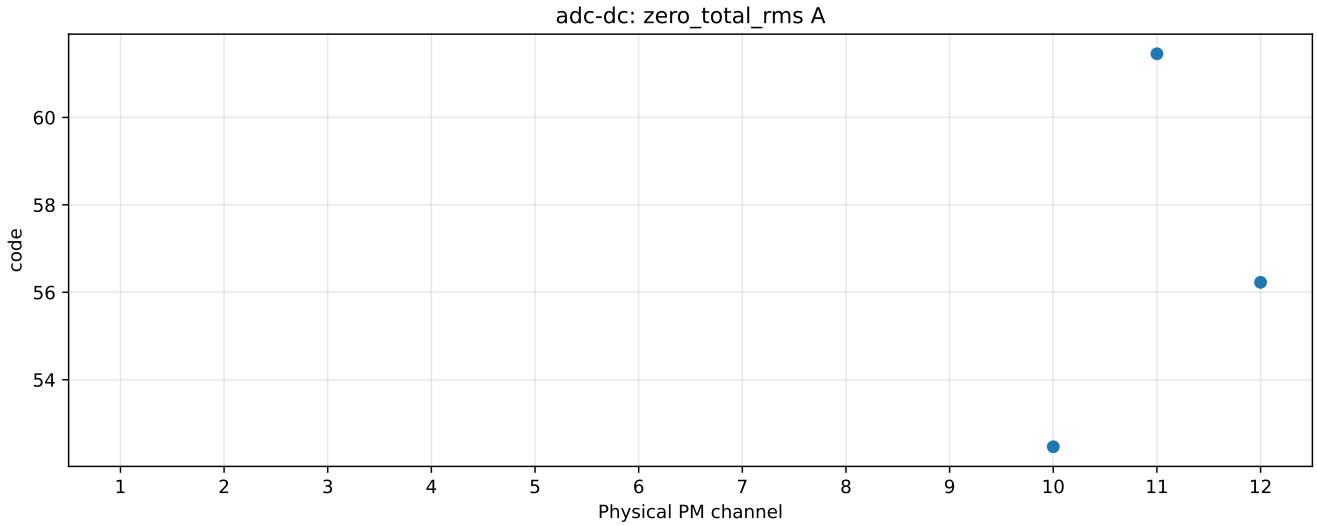

gate-phase: gate_phase A
Channel 09
Channel 10
Channel 11
Channel 12
ns
0
−5
−10
−15
−20
0
10000
20000
30000
40000
50000
60000
DAC code

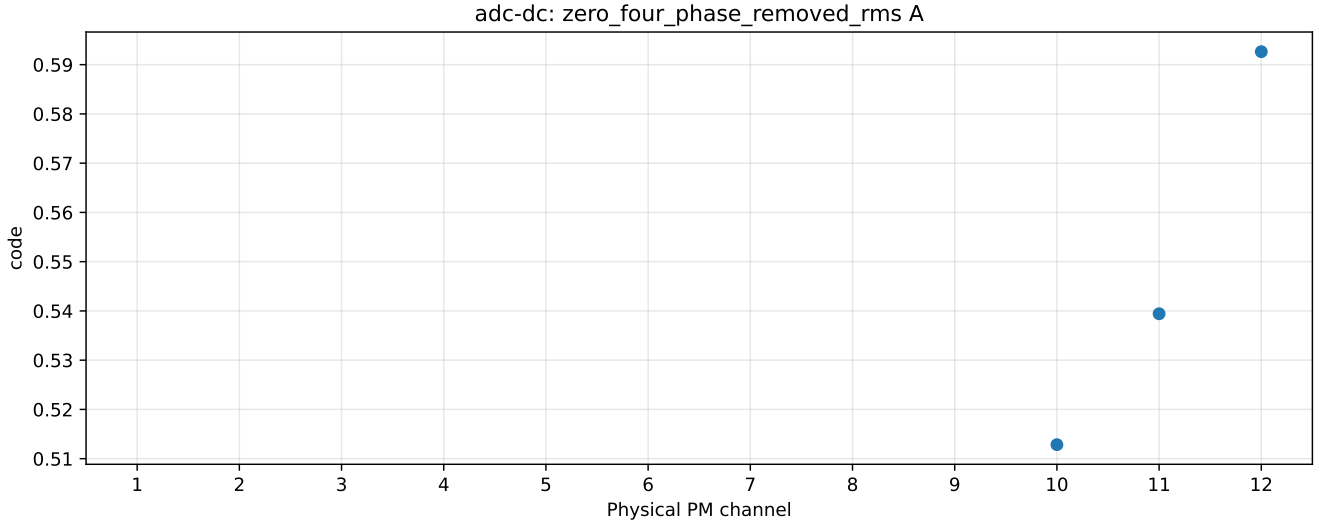

gate-phase: gate_phase B
Channel 09
Channel 10
Channel 11
Channel 12
ns
DAC code

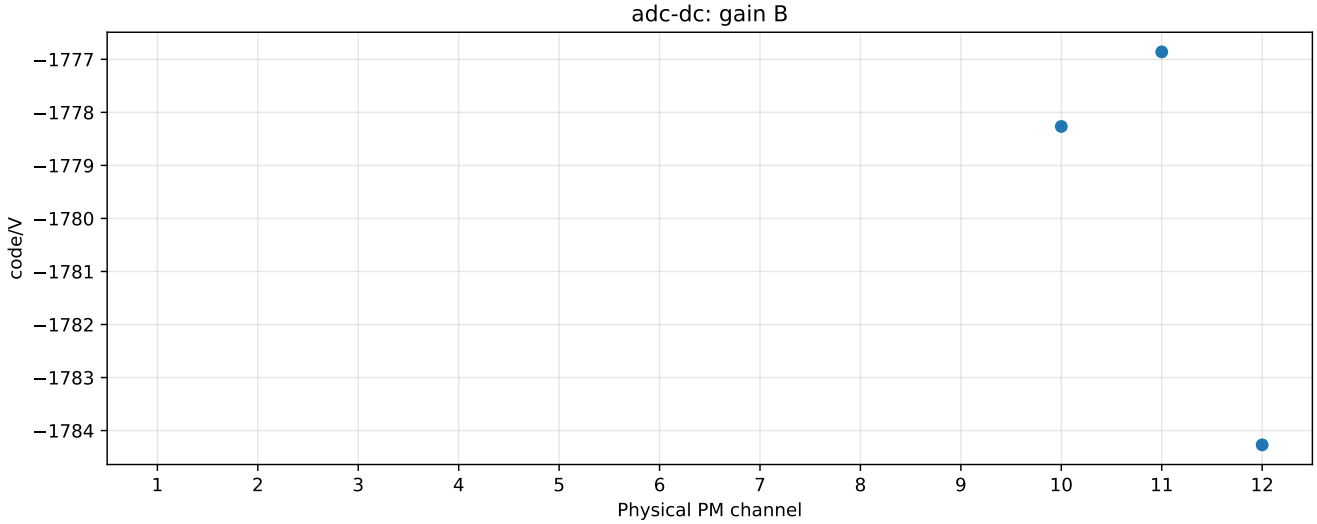

cfd-monitor: monitor_voltages
Channel 09
Channel 10
Channel 12
V
monitor input

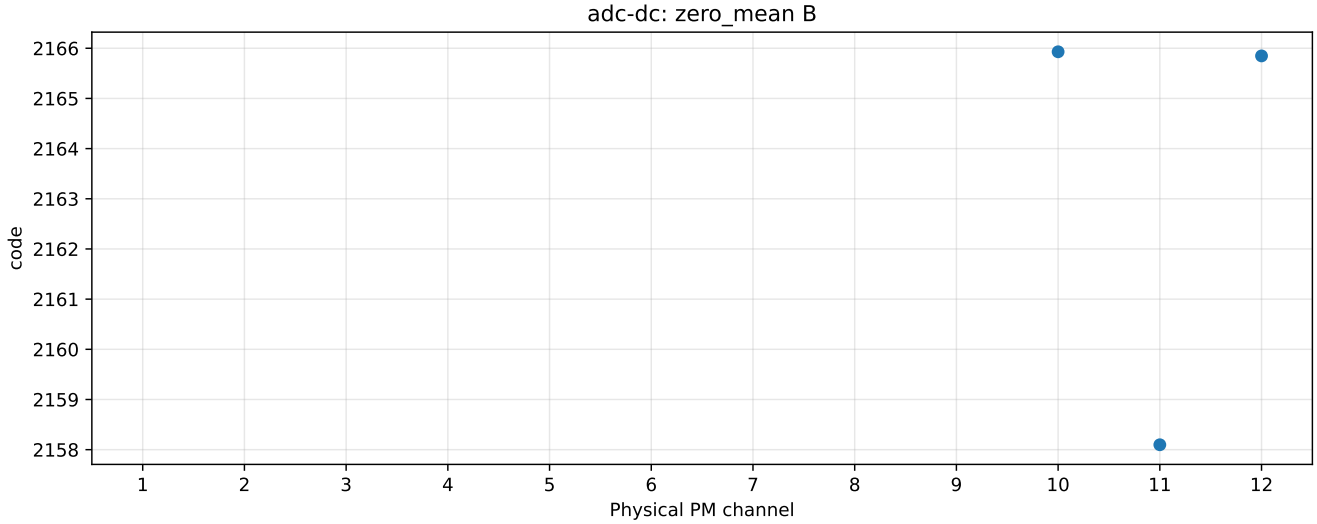

adc-dc: gain A
code/V
Physical PM channel

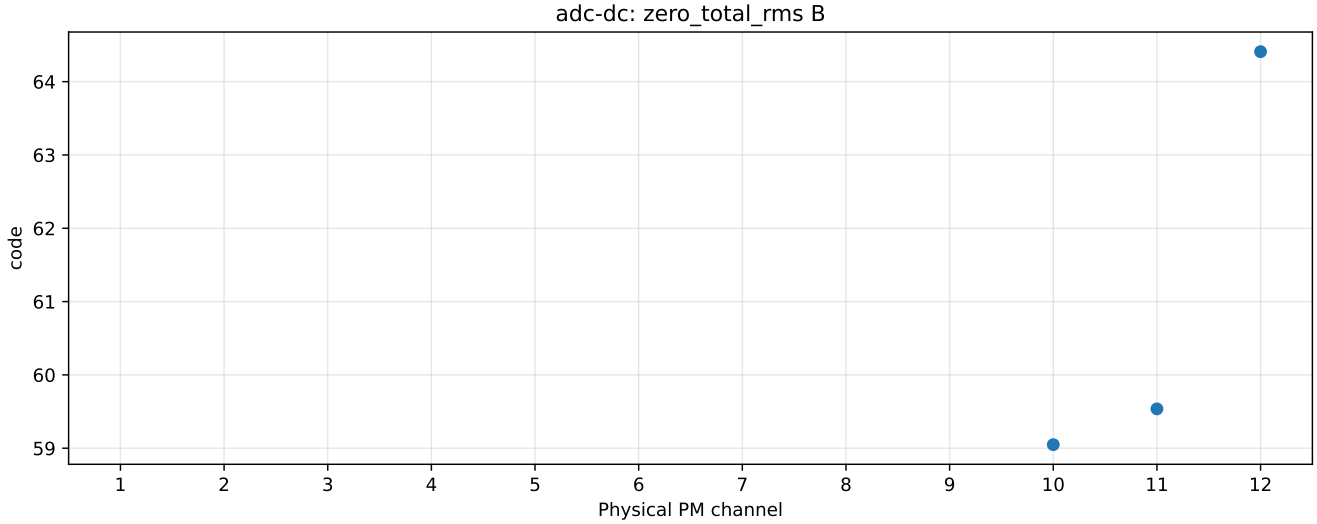

adc-dc: zero_mean A
code
2163
2162
2161
2160
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

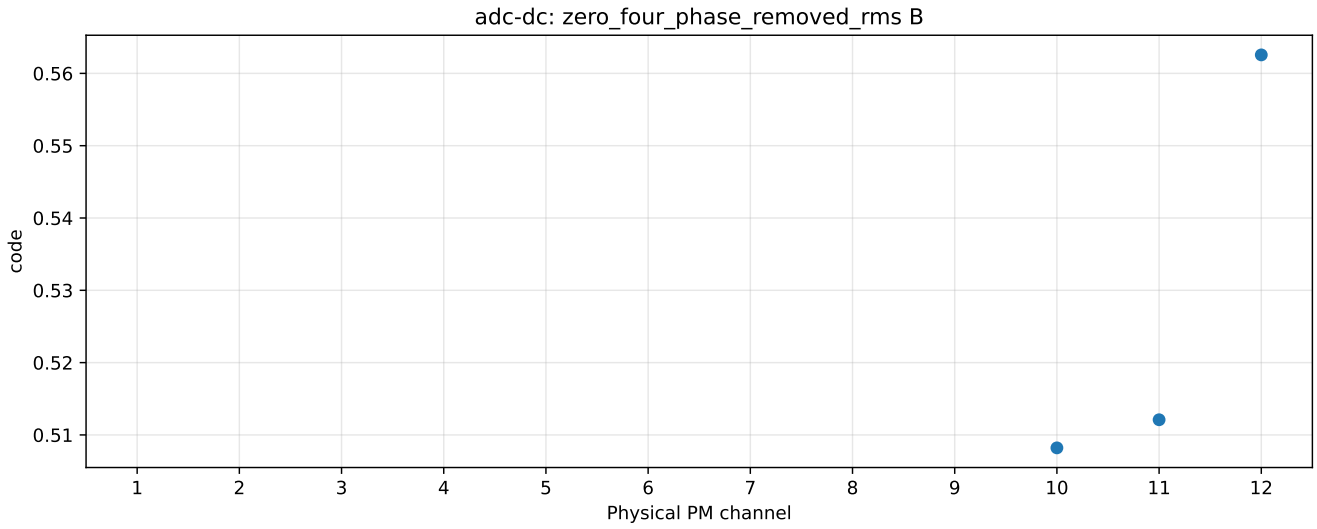

adc-dc: zero_total_rms A
code
60
58
56
54
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

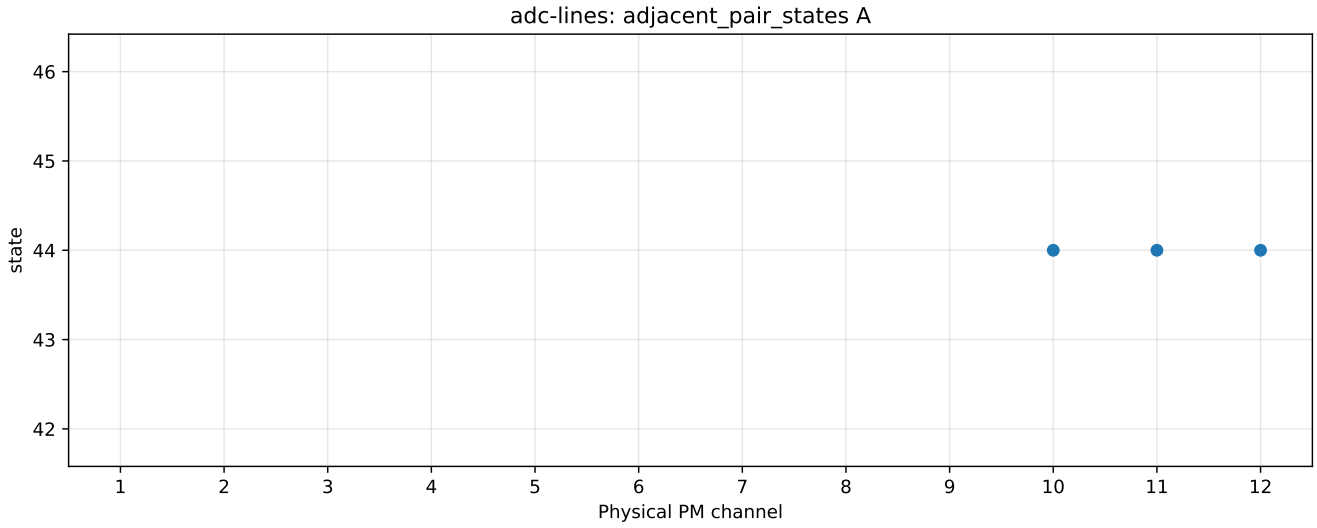

adc-dc: zero_four_phase_removed_rms A
code
0.59
0.58
0.57
0.56
0.55
0.54
0.53
0.52
0.51
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

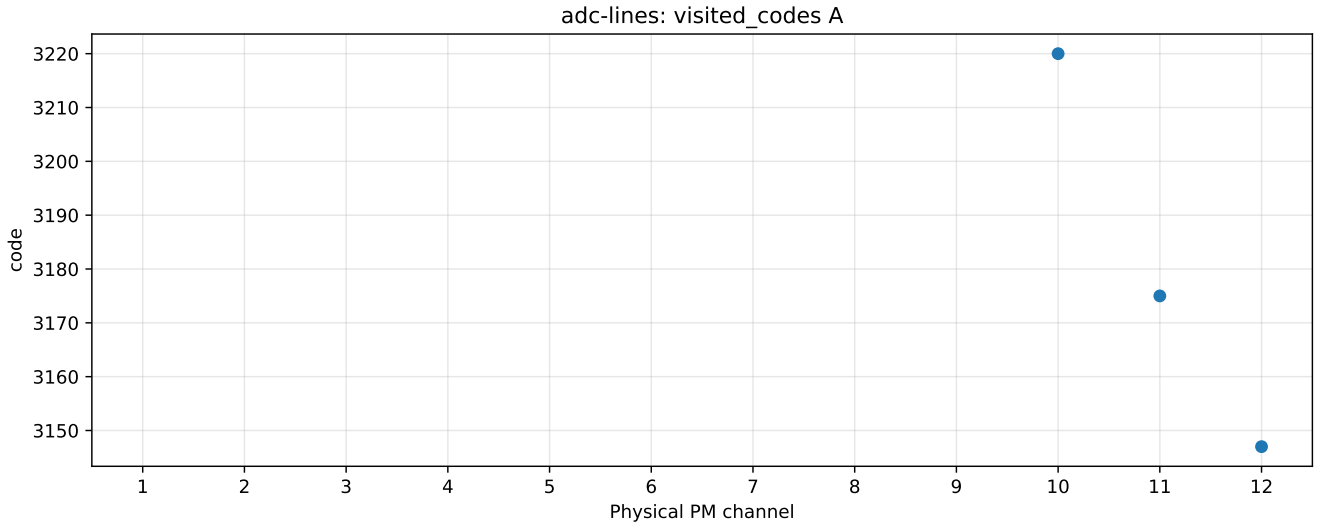

adc-dc: gain B
code/V
−1777
−1778
−1779
−1780
−1781
−1782
−1783
−1784
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

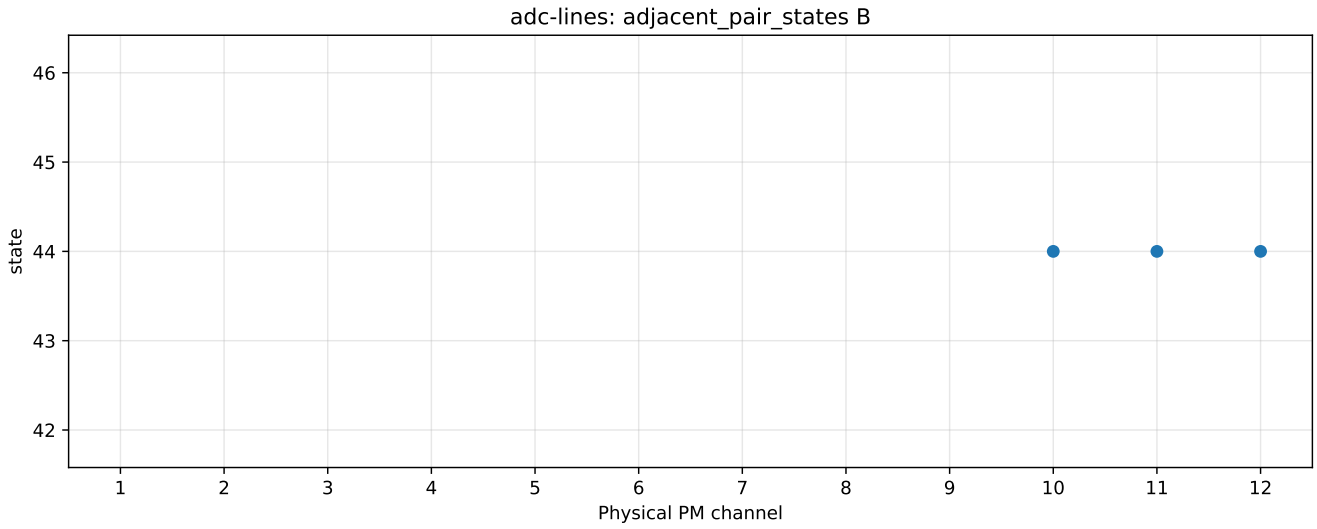

adc-dc: zero_mean B
code
2166
2165
2164
2163
2162
2161
2160
2159
2158
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

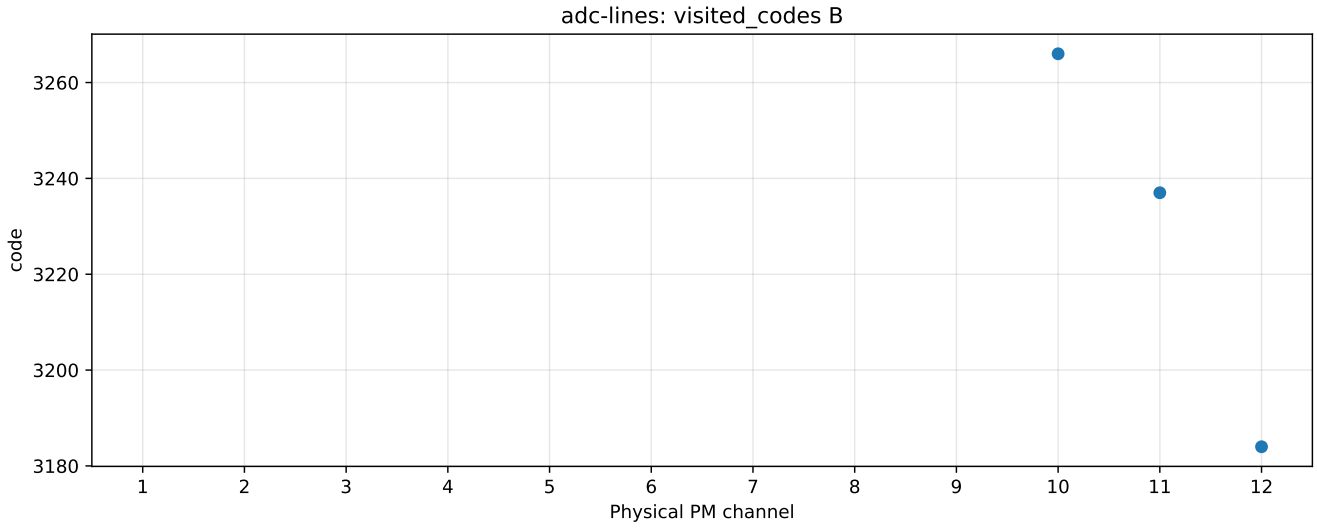

adc-dc: zero_total_rms B
code
64
63
62
61
60
59
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

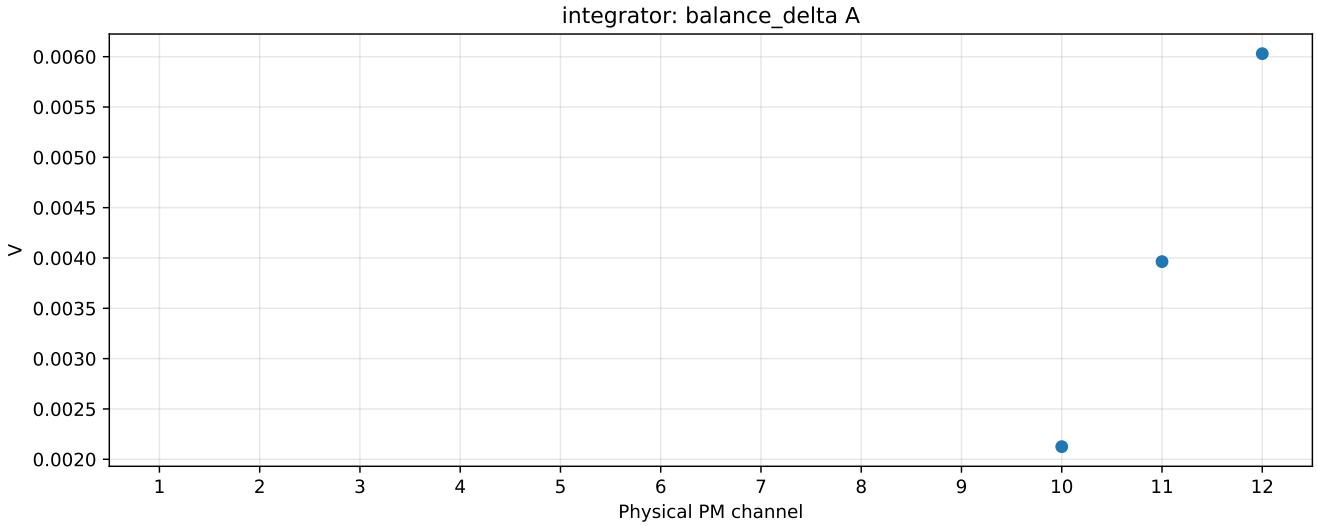

adc-dc: zero_four_phase_removed_rms B
code
0.56
0.55
0.54
0.53
0.52
0.51
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

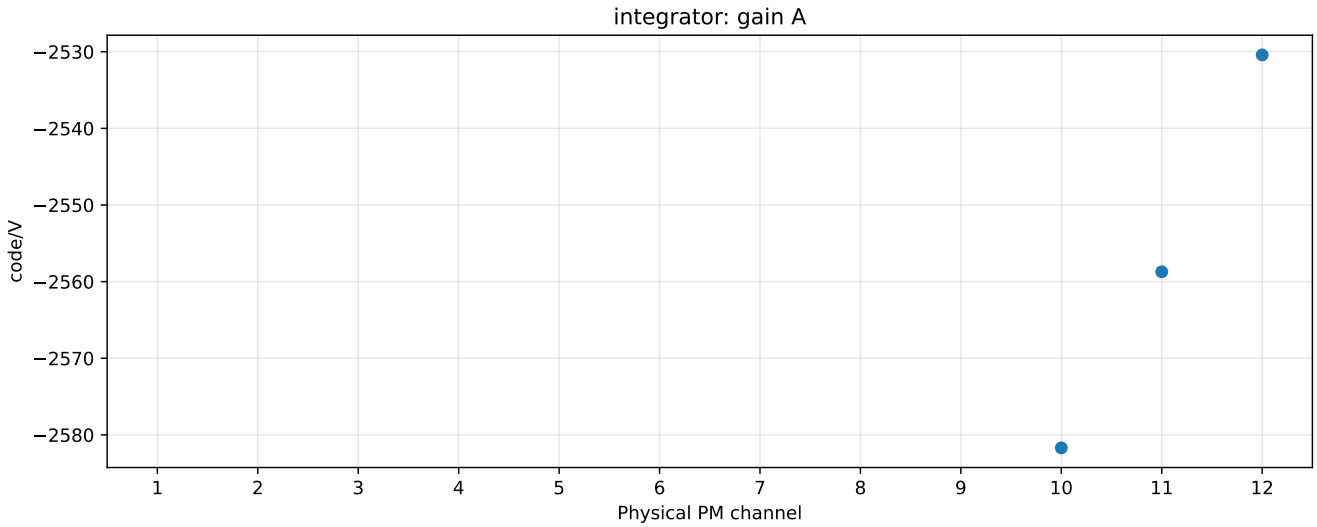

adc-lines: adjacent_pair_states A
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

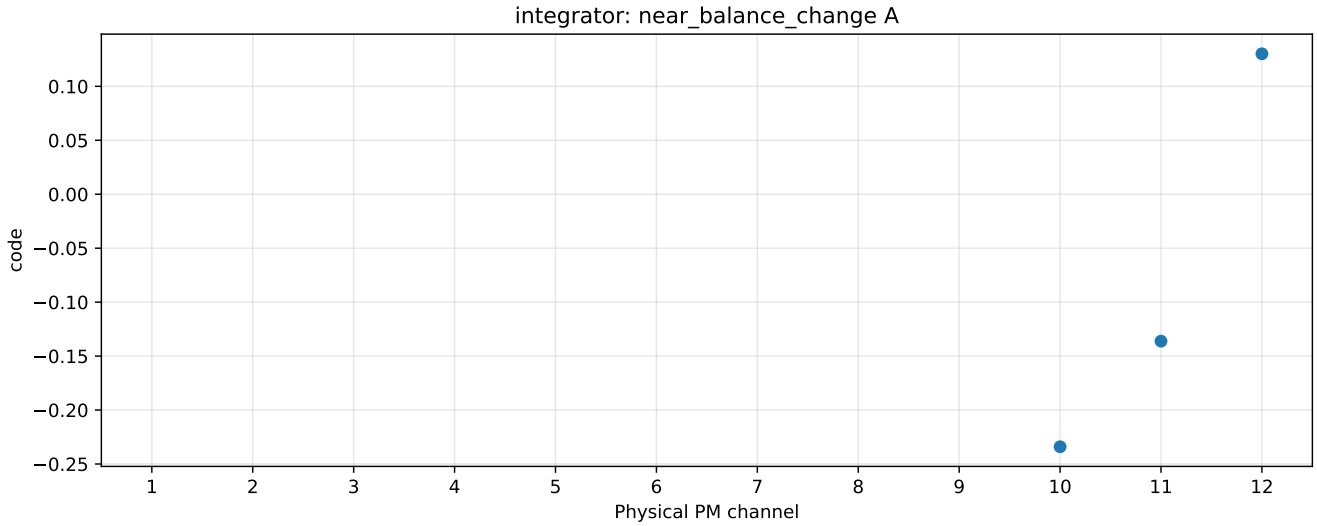

adc-lines: visited_codes A
code
3220
3210
3200
3190
3180
3170
3160
3150
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

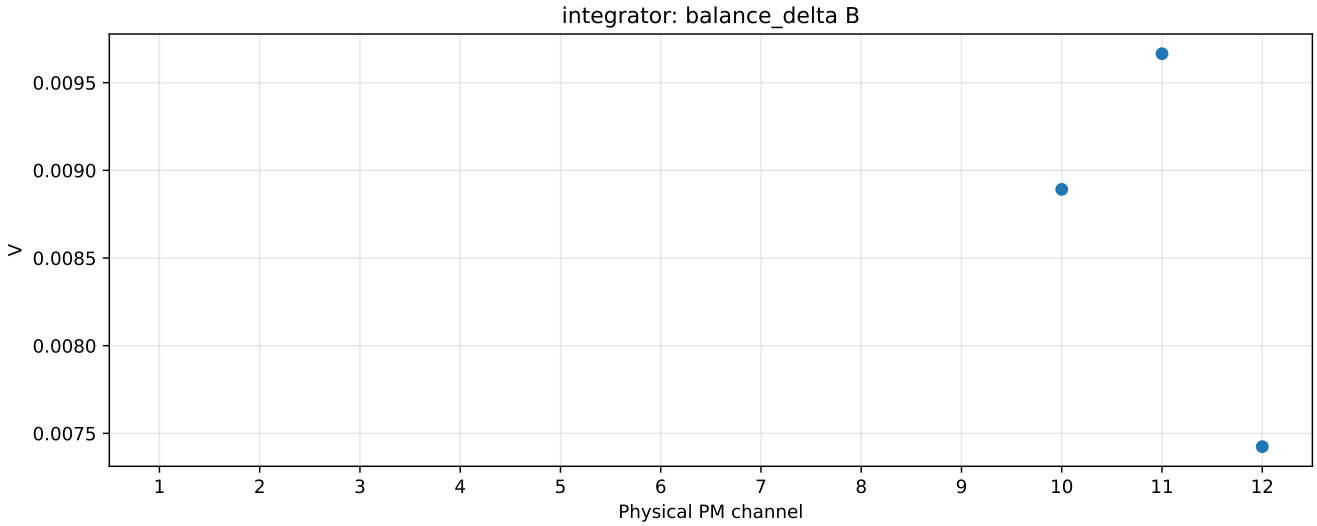

adc-lines: adjacent_pair_states B
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

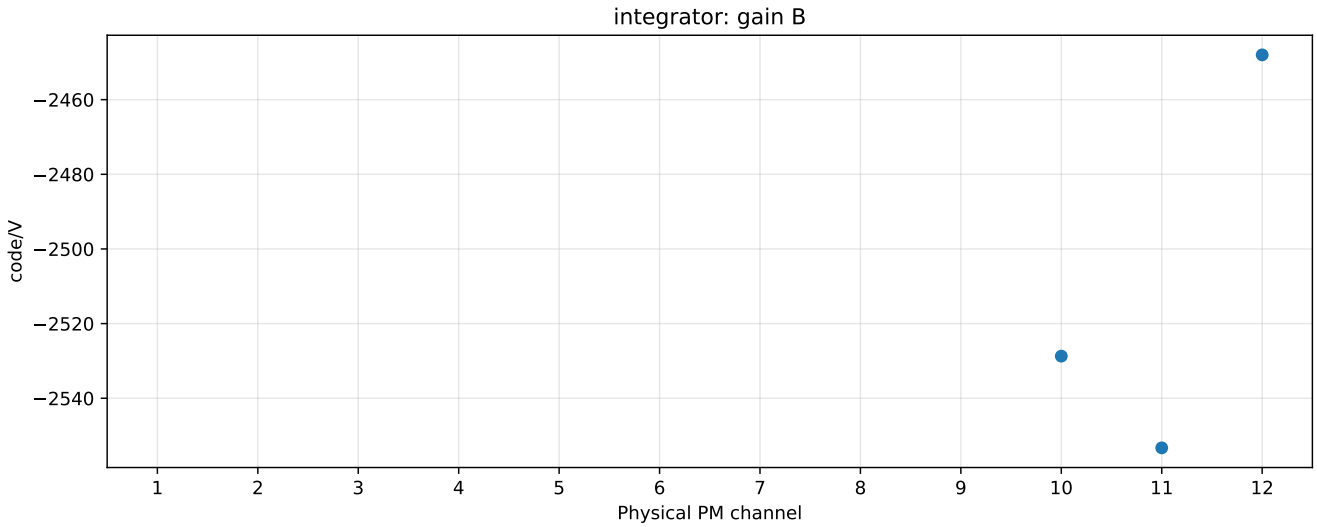

adc-lines: visited_codes B
code
3260
3240
3220
3200
3180
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

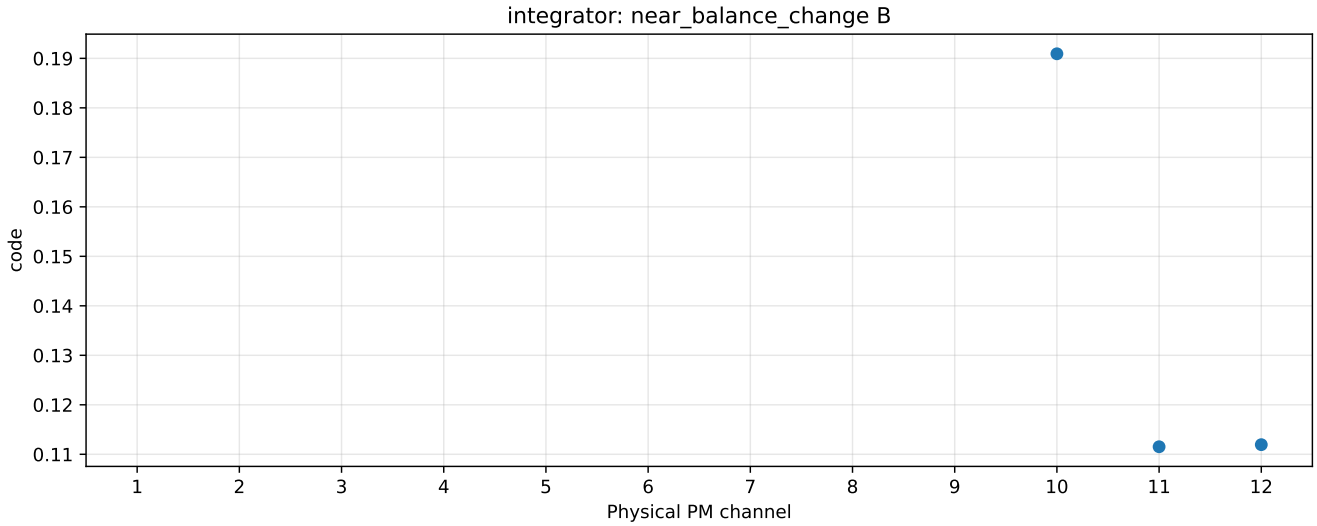

integrator: balance_delta A
V
0.0060
0.0055
0.0050
0.0045
0.0040
0.0035
0.0030
0.0025
0.0020
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

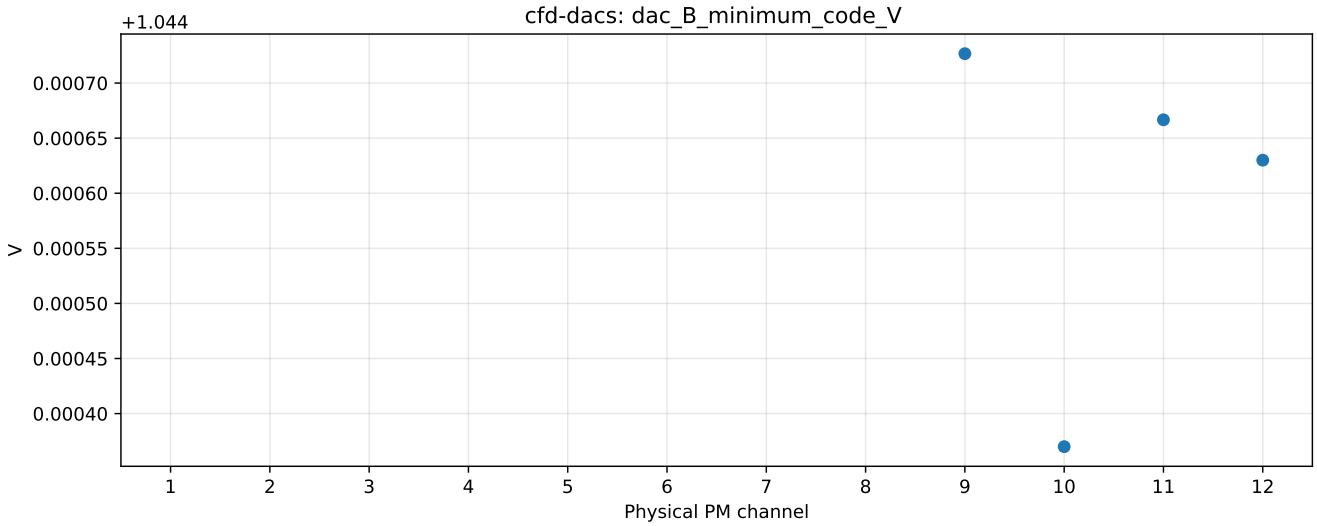

integrator: gain A
code/V
−2530
−2540
−2550
−2560
−2570
−2580
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

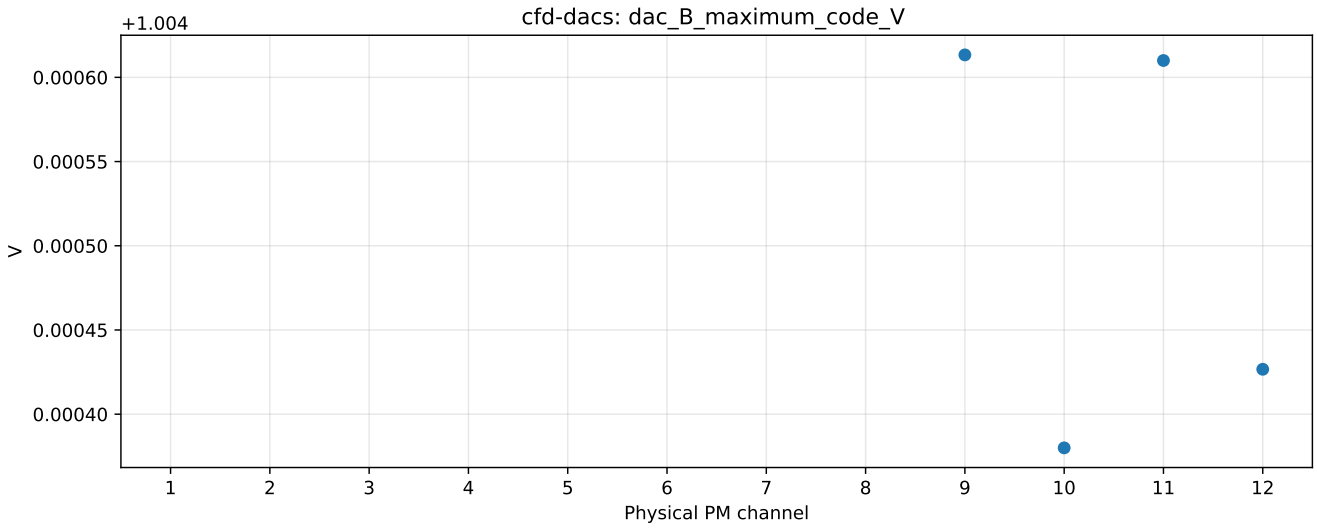

integrator: near_balance_change A
code
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

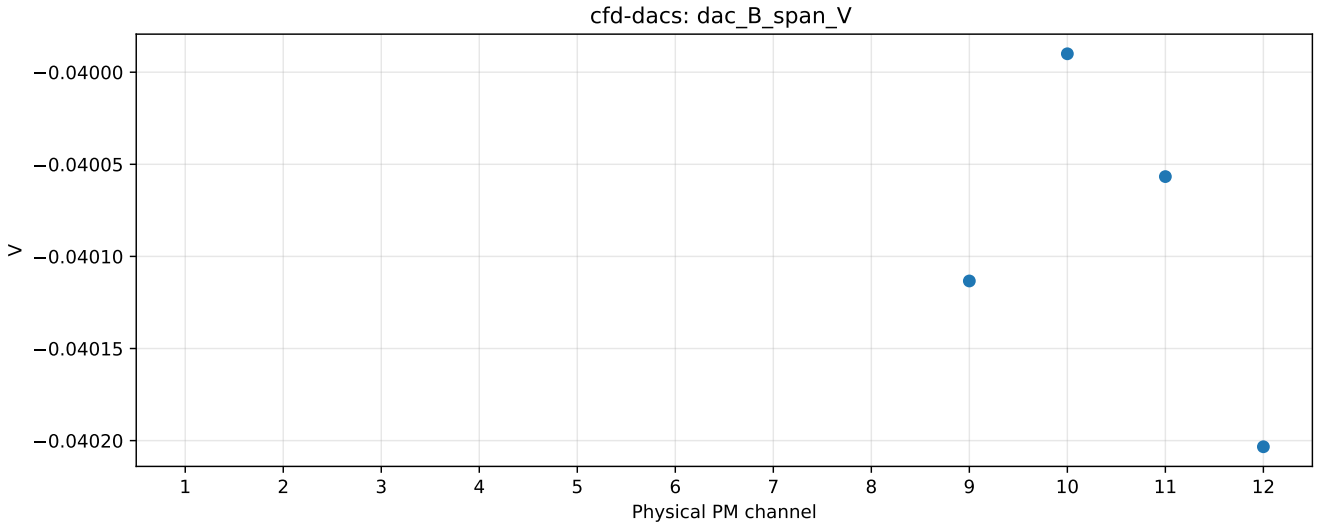

integrator: balance_delta B
V
0.0095
0.0090
0.0085
0.0080
0.0075
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

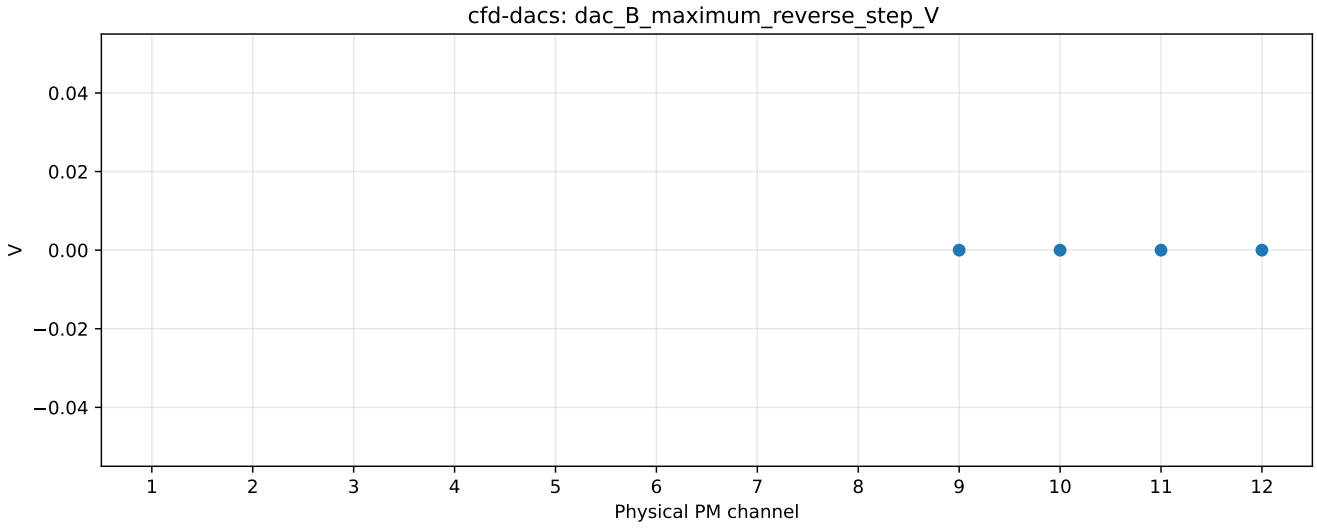

integrator: gain B
code/V
−2460
−2480
−2500
−2520
−2540
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

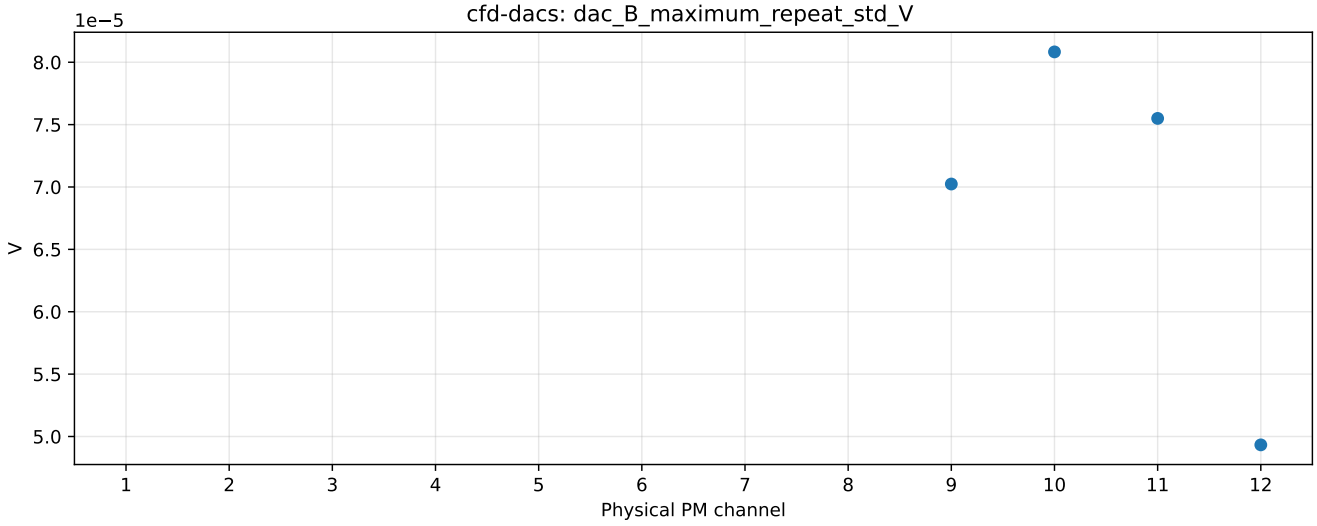

integrator: near_balance_change B
code
0.19
0.18
0.17
0.16
0.15
0.14
0.13
0.12
0.11
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

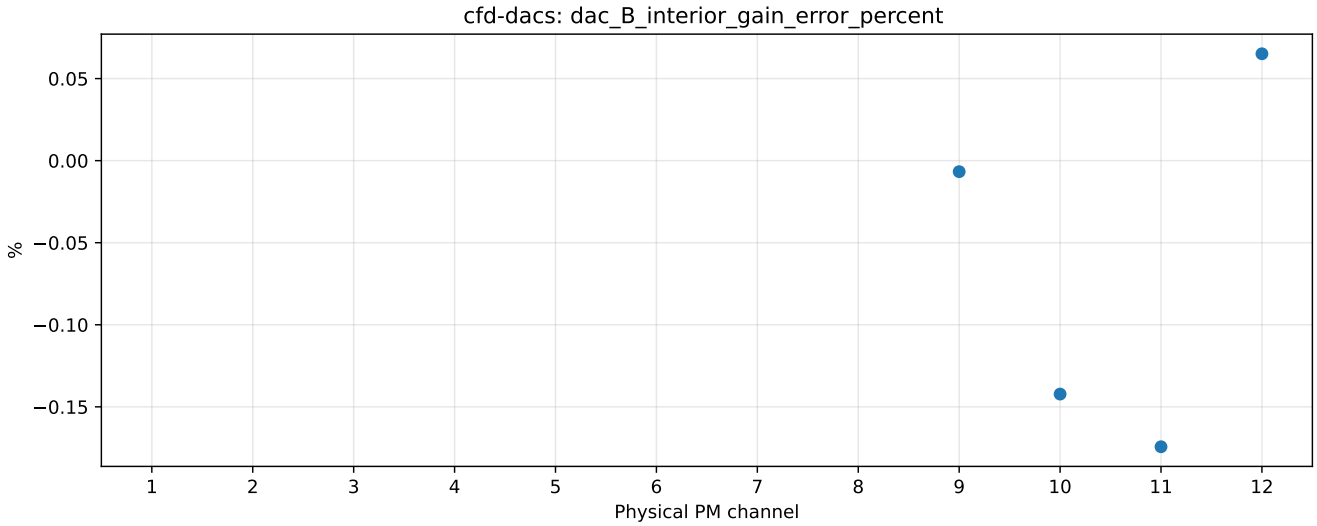

cfd-dacs: dac_B_minimum_code_V
+1.044
V
0.00070
0.00065
0.00060
0.00055
0.00050
0.00045
0.00040
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

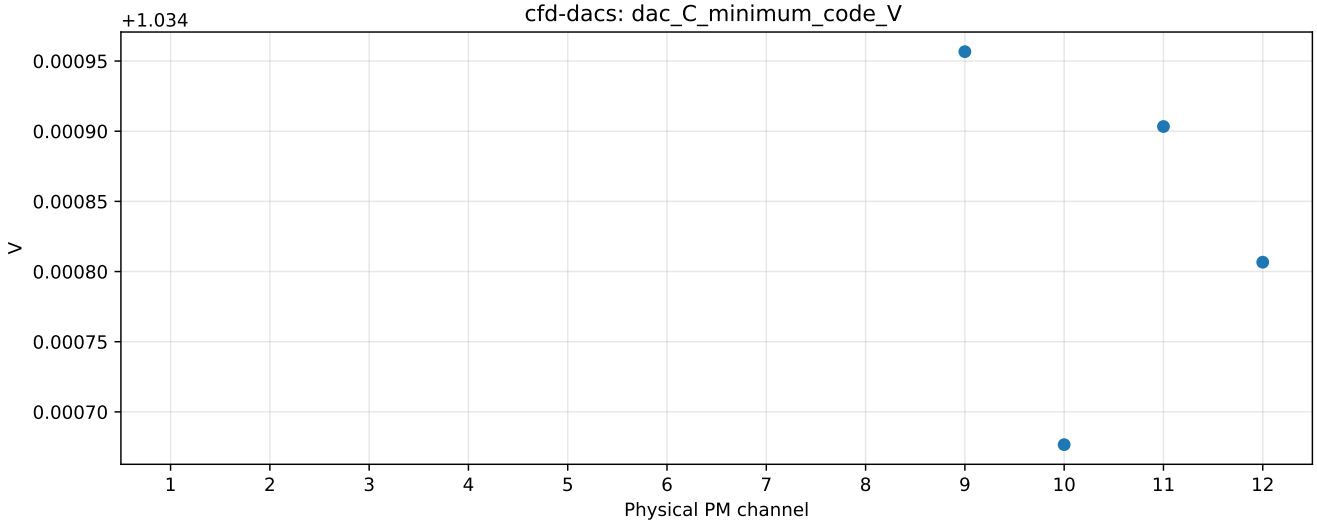

cfd-dacs: dac_B_maximum_code_V
+1.004
0.00060
0.00055
0.00050
0.00045
0.00040
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

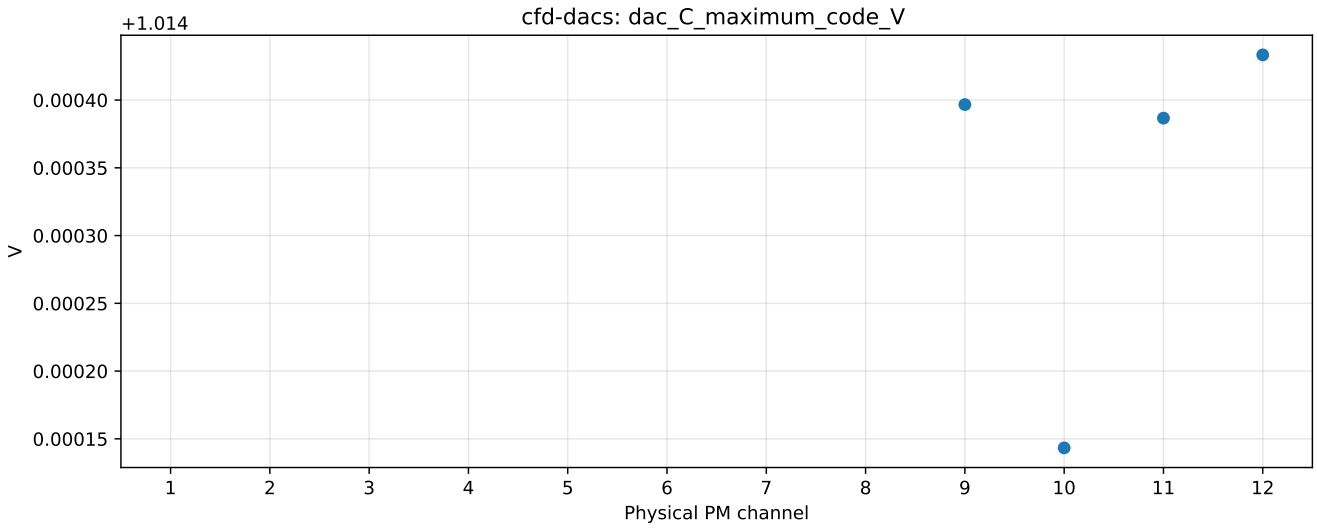

cfd-dacs: dac_B_span_V
−0.04000
−0.04005
−0.04010
−0.04015
−0.04020
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

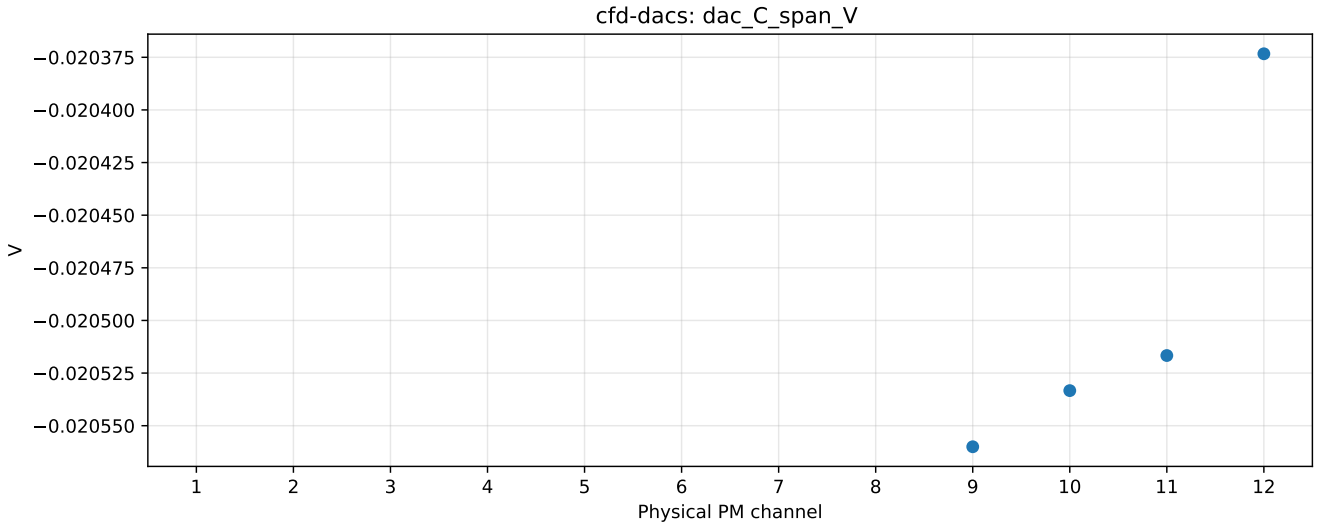

cfd-dacs: dac_B_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

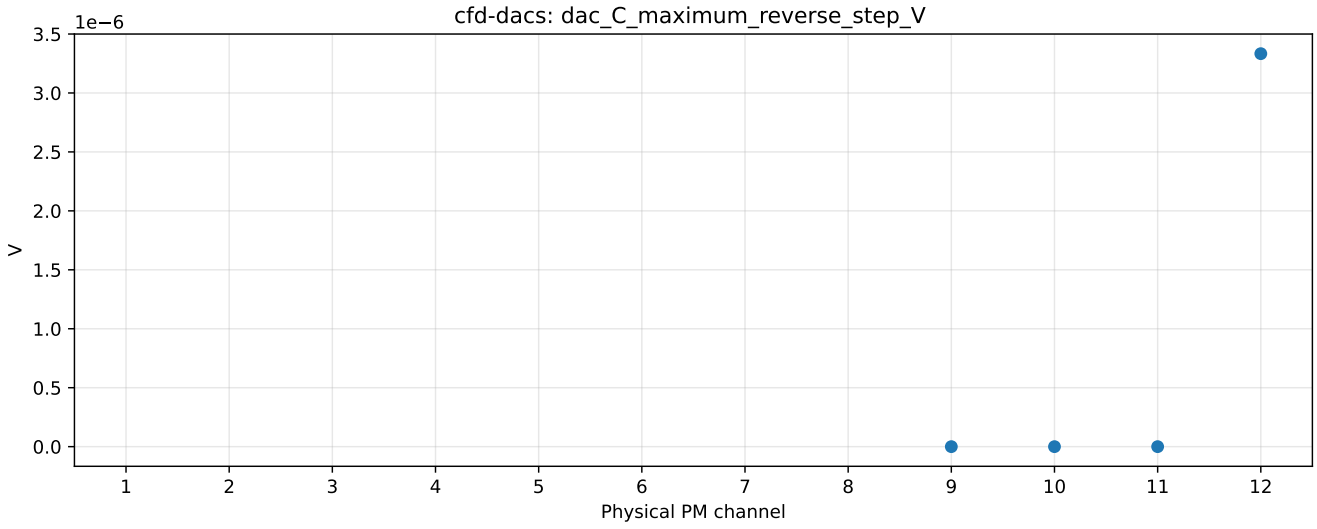

cfd-dacs: dac_B_maximum_repeat_std_V
1e−5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

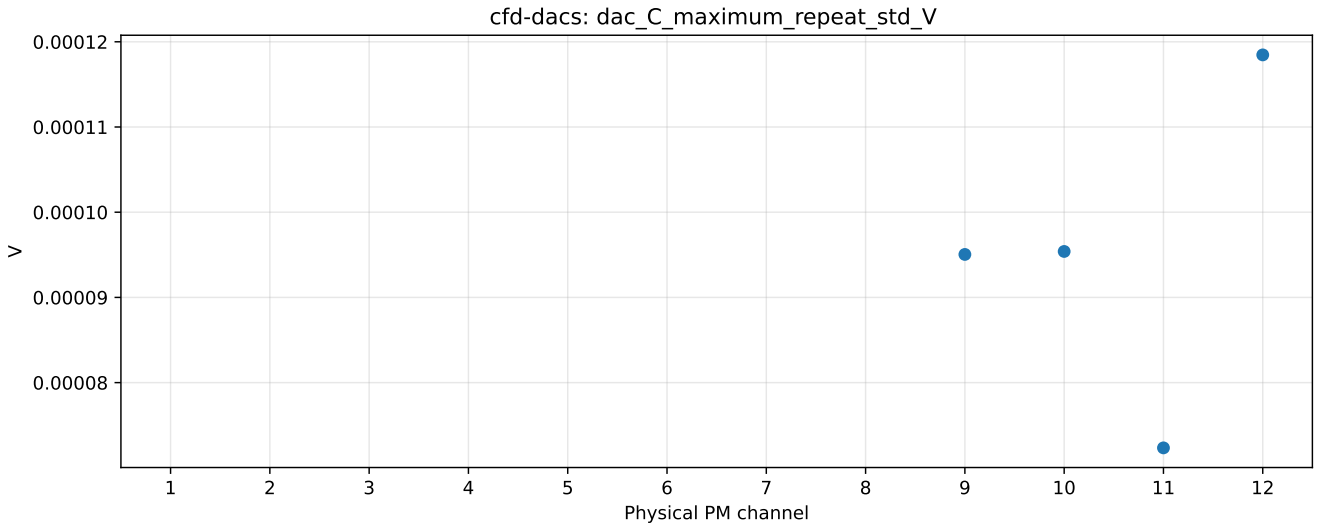

cfd-dacs: dac_B_interior_gain_error_percent
0.05
0.00
−0.05
−0.10
−0.15
%
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

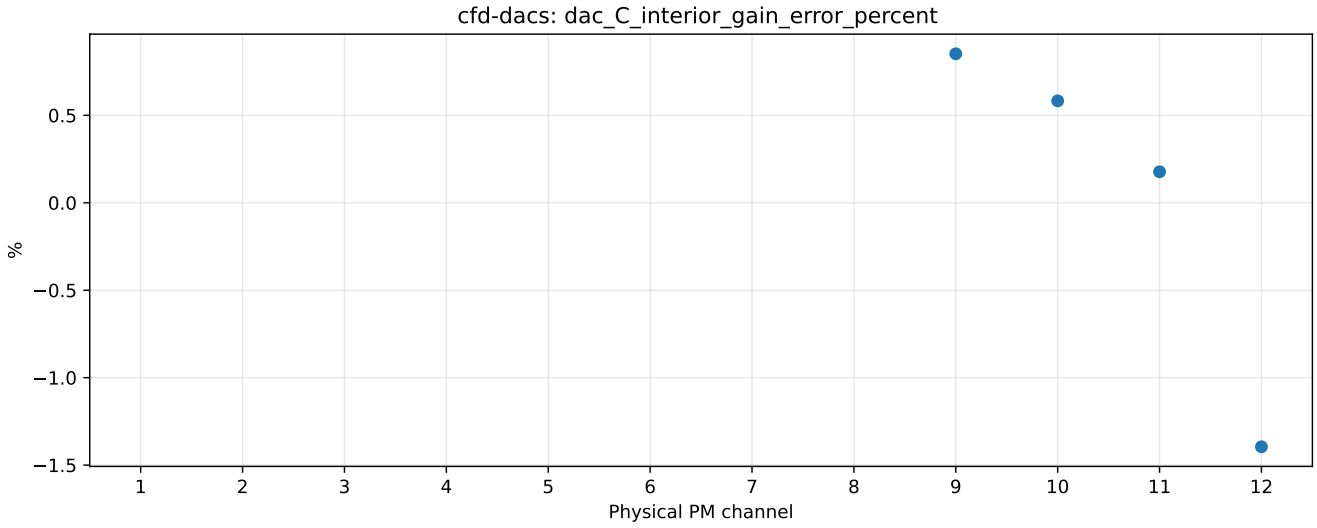

cfd-dacs: dac_C_minimum_code_V
+1.034
0.00095
0.00090
0.00085
0.00080
0.00075
0.00070
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

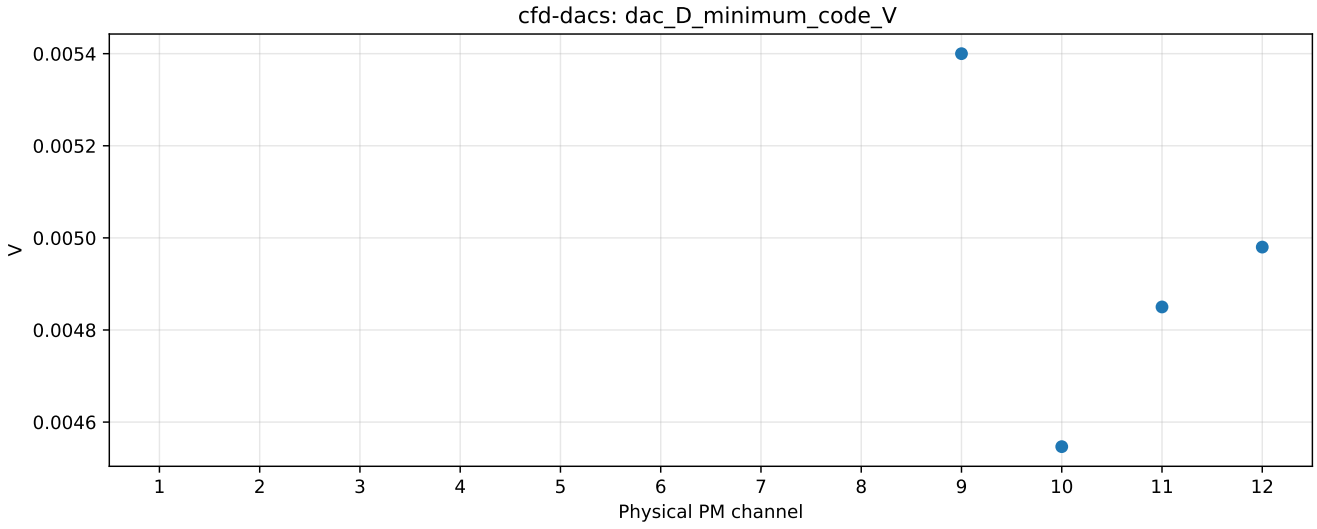

cfd-dacs: dac_C_maximum_code_V
+1.014
0.00040
0.00035
0.00030
0.00025
0.00020
0.00015
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

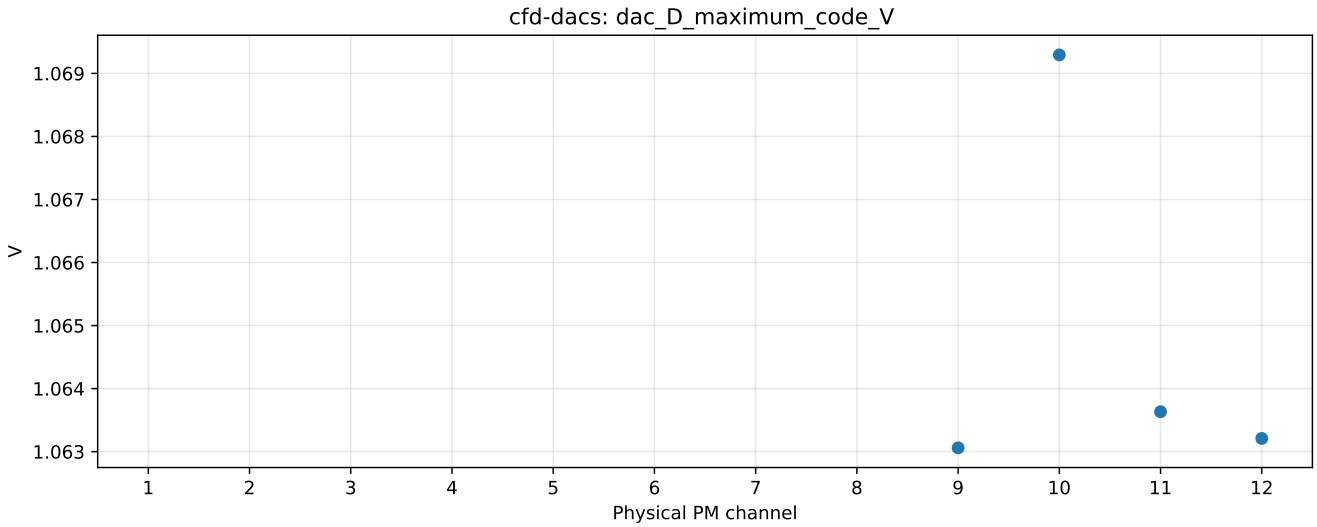

cfd-dacs: dac_C_span_V
−0.020375
−0.020400
−0.020425
−0.020450
−0.020475
−0.020500
−0.020525
−0.020550
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

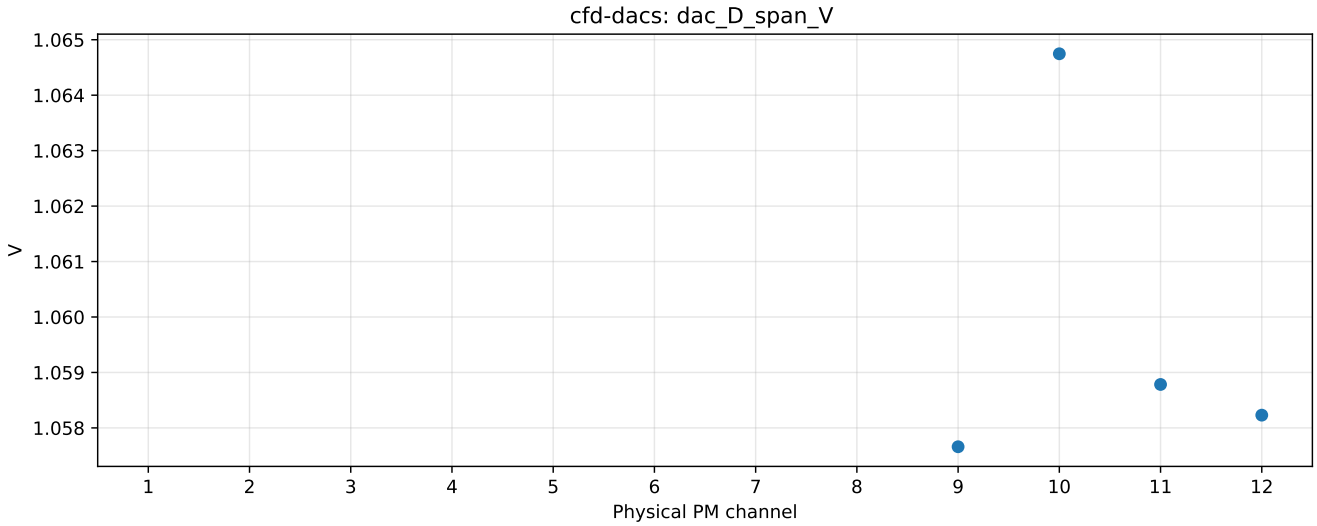

cfd-dacs: dac_C_maximum_reverse_step_V
1e−6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

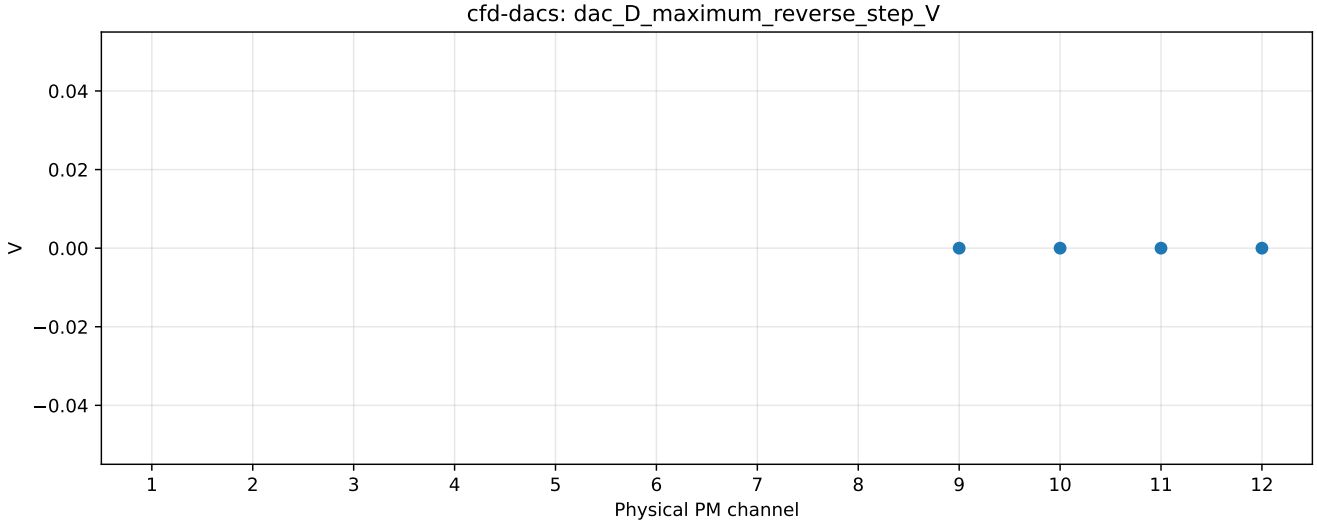

cfd-dacs: dac_C_maximum_repeat_std_V
0.00012
0.00011
0.00010
0.00009
0.00008
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

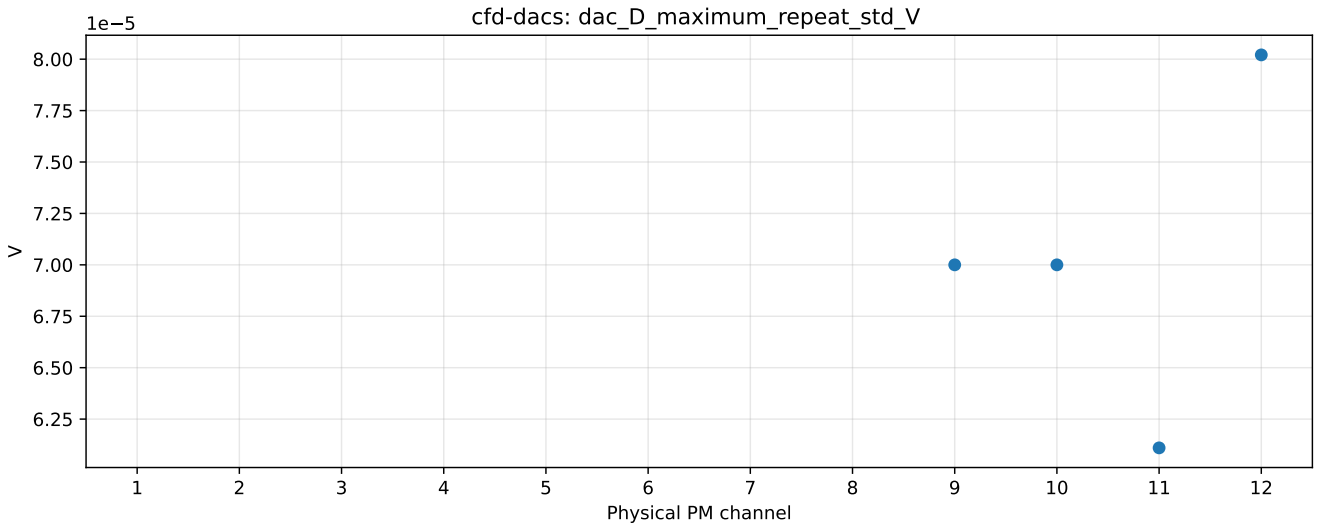

cfd-dacs: dac_C_interior_gain_error_percent
0.5
0.0
−0.5
−1.0
−1.5
%
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

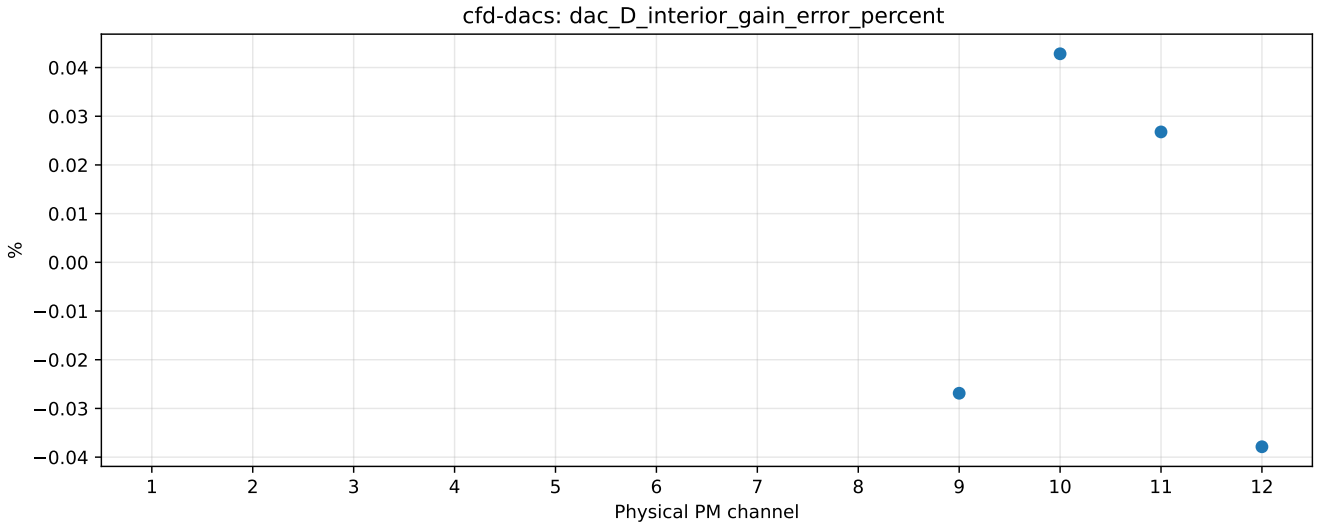

cfd-dacs: dac_D_minimum_code_V
0.0054
0.0052
0.0050
0.0048
0.0046
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

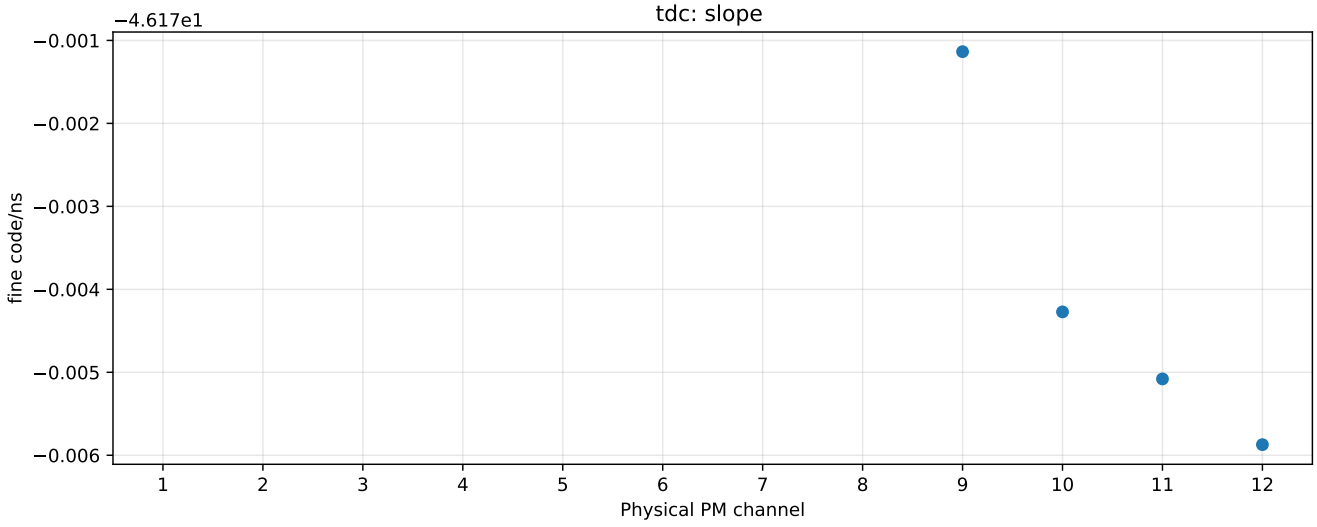

cfd-dacs: dac_D_maximum_code_V
1.069
1.068
1.067
1.066
1.065
1.064
1.063
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

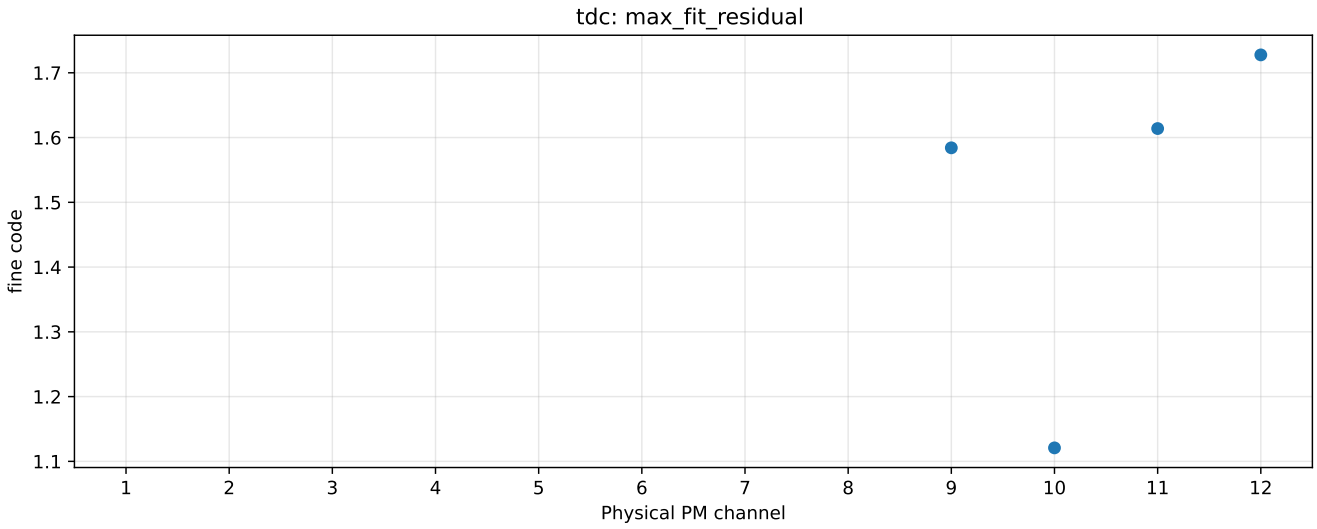

cfd-dacs: dac_D_span_V
1.065
1.064
1.063
1.062
1.061
1.060
1.059
1.058
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

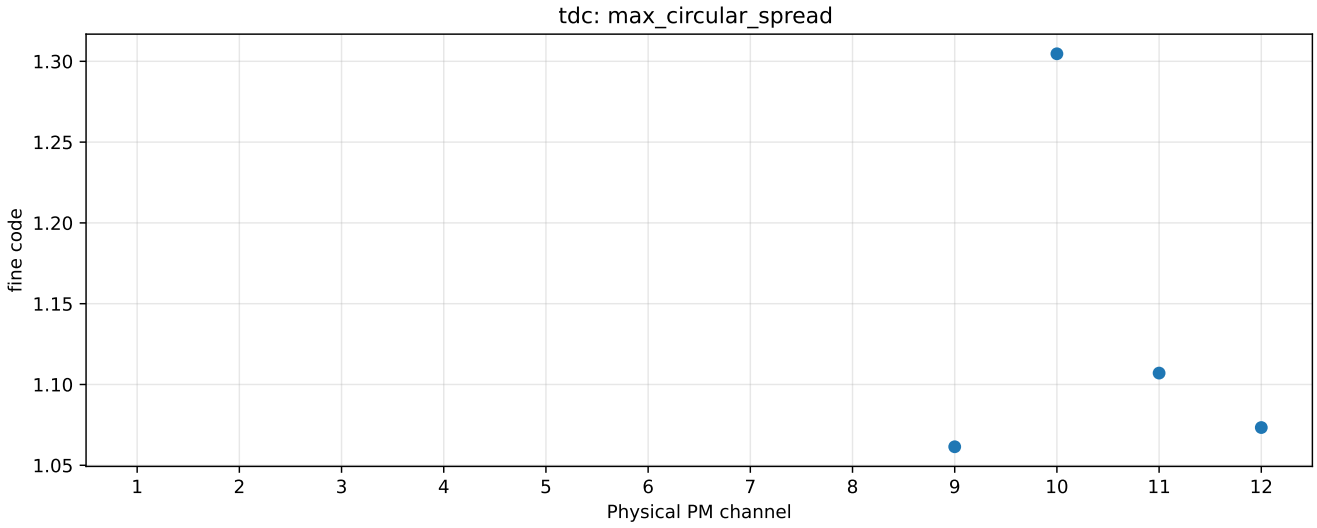

cfd-dacs: dac_D_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

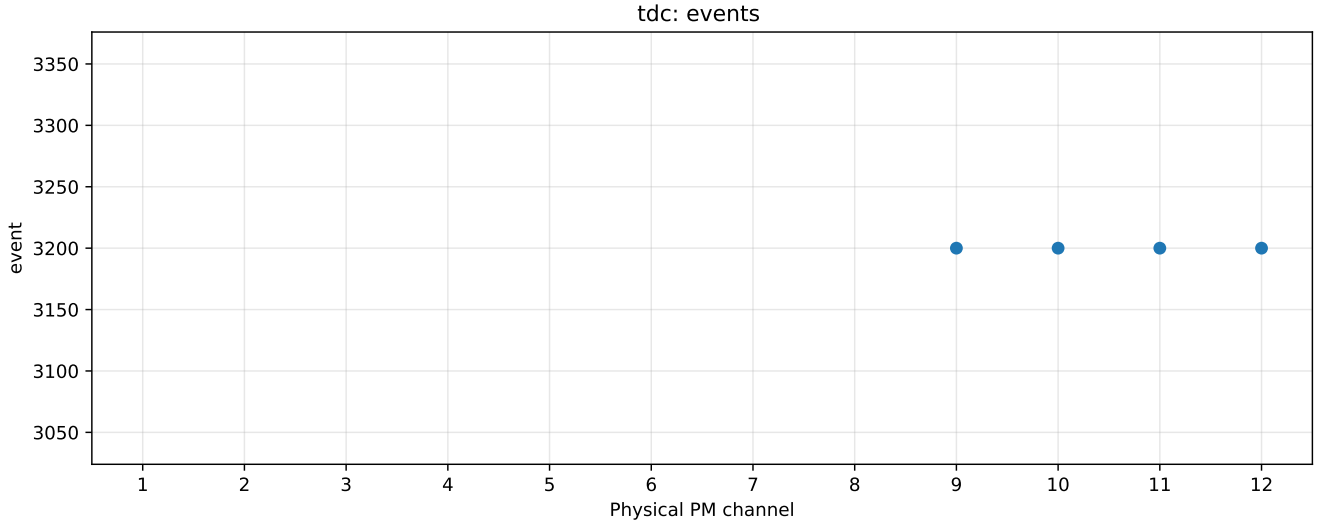

cfd-dacs: dac_D_maximum_repeat_std_V
1e−5
8.00
7.75
7.50
7.25
7.00
6.75
6.50
6.25
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

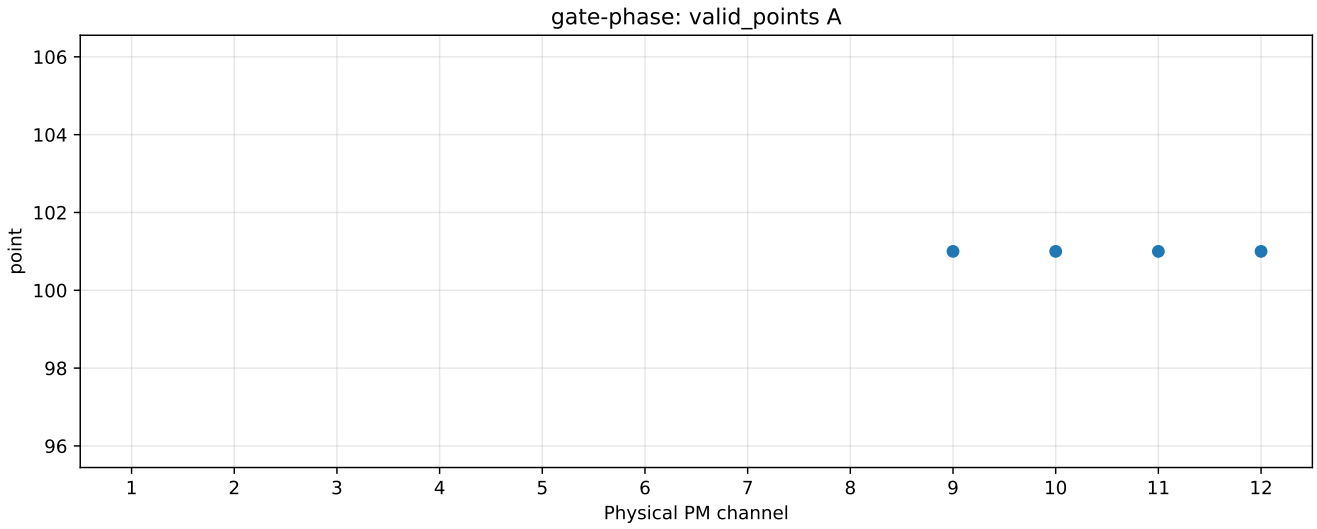

cfd-dacs: dac_D_interior_gain_error_percent
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
%
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

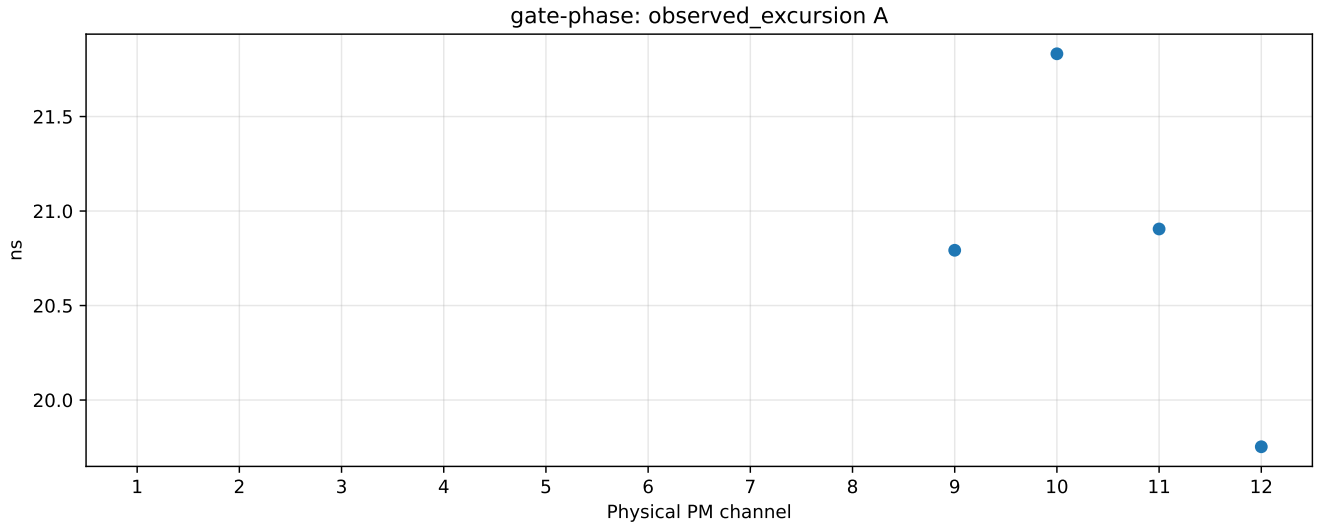

tdc: slope
−4.617e1
−0.001
−0.002
−0.003
−0.004
−0.005
−0.006
fine code/ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

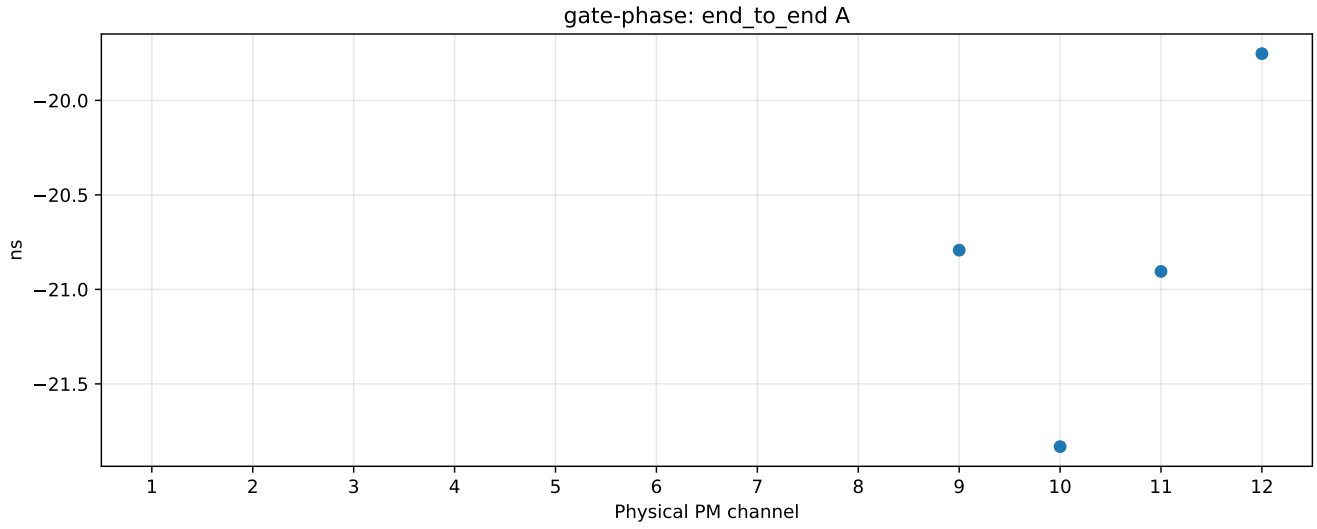

tdc: max_fit_residual
fine code
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

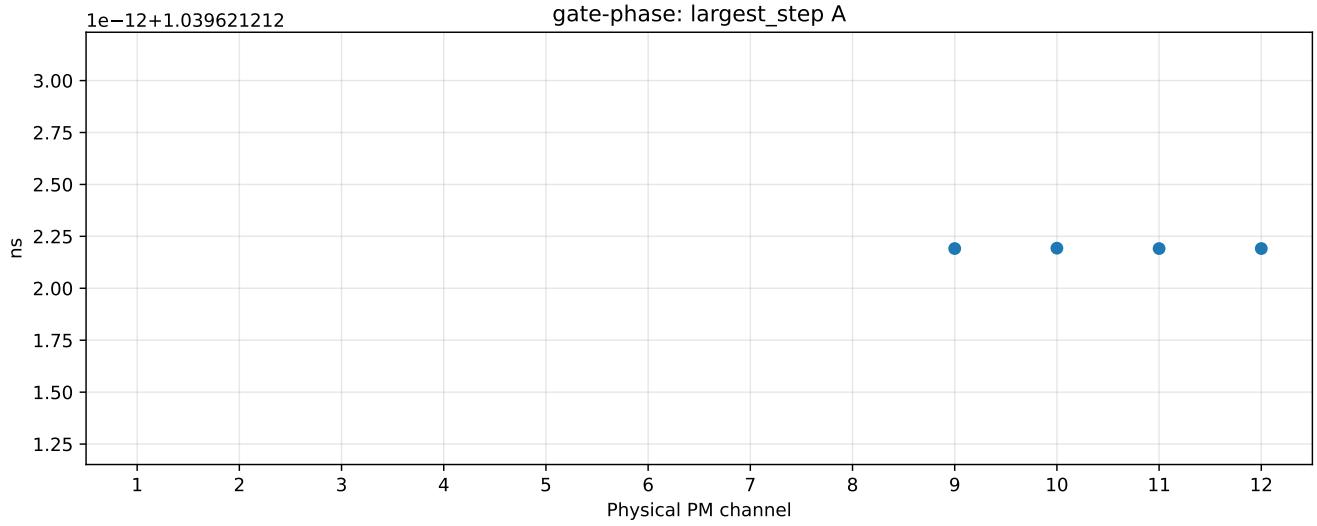

tdc: max_circular_spread
fine code
1.30
1.25
1.20
1.15
1.10
1.05
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

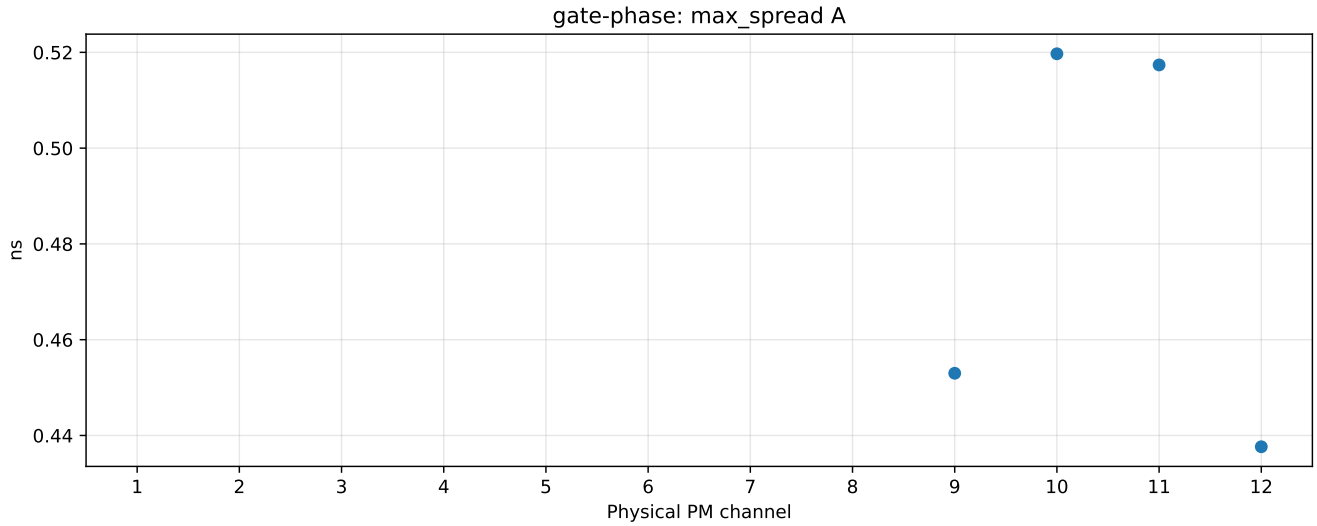

tdc: events
event
3350
3300
3250
3200
3150
3100
3050
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

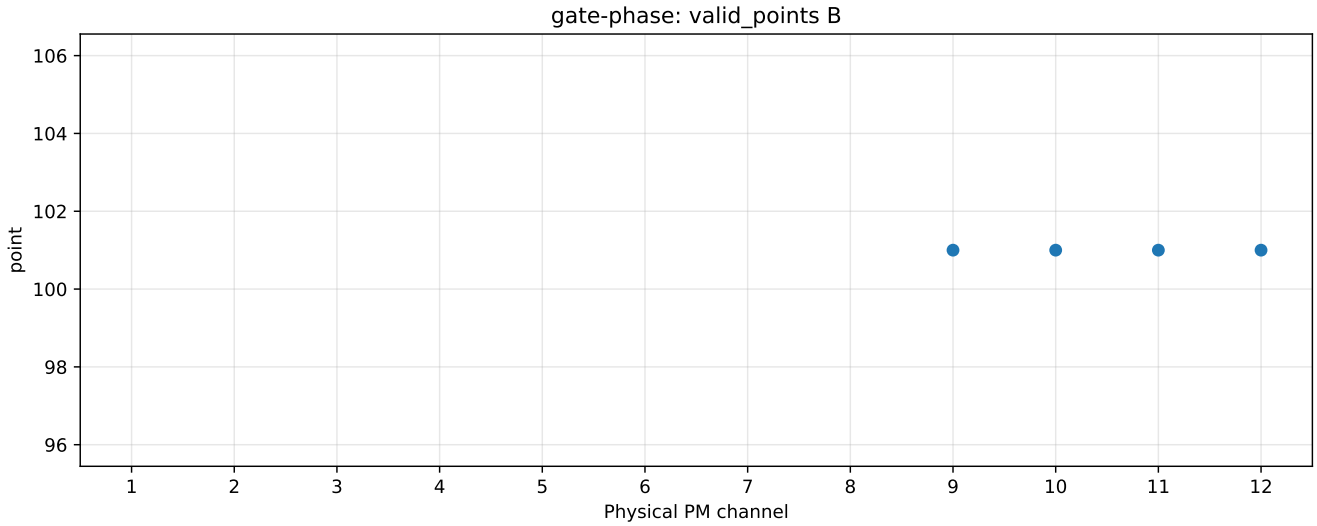

gate-phase: valid_points A
point
106
104
102
100
98
96
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

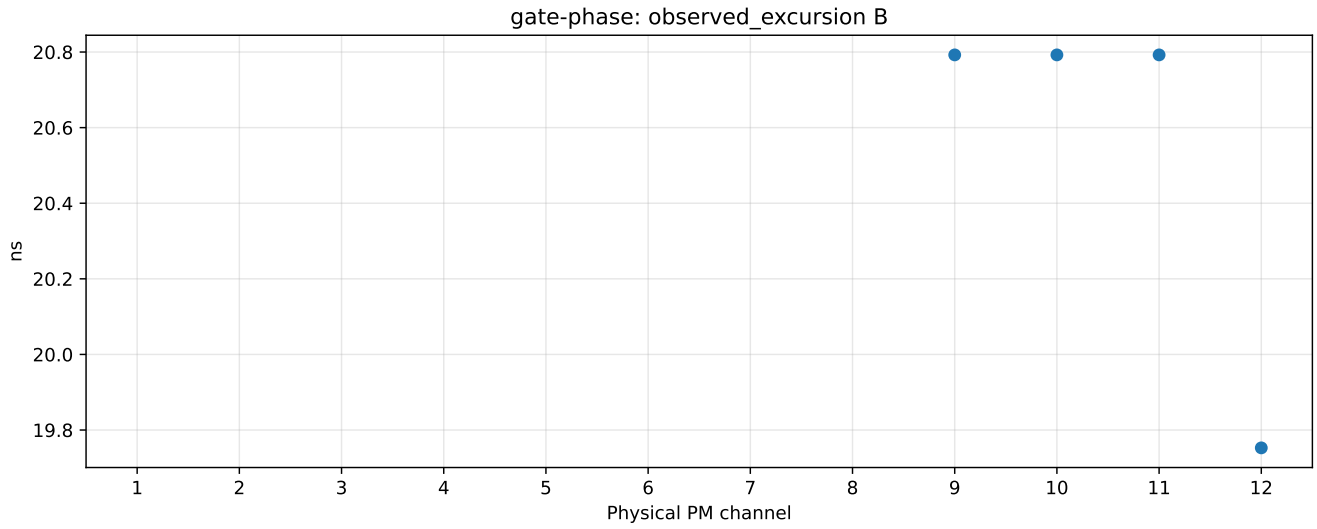

gate-phase: observed_excursion A
ns
21.5
21.0
20.5
20.0
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

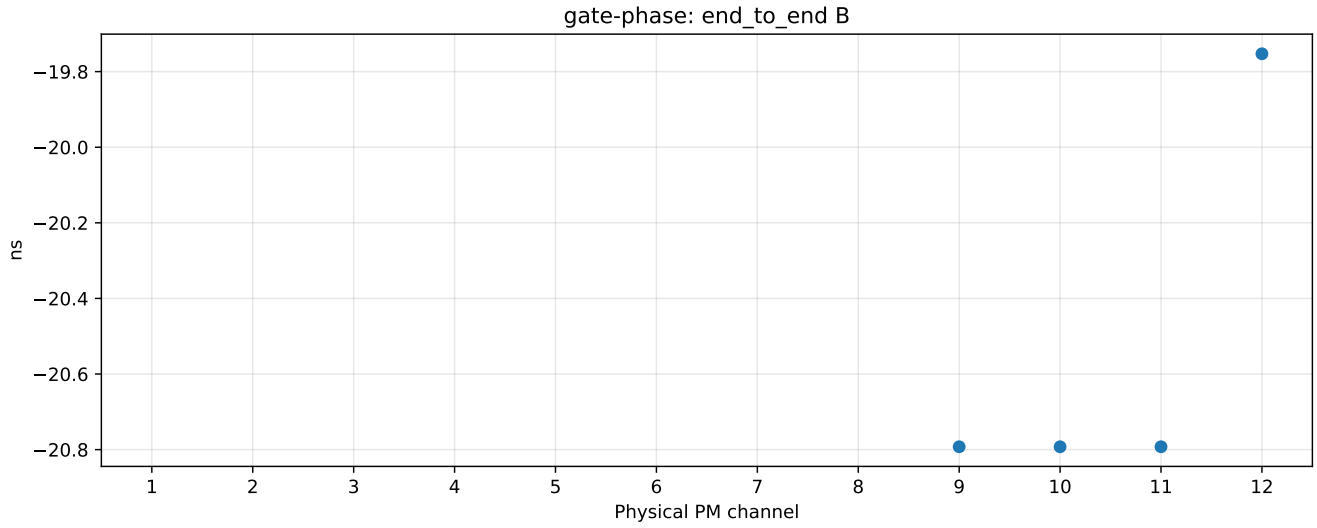

gate-phase: end_to_end A
ns
−20.0
−20.5
−21.0
−21.5
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

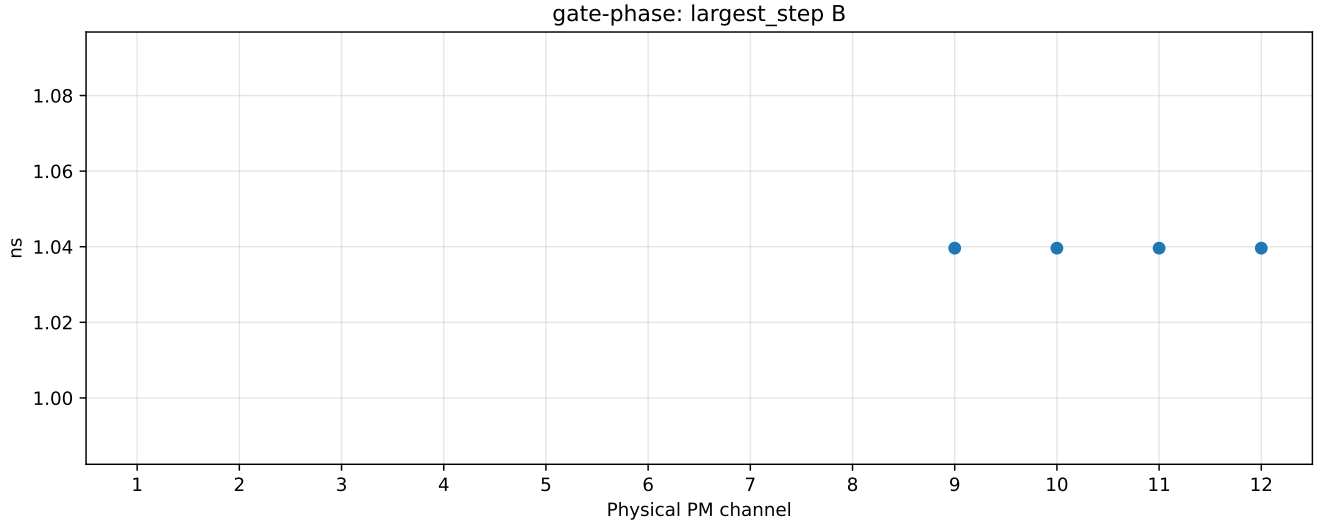

gate-phase: largest_step A
1e−12+1.039621212
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

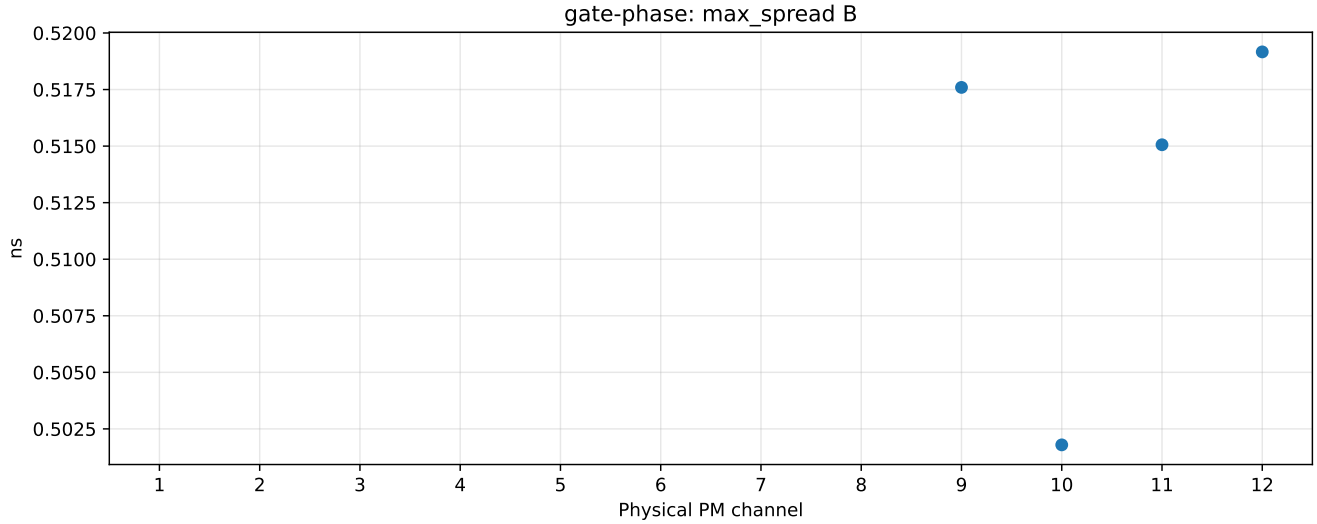

gate-phase: max_spread A
0.52
0.50
0.48
0.46
0.44
ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

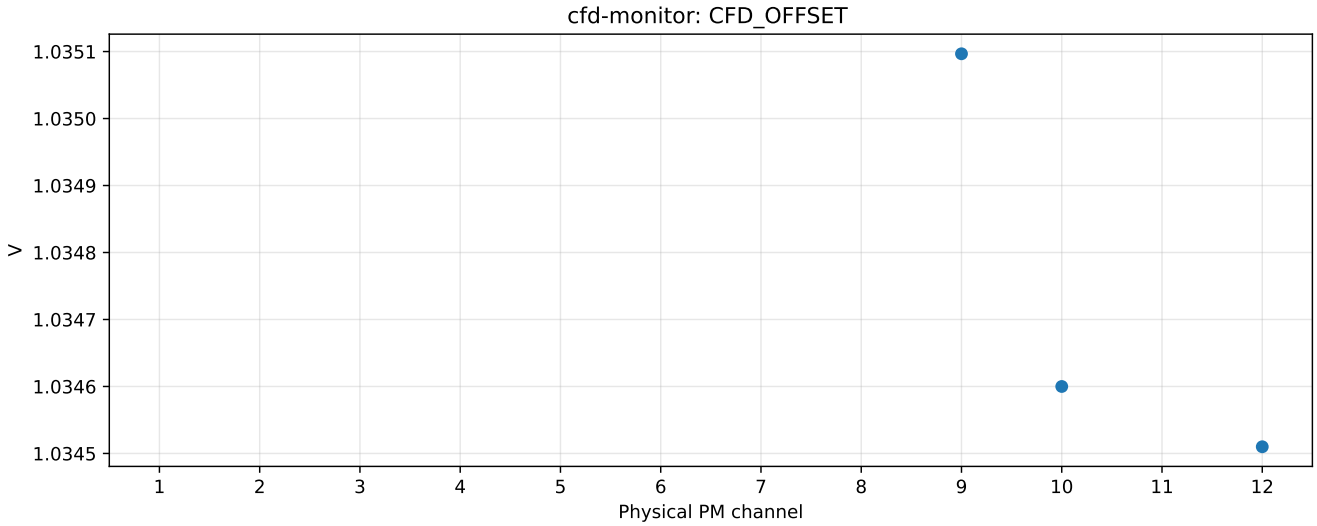

gate-phase: valid_points B
106
104
102
100
98
96
point
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

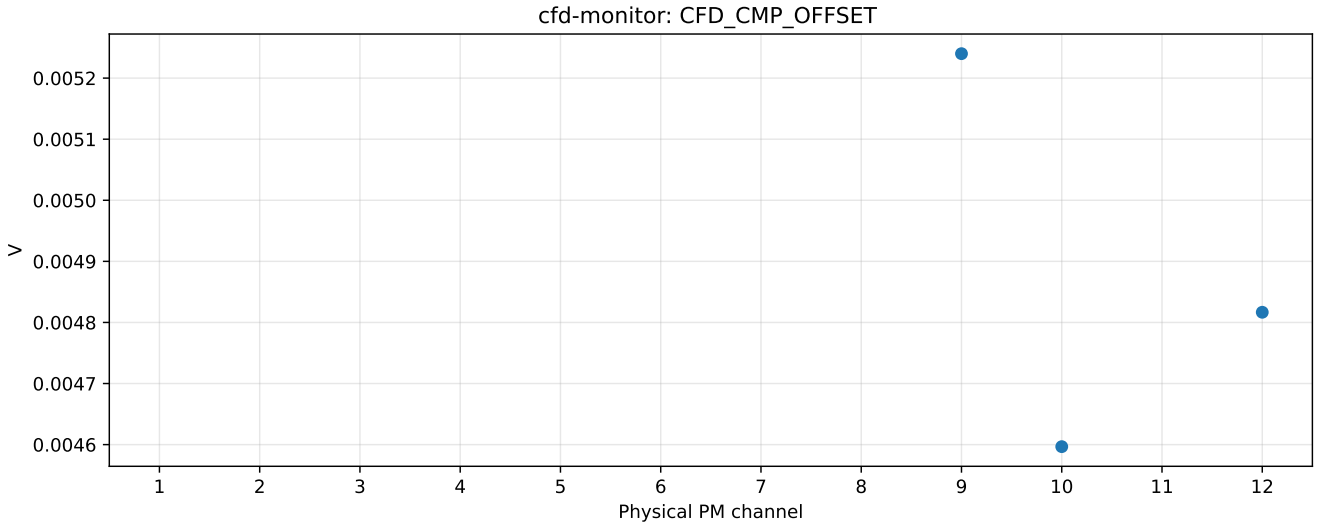

gate-phase: observed_excursion B
20.8
20.6
20.4
20.2
20.0
19.8
ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

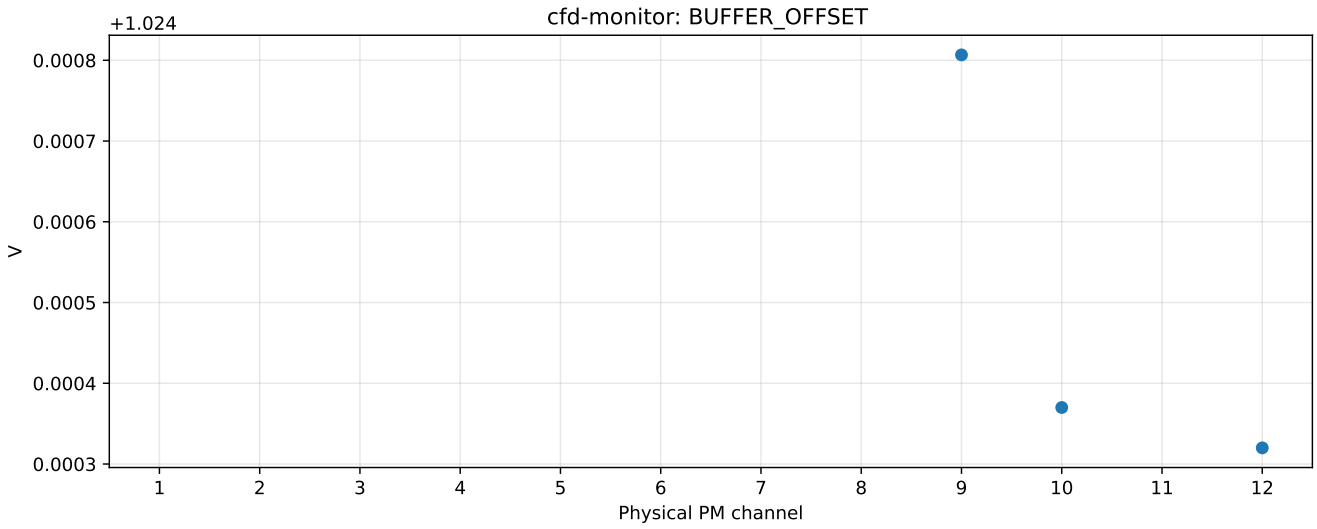

gate-phase: end_to_end B
−19.8
−20.0
−20.2
−20.4
−20.6
−20.8
ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

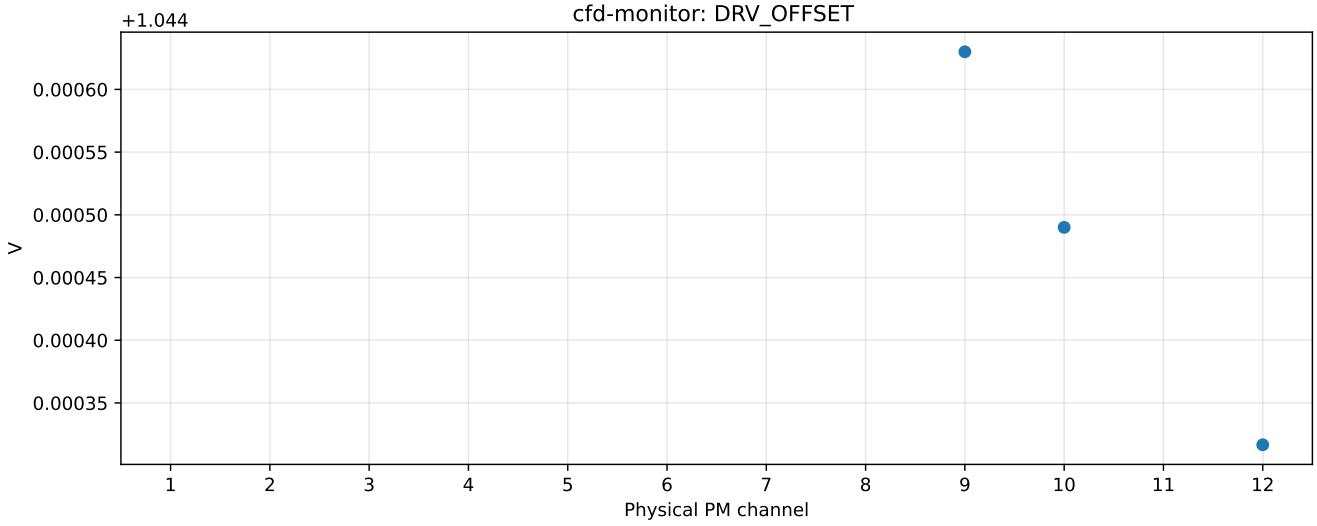

gate-phase: largest_step B
1.08
1.06
1.04
1.02
1.00
ns
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

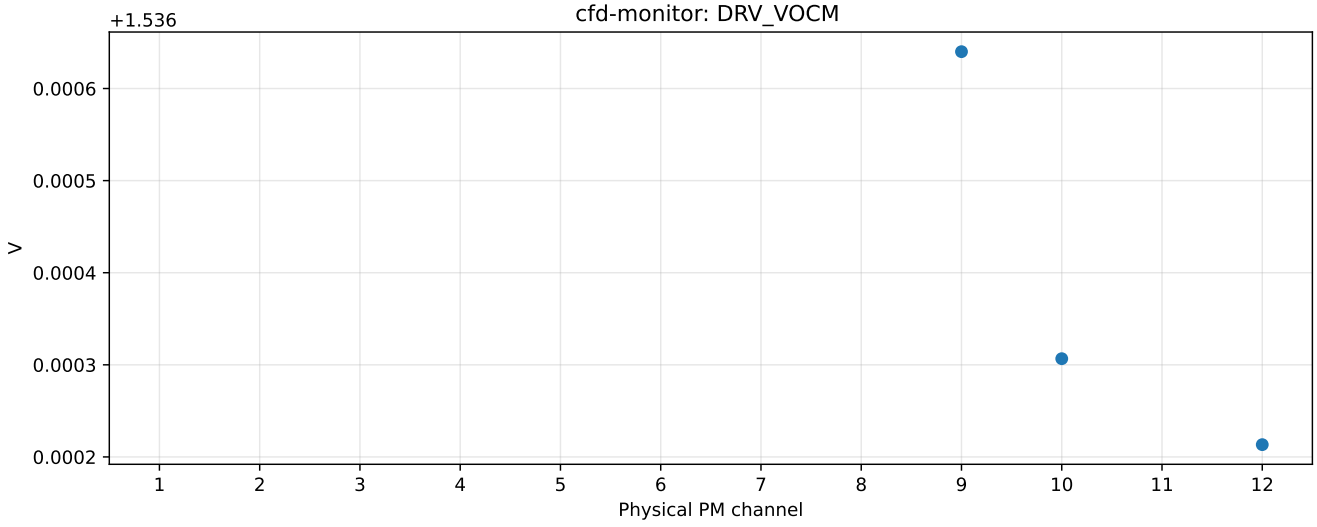

gate-phase: max_spread B
ns
0.5200
0.5175
0.5150
0.5125
0.5100
0.5075
0.5050
0.5025
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

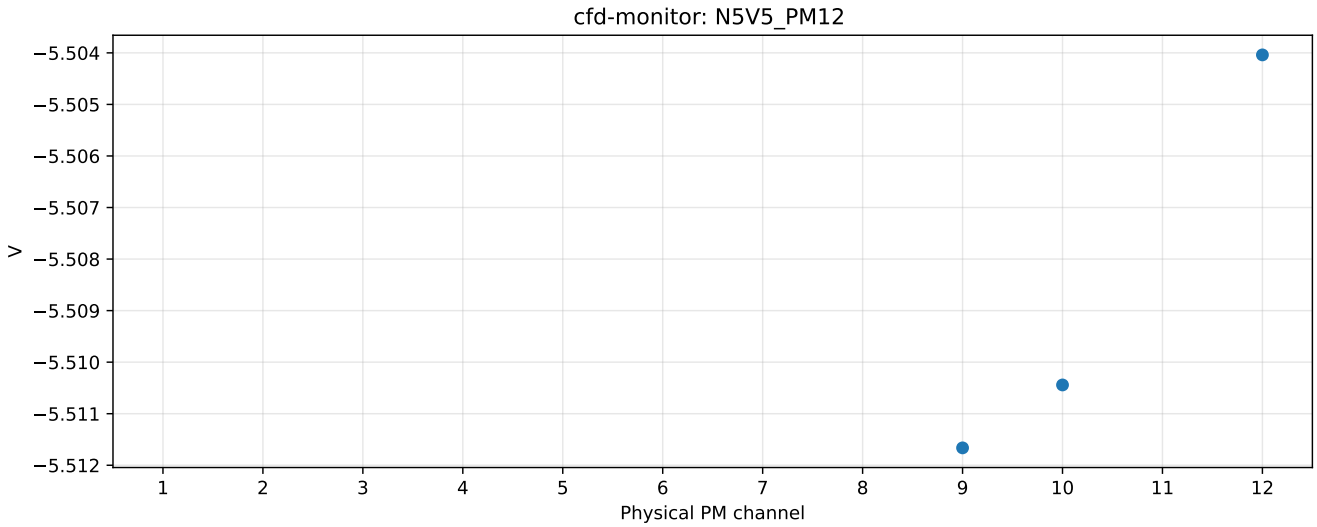

cfd-monitor: CFD_OFFSET
V
1.0351
1.0350
1.0349
1.0348
1.0347
1.0346
1.0345
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

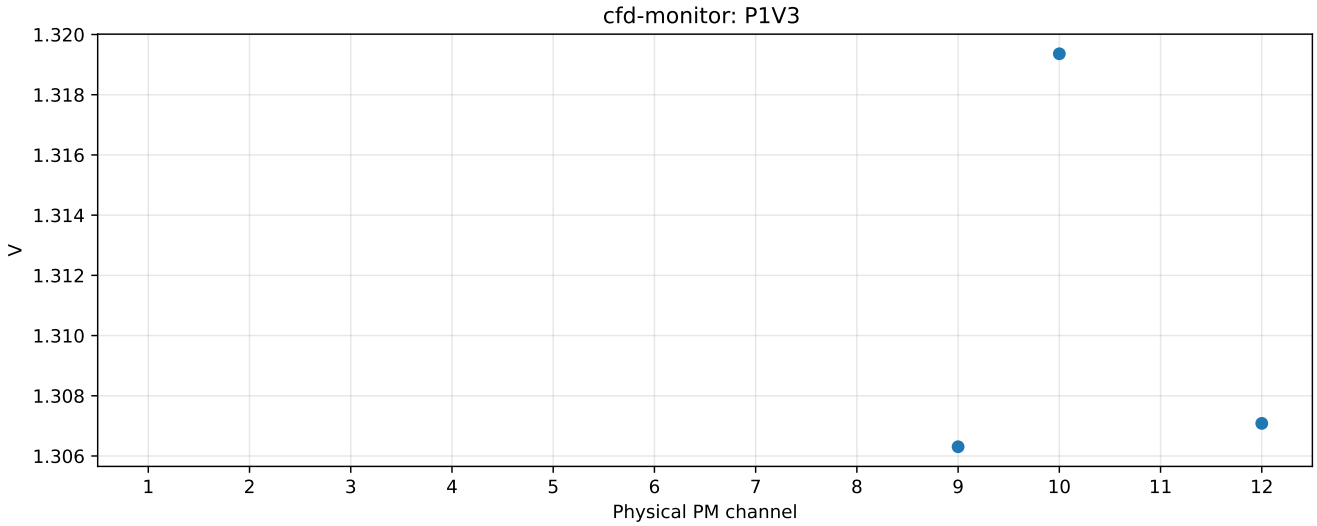

cfd-monitor: CFD_CMP_OFFSET
V
0.0052
0.0051
0.0050
0.0049
0.0048
0.0047
0.0046
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

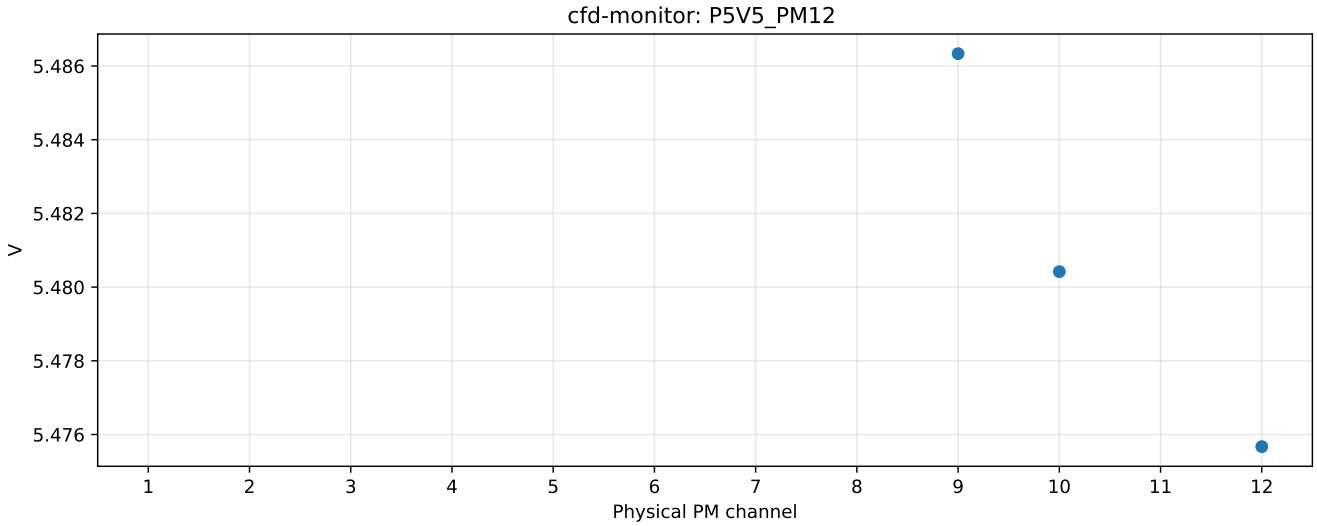

cfd-monitor: BUFFER_OFFSET
+1.024
0.0008
0.0007
0.0006
0.0005
0.0004
0.0003
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-monitor: DRV_OFFSET
+1.044
0.00060
0.00055
0.00050
0.00045
0.00040
0.00035
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-monitor: DRV_VOCM
+1.536
0.0006
0.0005
0.0004
0.0003
0.0002
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel







7 Static CPLD–FPGA connection discovery

Each path independently used338walking-one/zero vectors. All168expected sources match; one common table avoids repeated descriptions. This checks static connectivity, not at-speed margin or unconnected analog inputs.

PM channel	Signal	CPLD pin	FPGA ball	Gateware	Boundary
1	DI1[0]	99	T7	PASS	PASS
1	DI1[1]	97	U7	PASS	PASS
1	DI1[2]	96	U6	PASS	PASS
1	DI1[3]	95	U5	PASS	PASS
1	DI1[4]	94	U4	PASS	PASS
1	DI1[5]	93	U2	PASS	PASS
1	DI1[6]	92	U1	PASS	PASS
1	DI1[7]	91	V6	PASS	PASS
1	DI1[8]	90	V4	PASS	PASS
1	DI1[9]	89	W4	PASS	PASS
1	DI1[10]	87	V2	PASS	PASS
1	DI1[11]	86	V1	PASS	PASS
1	DI1[12]	85	W6	PASS	PASS
1	STR1	82	W5	PASS	PASS
2	DI2[0]	99	AF4	PASS	PASS
2	DI2[1]	97	AD5	PASS	PASS
2	DI2[2]	96	AE5	PASS	PASS
2	DI2[3]	95	AF5	PASS	PASS
2	DI2[4]	94	AB6	PASS	PASS
2	DI2[5]	93	AC6	PASS	PASS
2	DI2[6]	92	AD6	PASS	PASS
2	DI2[7]	91	AE6	PASS	PASS
2	DI2[8]	90	U9	PASS	PASS
2	DI2[9]	89	V7	PASS	PASS
2	DI2[10]	87	V8	PASS	PASS
2	DI2[11]	86	Y7	PASS	PASS
2	DI2[12]	85	Y8	PASS	PASS
2	STR2	82	AA7	PASS	PASS
3	DI3[0]	99	AA8	PASS	PASS
3	DI3[1]	97	AD11	PASS	PASS
3	DI3[2]	96	AE11	PASS	PASS
3	DI3[3]	95	V12	PASS	PASS
3	DI3[4]	94	Y12	PASS	PASS
3	DI3[5]	93	Y13	PASS	PASS
3	DI3[6]	92	AA12	PASS	PASS
3	DI3[7]	91	AA13	PASS	PASS
3	DI3[8]	90	V13	PASS	PASS
3	DI3[9]	89	W13	PASS	PASS
3	DI3[10]	87	V14	PASS	PASS
3	DI3[11]	86	W14	PASS	PASS
3	DI3[12]	85	AA14	PASS	PASS
3	STR3	82	AA15	PASS	PASS
4	DI4[0]	99	V18	PASS	PASS
4	DI4[1]	97	W18	PASS	PASS
4	DI4[2]	96	AA19	PASS	PASS
4	DI4[3]	95	AC19	PASS	PASS
4	DI4[4]	94	AE18	PASS	PASS
4	DI4[5]	93	AF18	PASS	PASS
4	DI4[6]	92	AB20	PASS	PASS
4	DI4[7]	91	V19	PASS	PASS
4	DI4[8]	90	W19	PASS	PASS
4	DI4[9]	89	AB19	PASS	PASS
4	DI4[10]	87	AD19	PASS	PASS
4	DI4[11]	86	AF19	PASS	PASS
4	DI4[12]	85	AA20	PASS	PASS
4	STR4	82	AD20	PASS	PASS
5	DI5[0]	99	AE20	PASS	PASS
5	DI5[1]	97	AF20	PASS	PASS
5	DI5[2]	96	AE21	PASS	PASS
5	DI5[3]	95	AF22	PASS	PASS

PM channel	Signal	CPLD pin	FPGA ball	Gateware	Boundary
5	DI5[4]	94	AE22	PASS	PASS
5	DI5[5]	93	AF23	PASS	PASS
5	DI5[6]	92	AE23	PASS	PASS
5	DI5[7]	91	AF24	PASS	PASS
5	DI5[8]	90	AF25	PASS	PASS
5	DI5[9]	89	AE25	PASS	PASS
5	DI5[10]	87	AE26	PASS	PASS
5	DI5[11]	86	AD21	PASS	PASS
5	DI5[12]	85	AD23	PASS	PASS
5	STR5	82	AB21	PASS	PASS
6	DI6[0]	99	AD25	PASS	PASS
6	DI6[1]	97	AD26	PASS	PASS
6	DI6[2]	96	AC21	PASS	PASS
6	DI6[3]	95	AC22	PASS	PASS
6	DI6[4]	94	AC23	PASS	PASS
6	DI6[5]	93	AD24	PASS	PASS
6	DI6[6]	92	AC26	PASS	PASS
6	DI6[7]	91	AC24	PASS	PASS
6	DI6[8]	90	AB22	PASS	PASS
6	DI6[9]	89	AB24	PASS	PASS
6	DI6[10]	87	AB25	PASS	PASS
6	DI6[11]	86	AB26	PASS	PASS
6	DI6[12]	85	AA22	PASS	PASS
6	STR6	82	AA23	PASS	PASS
7	DI7[0]	99	AA24	PASS	PASS
7	DI7[1]	97	AA25	PASS	PASS
7	DI7[2]	96	Y20	PASS	PASS
7	DI7[3]	95	Y21	PASS	PASS
7	DI7[4]	94	Y22	PASS	PASS
7	DI7[5]	93	Y23	PASS	PASS
7	DI7[6]	92	Y25	PASS	PASS
7	DI7[7]	91	Y26	PASS	PASS
7	DI7[8]	90	W20	PASS	PASS
7	DI7[9]	89	W21	PASS	PASS
7	DI7[10]	87	W23	PASS	PASS
7	DI7[11]	86	W24	PASS	PASS
7	DI7[12]	85	W25	PASS	PASS
7	STR7	82	W26	PASS	PASS
8	DI8[0]	99	V21	PASS	PASS
8	DI8[1]	97	V22	PASS	PASS
8	DI8[2]	96	V23	PASS	PASS
8	DI8[3]	95	V24	PASS	PASS
8	DI8[4]	94	V26	PASS	PASS
8	DI8[5]	93	U21	PASS	PASS
8	DI8[6]	92	U22	PASS	PASS
8	DI8[7]	91	U24	PASS	PASS
8	DI8[8]	90	U25	PASS	PASS
8	DI8[9]	89	U26	PASS	PASS
8	DI8[10]	87	U16	PASS	PASS
8	DI8[11]	86	U17	PASS	PASS
8	DI8[12]	85	U19	PASS	PASS
8	STR8	82	U20	PASS	PASS
9	DI9[0]	99	T17	PASS	PASS
9	DI9[1]	97	T18	PASS	PASS
9	DI9[2]	96	T19	PASS	PASS
9	DI9[3]	95	T20	PASS	PASS
9	DI9[4]	94	T22	PASS	PASS
9	DI9[5]	93	T23	PASS	PASS
9	DI9[6]	92	T24	PASS	PASS
9	DI9[7]	91	T25	PASS	PASS

PM channel	Signal	CPLD pin	FPGA ball	Gateware	Boundary
9	DI9[8]	90	R16	PASS	PASS
9	DI9[9]	89	R17	PASS	PASS
9	DI9[10]	87	R18	PASS	PASS
9	DI9[11]	86	R20	PASS	PASS
9	DI9[12]	85	R21	PASS	PASS
9	STR9	82	R22	PASS	PASS
10	DI10[0]	99	R23	PASS	PASS
10	DI10[1]	97	R25	PASS	PASS
10	DI10[2]	96	R26	PASS	PASS
10	DI10[3]	95	P16	PASS	PASS
10	DI10[4]	94	P18	PASS	PASS
10	DI10[5]	93	P19	PASS	PASS
10	DI10[6]	92	P20	PASS	PASS
10	DI10[7]	91	P21	PASS	PASS
10	DI10[8]	90	P23	PASS	PASS
10	DI10[9]	89	P24	PASS	PASS
10	DI10[10]	87	P25	PASS	PASS
10	DI10[11]	86	P26	PASS	PASS
10	DI10[12]	85	N16	PASS	PASS
10	STR10	82	N17	PASS	PASS
11	DI11[0]	99	N18	PASS	PASS
11	DI11[1]	97	N19	PASS	PASS
11	DI11[2]	96	N21	PASS	PASS
11	DI11[3]	95	N22	PASS	PASS
11	DI11[4]	94	N23	PASS	PASS
11	DI11[5]	93	N24	PASS	PASS
11	DI11[6]	92	N26	PASS	PASS
11	DI11[7]	91	M19	PASS	PASS
11	DI11[8]	90	M20	PASS	PASS
11	DI11[9]	89	M21	PASS	PASS
11	DI11[10]	87	M22	PASS	PASS
11	DI11[11]	86	M24	PASS	PASS
11	DI11[12]	85	M25	PASS	PASS
11	STR11	82	M26	PASS	PASS
12	DI12[0]	99	L24	PASS	PASS
12	DI12[1]	97	L25	PASS	PASS
12	DI12[2]	96	K25	PASS	PASS
12	DI12[3]	95	K26	PASS	PASS
12	DI12[4]	94	M16	PASS	PASS
12	DI12[5]	93	M17	PASS	PASS
12	DI12[6]	92	L17	PASS	PASS
12	DI12[7]	91	L18	PASS	PASS
12	DI12[8]	90	L19	PASS	PASS
12	DI12[9]	89	L20	PASS	PASS
12	DI12[10]	87	K18	PASS	PASS
12	DI12[11]	86	K16	PASS	PASS
12	DI12[12]	85	K17	PASS	PASS
12	STR12	82	J19	PASS	PASS